

# MERN Mastery – Complete 182-Day Roadmap

MERN MASTERY · 1 CR PRODUCT PREP · HARSH VARDHAN GUPTA

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## Cover Page

| <b>Program Title</b>       | MERN Mastery – 6-Month Day-by-Day Bootcamp                                                                                        |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Subtitle</b>            | From JavaScript Fundamentals to System Design Interviews                                                                          |
| <b>Duration</b>            | 182 Days (26 Weeks)                                                                                                               |
| <b>Daily Commitment</b>    | Mon-Fri: 7:30–8:30pm + 9–10pm · Sat-Sun: 9am–12pm + 3–5:30pm + 6–8:30pm                                                           |
| <b>Weekly Hours</b>        | ~26 hours/week                                                                                                                    |
| <b>Total Program Hours</b> | ~470 hours                                                                                                                        |
| <b>Primary Goal</b>        | ~1 CR at product companies (Google L4–L5 / Meta E4–E5 / Amazon SDE2–3)                                                            |
| <b>Target Employers</b>    | Google, Meta, Amazon, Microsoft, Uber, Atlassian, Adobe, Salesforce, Apple, Netflix, Stripe                                       |
| <b>Stack Covered</b>       | JavaScript → TypeScript → React → Next.js → Node.js → Express → MongoDB → PostgreSQL → Redis → Docker → Testing → DSA → LLD → HLD |
| <b>LeetCode Target</b>     | 220+ by Day 168 · 230+ by Day 182                                                                                                 |
| <b>Portfolio Target</b>    | 2 deployed MERN projects (TaskFlow + ShopLite) + Blog + 6 LLD implementations                                                     |
| <b>Start Date</b>          | <b>Mon, 17 Aug 2026</b> (Day 1)                                                                                                   |
| <b>Expected End Date</b>   | <b>Sun, 14 Feb 2027</b> (Day 182)                                                                                                 |

## How to Use This Roadmap

### Daily Workflow (Non-Negotiable)

1. Open this file → find today's day number (or calculate from your start date).
2. Read the theory section linked in the Theory File row (30–60 min weekdays, 1–2 hr weekends).

3. **Complete specific exercises** listed in the Practice File row — type every line, never copy-paste blind.
4. **Solve the LeetCode problem** if scheduled — timed 35–45 min, JavaScript/TypeScript only.
5. **Work on weekend project tasks** if applicable (Sat–Sun blocks).
6. **Check all four boxes** at the bottom of each day before marking complete.
7. **Git commit daily** — even small exercises. Employers audit GitHub activity.
8. **Explain aloud** — 2-minute verbal summary pretends you're in an interview.

## Weekly Rhythm

| Day Type | Focus                                        | Hours  |
|----------|----------------------------------------------|--------|
| Mon–Thu  | New theory + exercises + occasional LC       | 2h/day |
| Friday   | Review + LeetCode heavy                      | 2h/day |
| Saturday | LC batch + project build                     | 8h     |
| Sunday   | Project finish + review + LLD/SD (Month 5–6) | 8h     |

## File Map

| Folder                        | Purpose                                                        |
|-------------------------------|----------------------------------------------------------------|
| <code>detailed/theory/</code> | <b>Complete theory guides</b> (use these, not condensed notes) |
| <code>practice/</code>        | Weekly exercise files with checklists                          |
| <code>notes/</code>           | Condensed reference notes (secondary)                          |
| <code>ROADMAP.md</code>       | Quick-reference summary roadmap                                |
| <b>This file</b>              | Most detailed day-by-day plan for PDF export                   |

## Rules

1. **Never skip LeetCode** — even 1 problem on busy days.
2. **Weekends = build** — theory on weekdays, projects on weekends.
3. **No tutorial hell** — read theory, then code immediately.

4. **One GitHub repo** ( `mern-mastery` ) with weekly folders.
  5. **Track failures** in `REVIEW.md` — revisit on revision days.
  6. **Mark completion** with `[x]` in milestone tracker below.
-

## Time Budget Tables

### Weekly Time Budget (26 weeks)

| Week | Days  | Theme                        | Weekday hrs | Weekend hrs | Total |
|------|-------|------------------------------|-------------|-------------|-------|
| 1    | 1-7   | JavaScript Basics            | 10          | 16          | 26    |
| 2    | 8-14  | Prototypes, Classes, Objects | 10          | 16          | 26    |
| 3    | 15-21 | Async JavaScript             | 10          | 16          | 26    |
| 4    | 22-28 | TypeScript + Git             | 10          | 16          | 26    |
| 5    | 29-35 | React Fundamentals           | 10          | 16          | 26    |
| 6    | 36-42 | React Hooks                  | 10          | 16          | 26    |
| 7    | 43-49 | React Advanced + Router      | 10          | 16          | 26    |
| 8    | 50-56 | Next.js + Deploy             | 10          | 16          | 26    |
| 9    | 57-63 | Node.js + Express            | 10          | 16          | 26    |
| 10   | 64-70 | Authentication (JWT)         | 10          | 16          | 26    |
| 11   | 71-77 | MongoDB                      | 10          | 16          | 26    |
| 12   | 78-84 | PostgreSQL + Redis           | 10          | 16          | 26    |
| 13   | 85-91 | TaskFlow Project (Backend)   | 10          | 16          | 26    |
| 14   | 92-98 | TaskFlow Project             | 10          | 16          | 26    |

|              |            |                                   |            |            |            |
|--------------|------------|-----------------------------------|------------|------------|------------|
|              |            | (Frontend + Docker)               |            |            |            |
| 15           | 99-105     | ShopLite E-commerce               | 10         | 16         | 26         |
| 16           | 106-112    | Testing + CI + Deploy             | 10         | 16         | 26         |
| 17           | 113-119    | Product DSA refresh + easy LLD    | 10         | 16         | 26         |
| 18           | 120-126    | Trees/graphs + LRU/Rate Limiter   | 10         | 16         | 26         |
| 19           | 127-133    | DP, intervals, company-tagged DSA | 10         | 16         | 26         |
| 20           | 134-140    | Hard DSA + backend/frontend MC    | 10         | 16         | 26         |
| 21           | 141-147    | Product HLD foundations + URL     | 10         | 16         | 26         |
| 22           | 148-154    | Feed, notify, chat, payments      | 10         | 16         | 26         |
| 23           | 155-161    | Distributed + YouTube/Drive start | 10         | 16         | 26         |
| 24           | 162-168    | Typeahead, CS pack, FE MC, loops  | 10         | 16         | 26         |
| 25           | 169-175    | Active Interviewing + Mocks       | 10         | 16         | 26         |
| 26           | 176-182    | Revision + Negotiation + Offer    | 10         | 16         | 26         |
| <b>TOTAL</b> | <b>182</b> | <b>Full Program</b>               | <b>260</b> | <b>208</b> | <b>468</b> |

## Monthly Time Budget

| Month | Days    | Focus Area                     | Approx Hours | LC Target EOM | Projects                                   |
|-------|---------|--------------------------------|--------------|---------------|--------------------------------------------|
| 1     | 1-28    | JavaScript, TypeScript, Git    | 104          | 20            | CLI Todo, TS Catalog API                   |
| 2     | 29-56   | React, Next.js                 | 104          | 60            | Form Wizard, User Dir, E-commerce UI, Blog |
| 3     | 57-84   | Backend, Auth, Databases       | 104          | 100           | Auth Microservice, Blog API (Mongo + PG)   |
| 4     | 85-112  | MERN Projects, DevOps          | 104          | 150           | TaskFlow, ShopLite (deployed)              |
| 5     | 113-140 | Product DSA, LLD, frontend MC  | 104          | 200           | 6 LLD + autocomplete widget                |
| 6     | 141-168 | Product HLD, CS, company loops | 104          | 220           | 10 product HLDs, CS pack, STAR             |
| Ext   | 169-182 | Active Interviewing            | 52           | 230+          | Offer pipeline                             |

## Daily Time Split Templates

### Weekday Template (2 hours)

```

0:00 - 0:25  Review yesterday's notes + flashcards
0:25 - 1:15  Read today's theory section (detailed/theory/)
1:15 - 1:45  Complete specific practice exercises
1:45 - 2:00  Git commit + 2-min aloud explanation

```

Friday variant (LC-heavy):

```
0:00 - 0:30    Review week's theory
0:30 - 1:45    Timed LeetCode (1-2 problems)
1:45 - 2:00    Commit + log result in REVIEW.md
```

### Weekend Template (8 hours)

```
0:00 - 2:00    LeetCode batch (2-3 problems, 35 min each + review)
2:00 - 6:00    Project / LLD / System Design build
6:00 - 7:00    Write summary notes in own words
7:00 - 8:00    Plan next week + Git commit + update tracker
```

### Month 5-6 weekend variant (DSA + LLD/SD):

```
0:00 - 3:00    Timed LC batch (3 mediums)
3:00 - 6:00    LLD implementation OR HLD blind design
6:00 - 7:00    Record yourself explaining design/solution
7:00 - 8:00    Update REVIEW.md + milestone tracker
```

## Milestone Tracker

| #  | Milestone                                                   | Target Day | Target Date | Done? |
|----|-------------------------------------------------------------|------------|-------------|-------|
| 1  | GitHub repo live with Week 1 code                           | 7          | 7 Aug 2026  | [ ]   |
| 2  | JavaScript fundamentals solid (Month 1 review $\geq 80\%$ ) | 28         | 28 Aug 2026 | [ ]   |
| 3  | TypeScript Catalog API complete                             | 28         | 28 Aug 2026 | [ ]   |
| 4  | React + Next.js blog deployed (live URL)                    | 56         | 25 Sep 2026 | [ ]   |
| 5  | 60+ LeetCode solved                                         | 56         | 25 Sep 2026 | [ ]   |
| 6  | Auth microservice complete (JWT + refresh)                  | 70         | 9 Oct 2026  | [ ]   |
| 7  | MongoDB + PostgreSQL both implemented                       | 84         | 23 Oct 2026 | [ ]   |
| 8  | 100+ LeetCode solved                                        | 84         | 23 Oct 2026 | [ ]   |
| 9  | TaskFlow deployed (Docker + live URL)                       | 98         | 6 Nov 2026  | [ ]   |
| 10 | ShopLite deployed with payment test                         | 112        | 20 Nov 2026 | [ ]   |
| 11 | 150+ LeetCode solved                                        | 112        | 20 Nov 2026 | [ ]   |
| 12 | CI/CD pipeline green on both projects                       | 112        | 20 Nov 2026 | [ ]   |
| 13 | 6 LLD problems coded from scratch                           | 140        | 18 Dec 2026 | [ ]   |

|    |                                        |     |             |     |
|----|----------------------------------------|-----|-------------|-----|
| 14 | 200+ LeetCode solved                   | 140 | 18 Dec 2026 | [ ] |
| 15 | 10 HLD systems designed on paper       | 168 | 15 Jan 2027 | [ ] |
| 16 | 8 STAR behavioral stories memorized    | 168 | 15 Jan 2027 | [ ] |
| 17 | 220+ LeetCode + applications submitted | 168 | 15 Jan 2027 | [ ] |
| 18 | 10+ company applications tracked       | 173 | 20 Jan 2027 | [ ] |
| 19 | 4+ mock interviews completed           | 175 | 22 Jan 2027 | [ ] |
| 20 | Offer received or final-round pipeline | 182 | 29 Jan 2027 | [ ] |

## Company Application Schedule (Foreign Product Companies)

### Phase 1 – Portfolio Ready (Days 160-168)

| Day | Action                                   | Companies                                  |
|-----|------------------------------------------|--------------------------------------------|
| 167 | Submit first batch (5)                   | Adobe, Salesforce, Microsoft, Amazon, Uber |
| 168 | GitHub + LinkedIn audit before more apps | –                                          |

## Phase 2 – Active Blitz (Days 169-173)

| Day | Apply (2/day)          | Notes                                            |
|-----|------------------------|--------------------------------------------------|
| 169 | Google, Atlassian      | Emphasize DSA count + open source                |
| 170 | Meta, Apple            | Tailor resume: React/Next for Meta, TS for Apple |
| 171 | Netflix, Spotify       | Highlight streaming SD + ShopLite                |
| 172 | Stripe, Razorpay       | Highlight payment integration in ShopLite        |
| 173 | Goldman Sachs, DE Shaw | FinTech angle + DSA strength                     |

## Phase 3 – Sustained Pipeline (Days 174-182)

| Action                        | Frequency                              |
|-------------------------------|----------------------------------------|
| Respond to recruiter InMails  | Within 24 hours                        |
| Complete Online Assessments   | Within 48 hours of invite              |
| Mock interviews (Pramp/peers) | 2-4 during Days 174-175                |
| Follow up on applications     | Weekly email if no response in 2 weeks |
| Referrals                     | Ask 1 connection per target company    |

## Application Tracker Template

| Company       | Tier | Applied | OA | Phone | Onsite | Offer | Notes |
|---------------|------|---------|----|-------|--------|-------|-------|
| Adobe         | A    |         |    |       |        |       |       |
| Salesforce    | A    |         |    |       |        |       |       |
| Microsoft     | A    |         |    |       |        |       |       |
| Google        | S    |         |    |       |        |       |       |
| Amazon        | A    |         |    |       |        |       |       |
| Atlassian     | A    |         |    |       |        |       |       |
| Uber          | A    |         |    |       |        |       |       |
| Meta          | S    |         |    |       |        |       |       |
| Apple         | S    |         |    |       |        |       |       |
| Netflix       | S    |         |    |       |        |       |       |
| Spotify       | A    |         |    |       |        |       |       |
| Stripe        | S    |         |    |       |        |       |       |
| Goldman Sachs | A    |         |    |       |        |       |       |
| DE Shaw       | S    |         |    |       |        |       |       |

Tiers: S = highly selective, A = achievable with this roadmap

## Resource Links – Master List

### Official Documentation

| Resource            | URL                                                                                                                           | Used In |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------|---------|
| MDN JavaScript      | <a href="https://developer.mozilla.org/en-US/docs/Web/JavaScript">https://developer.mozilla.org/en-US/docs/Web/JavaScript</a> | Month 1 |
| TypeScript Handbook | <a href="https://www.typescriptlang.org/docs/handbook/">https://www.typescriptlang.org/docs/handbook/</a>                     | Month 1 |
| React Docs          | <a href="https://react.dev/learn">https://react.dev/learn</a>                                                                 | Month 2 |
| Next.js Docs        | <a href="https://nextjs.org/docs">https://nextjs.org/docs</a>                                                                 | Month 2 |
| Node.js Docs        | <a href="https://nodejs.org/docs/latest/api/">https://nodejs.org/docs/latest/api/</a>                                         | Month 3 |
| Express Guide       | <a href="https://expressjs.com/en/guide/routing.html">https://expressjs.com/en/guide/routing.html</a>                         | Month 3 |
| MongoDB Manual      | <a href="https://www.mongodb.com/docs/manual/">https://www.mongodb.com/docs/manual/</a>                                       | Month 3 |
| PostgreSQL Tutorial | <a href="https://www.postgresqltutorial.com/">https://www.postgresqltutorial.com/</a>                                         | Month 3 |
| Prisma Docs         | <a href="https://www.prisma.io/docs">https://www.prisma.io/docs</a>                                                           | Month 3 |
| Redis Docs          | <a href="https://redis.io/docs/">https://redis.io/docs/</a>                                                                   | Month 3 |
| Docker Docs         | <a href="https://docs.docker.com/get-started/">https://docs.docker.com/get-started/</a>                                       | Month 4 |
| Jest Docs           | <a href="https://jestjs.io/docs/getting-started">https://jestjs.io/docs/getting-started</a>                                   | Month 4 |

## Practice Platforms

| Resource         | URL                                                                             | Purpose                |
|------------------|---------------------------------------------------------------------------------|------------------------|
| LeetCode         | <a href="https://leetcode.com/problemset/">https://leetcode.com/problemset/</a> | DSA daily              |
| NeetCode Roadmap | <a href="https://neetcode.io/roadmap">https://neetcode.io/roadmap</a>           | Pattern reference      |
| Pramp            | <a href="https://www.pramp.com/">https://www.pramp.com/</a>                     | Free mock interviews   |
| Interviewing.io  | <a href="https://interviewing.io/">https://interviewing.io/</a>                 | Anonymous mocks        |
| Frontend Mentor  | <a href="https://www.frontendmentor.io/">https://www.frontendmentor.io/</a>     | UI practice (optional) |

## Deployment & Cloud

| Resource      | URL                                                                       | Purpose              |
|---------------|---------------------------------------------------------------------------|----------------------|
| Vercel        | <a href="https://vercel.com/">https://vercel.com/</a>                     | Frontend deploy      |
| Render        | <a href="https://render.com/">https://render.com/</a>                     | Backend deploy       |
| MongoDB Atlas | <a href="https://www.mongodb.com/atlas">https://www.mongodb.com/atlas</a> | Free MongoDB cluster |
| Neon          | <a href="https://neon.tech/">https://neon.tech/</a>                       | Free PostgreSQL      |
| Upstash       | <a href="https://upstash.com/">https://upstash.com/</a>                   | Free Redis           |
| AWS Free Tier | <a href="https://aws.amazon.com/free/">https://aws.amazon.com/free/</a>   | S3, optional         |

## Salary & Interview Research

| Resource   | URL                                                                     | Purpose               |
|------------|-------------------------------------------------------------------------|-----------------------|
| levels.fyi | <a href="https://www.levels.fyi/">https://www.levels.fyi/</a>           | Compensation data     |
| Glassdoor  | <a href="https://www.glassdoor.co.in/">https://www.glassdoor.co.in/</a> | Company reviews       |
| Blind      | <a href="https://www.teamblind.com/">https://www.teamblind.com/</a>     | Interview experiences |

## Theory Files (This Repo)

| Module            | Complete Guide                                |
|-------------------|-----------------------------------------------|
| Git               | <a href="#">00-git-complete.md</a>            |
| JavaScript        | <a href="#">01-javascript-complete.md</a>     |
| TypeScript        | <a href="#">02-typescript-complete.md</a>     |
| React             | <a href="#">03-react-complete.md</a>          |
| Next.js           | <a href="#">04-nextjs-complete.md</a>         |
| Node.js & Express | <a href="#">05-nodejs-express-complete.md</a> |
| Databases         | <a href="#">06-databases-complete.md</a>      |
| Auth & DevOps     | <a href="#">07-auth-devops-complete.md</a>    |
| DSA Patterns      | <a href="#">08-dsa-patterns-complete.md</a>   |
| Low-Level Design  | <a href="#">09-llid-complete.md</a>           |
| System Design     | <a href="#">10-system-design-complete.md</a>  |

## Month Summaries

### Month 1 – JavaScript, TypeScript, Git (Days 1-28)

**Goal:** Solid JavaScript fundamentals + TypeScript fluency + Git workflow.

| Week | Days  | Key Deliverables                                                 |
|------|-------|------------------------------------------------------------------|
| 1    | 1-7   | Variables → closures → this; 5 LC; GitHub repo                   |
| 2    | 8-14  | Prototypes, classes, map/filter/reduce polyfills; JS cheat sheet |
| 3    | 15-21 | Event loop, promises, async/await; CLI Todo App                  |
| 4    | 22-28 | TypeScript types/generics; Git branching; TS Catalog API         |

**End-of-month exam:** Rewrite Week 1–3 exercises blind; build typed API in 4 hours.

## Month 2 – React + Next.js (Days 29-56)

**Goal:** Production React patterns + Next.js App Router + deployed blog.

| Week | Days  | Key Deliverables                              |
|------|-------|-----------------------------------------------|
| 5    | 29-35 | React fundamentals; form wizard project       |
| 6    | 36-42 | Hooks, custom hooks, context; user directory  |
| 7    | 43-49 | Router, Zustand, performance; e-commerce UI   |
| 8    | 50-56 | Next.js SSR/ISR; blog with MDX; Vercel deploy |

**End-of-month exam:** Build CRUD page with React Hook Form + API in 3 hours.

## Month 3 – Backend + Databases (Days 57-84)

**Goal:** Production Express APIs + JWT auth + MongoDB + PostgreSQL + Redis.

| Week | Days  | Key Deliverables                            |
|------|-------|---------------------------------------------|
| 9    | 57-63 | Express layered architecture; auth skeleton |
| 10   | 64-70 | JWT + refresh rotation; auth microservice   |
| 11   | 71-77 | MongoDB blog API with aggregation           |
| 12   | 78-84 | PostgreSQL rebuild; Redis cache; 100 LC     |

**End-of-month exam:** Explain auth flow + draw Mongo vs SQL tradeoffs in 30 min.

## Month 4 – Projects + DevOps (Days 85-112)

**Goal:** Two deployed flagship projects with CI/CD.

| Week | Days    | Key Deliverables                                     |
|------|---------|------------------------------------------------------|
| 13   | 85-91   | TaskFlow backend: auth, teams, tasks, Socket.io      |
| 14   | 92-98   | TaskFlow frontend + Docker Compose                   |
| 15   | 99-105  | ShopLite e-commerce + Stripe/Razorpay                |
| 16   | 106-112 | Jest + Supertest + RTL + GitHub Actions; deploy both |

**End-of-month exam:** Demo both live URLs; walk through architecture in 15 min.

## Month 5 – DSA + LLD (Days 113-140)

**Goal:** 200+ LC; 6 LLD problems; medium in 35 min average.

| Week | Days    | Key Deliverables                                                  |
|------|---------|-------------------------------------------------------------------|
| 17   | 113-119 | Two pointers, sliding window, hash map; LRU Cache LLD             |
| 18   | 120-126 | Binary search, stack, linked list; Rate Limiter + Parking Lot LLD |
| 19   | 127-133 | Trees, graphs, topo sort; Splitwise + Vending Machine LLD         |
| 20   | 134-140 | Dynamic programming intensive; BookMyShow LLD; 200 LC             |

**End-of-month exam:** 2 timed mediums + 1 LLD in 90 min total.

## Month 6 – System Design + Interviews (Days 141-168)

**Goal:** 10 HLD designs; 8 STAR stories; 220 LC; applications live.

| Week | Days    | Key Deliverables                                       |
|------|---------|--------------------------------------------------------|
| 21   | 141-147 | SD framework; URL Shortener; Paste Bin; feed intro     |
| 22   | 148-154 | Twitter, Instagram, Notification; Kafka basics         |
| 23   | 155-161 | WhatsApp chat; YouTube streaming; CDN                  |
| 24   | 162-168 | Uber, E-commerce, Payment; mocks; apply to 5 companies |

**End-of-month exam:** 45-min blind HLD + full mock interview.

## Weeks 25-26 – Active Interviewing (Days 169-182)

**Goal:** Convert pipeline to offers; negotiate 30+ LPA.

| Days    | Focus                                               |
|---------|-----------------------------------------------------|
| 169-173 | 2 LC/day + 1 SD/day + 2 applications/day            |
| 174-175 | Mock interviews (Pramp/peers)                       |
| 176-178 | Weak pattern + SD revision                          |
| 179-180 | Offer negotiation research + mock HR                |
| 181-182 | Continue interviews; final audit; post-program plan |

# Complete Day-by-Day Plan (Days 1-182)

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# MONTH 1 – JavaScript, TypeScript, Git (Days 1-28)

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Day 1 – Mon |  17 Aug 2026 | 2 hours

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Mon, 17 Aug 2026                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 1 in <a href="#">calendar/mern-mastery-study.ics</a>  |
| Calendar Week           | Week 1 of 26                                              |
| Month Phase             | Month 1 – JS/TS/Git                                       |
| Exact Topic             | Variables, Data Types, Coercion, and Type Checking        |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a> |
| Theory Section          | Sections 2-3: Variables, Data Types, Coercion & Equality  |
| Practice File           | <a href="#">practice/week-01-exercises.md</a>             |
| Specific Exercises      | Exercise 1 (Variable quiz), Exercise 2 (FizzBuzz)         |
| LeetCode                | <a href="#">LC 1: Two Sum</a>                             |
| LeetCode Note           | LC scheduled Day 5 – preview problem statement today      |
| Weekend / Project Tasks | None (weekday)                                            |
| Time Split Today        | 0:00-0:30 review setup                                    |

## What Success Looks Like Today

You can explain var vs let vs const, predict coercion outputs (`[] + {}`, `typeof null`), and complete FizzBuzz without Googling syntax.

- ☐ Theory section read and highlighted

- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

## Day 2 – Tue | 18 Aug 2026 | 2 hours

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Tue, 18 Aug 2026                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 2 in <a href="#">calendar/mern-mastery-study.ics</a>  |
| Calendar Week           | Week 1 of 26                                              |
| Month Phase             | Month 1 – JS/TS/Git                                       |
| Exact Topic             | Functions, Scope, Hoisting, and Function Expressions      |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a> |
| Theory Section          | Sections 5-6: Functions, Scope                            |
| Practice File           | <a href="#">practice/week-01-exercises.md</a>             |
| Specific Exercises      | Exercise 3 setup – closure counter skeleton               |
| LeetCode                | None (save LC for Fri/Sat)                                |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:25 review Day 1                                    |

### What Success Looks Like Today

You can draw scope chain for nested functions and explain temporal dead zone for let/const.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 3 – Wed | 17 19 Aug 2026 | 2 hours**

| Field                   | Detail                                                                    |
|-------------------------|---------------------------------------------------------------------------|
| Study Date              | Wed, 19 Aug 2026                                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                 |
| Calendar                | Day 3 in <a href="#">calendar/mern-mastery-study.ics</a>                  |
| Calendar Week           | Week 1 of 26                                                              |
| Month Phase             | Month 1 – JS/TS/Git                                                       |
| Exact Topic             | Closures – lexical environment, practical patterns                        |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a>                 |
| Theory Section          | Section 7: Closures                                                       |
| Practice File           | <a href="#">practice/week-01-exercises.md</a>                             |
| Specific Exercises      | Exercise 3 (Closure counter – full implementation),<br>Exercise 4 preview |
| LeetCode                | None                                                                      |
| Weekend / Project Tasks | None                                                                      |
| Time Split Today        | 0:00-0:20 review scope                                                    |

**What Success Looks Like Today**

createCounter works with increment/decrement/reset/getValue; you can explain closure in one sentence to an interviewer.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 4 – Thu | 20 Aug 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Thu, 20 Aug 2026                                            |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 4 in <a href="#">calendar/mern-mastery-study.ics</a>    |
| Calendar Week           | Week 1 of 26                                                |
| Month Phase             | Month 1 – JS/TS/Git                                         |
| Exact Topic             | The <b>this</b> Keyword – 4 binding rules, arrow functions  |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a>   |
| Theory Section          | Section 8: The this Keyword                                 |
| Practice File           | <a href="#">practice/week-01-exercises.md</a>               |
| Specific Exercises      | Exercise 4 (this prediction – write outputs before running) |
| LeetCode                | None                                                        |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:25 review closures                                   |

**What Success Looks Like Today**

You predict all Exercise 4 outputs correctly and explain implicit vs explicit binding.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 5 – Fri | 21 Aug 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Fri, 21 Aug 2026                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 5 in <a href="#">calendar/mern-mastery-study.ics</a>  |
| Calendar Week           | Week 1 of 26                                              |
| Month Phase             | Month 1 – JS/TS/Git                                       |
| Exact Topic             | Week 1 Review + First LeetCode Problem                    |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a> |
| Theory Section          | Sections 2-8 review – skim weak sections                  |
| Practice File           | <a href="#">practice/week-01-exercises.md</a>             |
| Specific Exercises      | Redo any failed exercises from Ex 1-4                     |
| LeetCode                | <a href="#">LC 1: Two Sum</a>                             |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:30 review notes                                    |

**What Success Looks Like Today**

LC 1 solved in JavaScript with  $O(n)$  hash map approach; you can explain brute force vs optimal.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 6 – Sat | 22 Aug 2026 | 8 hours**

| Field                   | Detail                                                                                                                  |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Study Date              | Sat, 22 Aug 2026                                                                                                        |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                                                 |
| Calendar                | Day 6 in <a href="#">calendar/mern-mastery-study.ics</a>                                                                |
| Calendar Week           | Week 1 of 26                                                                                                            |
| Month Phase             | Month 1 – JS/TS/Git                                                                                                     |
| Exact Topic             | Prototypes Intro + Array Fundamentals (Weekend Block 1)                                                                 |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a>                                                               |
| Theory Section          | Sections 9-10: Prototypes & Classes, Arrays                                                                             |
| Practice File           | <a href="#">practice/week-01-exercises.md</a>                                                                           |
| Specific Exercises      | Exercise 5 (Array manipulation: dedupe, 2nd largest, rotate)                                                            |
| LeetCode                | <a href="#">LC 242: Valid Anagram</a> , <a href="#">LC 121: Best Time to Buy and Sell Stock</a>                         |
| Weekend / Project Tasks | Block 3 (2h): Create GitHub repo <a href="#">mern-mastery</a> , push week-01 exercises, add README with Day 1-5 summary |
| Time Split Today        | 0:00-2:00 theory + Ex 5                                                                                                 |

**What Success Looks Like Today**

Ex 5 complete; LC 242 & 121 solved; GitHub repo live with week-01 folder committed.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 7 – Sun | 23 Aug 2026 | 8 hours**

| Field                   | Detail                                                                                                        |
|-------------------------|---------------------------------------------------------------------------------------------------------------|
| Study Date              | Sun, 23 Aug 2026                                                                                              |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                                       |
| Calendar                | Day 7 in <a href="#">calendar/mern-mastery-study.ics</a>                                                      |
| Calendar Week           | Week 1 of 26                                                                                                  |
| Month Phase             | Month 1 – JS/TS/Git                                                                                           |
| Exact Topic             | Array Methods Deep Dive + Week 1 Consolidation                                                                |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a>                                                     |
| Theory Section          | Section 10: Arrays – map, filter, reduce, find, some, every                                                   |
| Practice File           | <a href="#">practice/week-01-exercises.md</a>                                                                 |
| Specific Exercises      | 10 map/filter/reduce drills (custom arrays of objects)                                                        |
| LeetCode                | <a href="#">LC 217: Contains Duplicate</a> , <a href="#">LC 53: Maximum Subarray</a>                          |
| Weekend / Project Tasks | Block 3 (3h): Rewrite ALL Week 1 exercises from memory without looking – self-test; log failures in REVIEW.md |
| Time Split Today        | 0:00-3:00 array drills                                                                                        |

**What Success Looks Like Today**

Can implement map/filter/reduce from memory; 5 LC total this week; Week 1 self-test score  $\geq 80\%$ .

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 8 – Mon | 24 Aug 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Mon, 24 Aug 2026                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 8 in <a href="#">calendar/mern-mastery-study.ics</a>  |
| Calendar Week           | Week 2 of 26                                              |
| Month Phase             | Month 1 – JS/TS/Git                                       |
| Exact Topic             | Prototype Chain and the <b>new</b> Keyword                |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a> |
| Theory Section          | Section 9: Prototypes – chain, Object.create, new keyword |
| Practice File           | <a href="#">practice/week-02-exercises.md</a>             |
| Specific Exercises      | Exercise 3 (Class inheritance – Animal → Dog → GuideDog)  |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:25 review                                          |

**What Success Looks Like Today**

You can diagram prototype chain and explain what **new** does step-by-step.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 9 – Tue | 25 Aug 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Tue, 25 Aug 2026                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 9 in <a href="#">calendar/mern-mastery-study.ics</a>  |
| Calendar Week           | Week 2 of 26                                              |
| Month Phase             | Month 1 – JS/TS/Git                                       |
| Exact Topic             | ES6 Classes, Inheritance, and instanceof                  |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a> |
| Theory Section          | Section 9: ES6 Classes (syntactic sugar over prototypes)  |
| Practice File           | <a href="#">practice/week-02-exercises.md</a>             |
| Specific Exercises      | Exercise 3 complete – speak(), fetch(), guide() methods   |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:20 review                                          |

**What Success Looks Like Today**

Class hierarchy passes instanceof checks; you explain classes vs prototypes.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 10 – Wed | 26 Aug 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Wed, 26 Aug 2026                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 10 in <a href="#">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 2 of 26                                              |
| Month Phase             | Month 1 – JS/TS/Git                                       |
| Exact Topic             | Objects, Destructuring, Spread, and Rest                  |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a> |
| Theory Section          | Section 11: Objects – Complete Reference                  |
| Practice File           | <a href="#">practice/week-02-exercises.md</a>             |
| Specific Exercises      | Exercise 4 (groupBy function)                             |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:25 review                                          |

**What Success Looks Like Today**

groupBy works for any key; you use destructuring/spread fluently in 5 examples.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 11 – Thu | 27 Aug 2026 | 2 hours**

| Field                   | Detail                                                     |
|-------------------------|------------------------------------------------------------|
| Study Date              | Thu, 27 Aug 2026                                           |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                  |
| Calendar                | Day 11 in <a href="#">calendar/mern-mastery-study.ics</a>  |
| Calendar Week           | Week 2 of 26                                               |
| Month Phase             | Month 1 – JS/TS/Git                                        |
| Exact Topic             | Implement map, filter, reduce from Scratch                 |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a>  |
| Theory Section          | Section 10: Arrays – internal iteration patterns           |
| Practice File           | <a href="#">practice/week-02-exercises.md</a>              |
| Specific Exercises      | Exercise 1 (myMap, myFilter, myReduce – no native methods) |
| LeetCode                | None                                                       |
| Weekend / Project Tasks | None                                                       |
| Time Split Today        | 0:00-0:20 review                                           |

**What Success Looks Like Today**

All three polyfills pass test cases including edge cases (empty array, no initial in reduce).

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 12 – Fri | 28 Aug 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Fri, 28 Aug 2026                                            |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 12 in <a href="#">calendar/mern-mastery-study.ics</a>   |
| Calendar Week           | Week 2 of 26                                                |
| Month Phase             | Month 1 – JS/TS/Git                                         |
| Exact Topic             | LeetCode + Two Pointer Preview                              |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a> |
| Theory Section          | Pattern 1: Two Pointers – intro section                     |
| Practice File           | <a href="#">practice/week-02-exercises.md</a>               |
| Specific Exercises      | Review Ex 1 implementations                                 |
| LeetCode                | <a href="#">LC 88: Merge Sorted Array</a>                   |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:30 two-pointer theory                                |

**What Success Looks Like Today**

LC 88 solved using two pointers from end; you explain why backward merge works.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 13 – Sat | 29 Aug 2026 | 8 hours**

| Field                   | Detail                                                                                 |
|-------------------------|----------------------------------------------------------------------------------------|
| Study Date              | Sat, 29 Aug 2026                                                                       |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                |
| Calendar                | Day 13 in <a href="#">calendar/mern-mastery-study.ics</a>                              |
| Calendar Week           | Week 2 of 26                                                                           |
| Month Phase             | Month 1 – JS/TS/Git                                                                    |
| Exact Topic             | Flatten Array, Curry, Optional Chaining Drills                                         |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a>                              |
| Theory Section          | Section 19: Utility Patterns – Curry, Deep Clone                                       |
| Practice File           | <a href="#">practice/week-02-exercises.md</a>                                          |
| Specific Exercises      | Exercise 2 (flatten depth N), Exercise 5 (curry), 10 optional chaining object drills   |
| LeetCode                | <a href="#">LC 27: Move Zeroes</a> , <a href="#">LC 977: Squares of a Sorted Array</a> |
| Weekend / Project Tasks | Push week-02 folder to GitHub                                                          |
| Time Split Today        | 0:00-4:00 Ex 2, 5, optional chaining                                                   |

**What Success Looks Like Today**

flatten works depth 5+; curry supports partial application; LC 27 & 977 done.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

## Day 14 – Sun | 30 Aug 2026 | 8 hours

| Field                   | Detail                                                                                                                                                      |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Study Date              | Sun, 30 Aug 2026                                                                                                                                            |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                                                                                     |
| Calendar                | Day 14 in <a href="#">calendar/mern-mastery-study.ics</a>                                                                                                   |
| Calendar Week           | Week 2 of 26                                                                                                                                                |
| Month Phase             | Month 1 – JS/TS/Git                                                                                                                                         |
| Exact Topic             | Week 2 Review + JS Cheat Sheet                                                                                                                              |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a>                                                                                                   |
| Theory Section          | Sections 9-11, 19 review                                                                                                                                    |
| Practice File           | <a href="#">practice/week-02-exercises.md</a>                                                                                                               |
| Specific Exercises      | Redo all failed exercises; write personal JS cheat sheet (1-2 pages)                                                                                        |
| LeetCode                | <a href="#">LC 238: Product of Array Except Self</a> , <a href="#">LC 9: Palindrome Number</a> , <a href="#">LC 26: Remove Duplicates from Sorted Array</a> |
| Weekend / Project Tasks | Create <a href="#">notes/personal-js-cheatsheet.md</a> in repo                                                                                              |
| Time Split Today        | 0:00-3:00 review + redo                                                                                                                                     |

### What Success Looks Like Today

Cheat sheet covers prototypes, closures, this, arrays; 3 LC solved; week-02 complete.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 15 – Mon | 31 Aug 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Mon, 31 Aug 2026                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 15 in <a href="#">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 3 of 26                                              |
| Month Phase             | Month 1 – JS/TS/Git                                       |
| Exact Topic             | Event Loop – Call Stack, Microtasks, Macrotasks           |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a> |
| Theory Section          | Section 14: The Event Loop                                |
| Practice File           | <a href="#">practice/week-03-exercises.md</a>             |
| Specific Exercises      | Exercise 1 (predict output A-F before running)            |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:25 review                                          |

**What Success Looks Like Today**

Exercise 1 output predicted correctly; you draw event loop diagram from memory.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 16 – Tue | 17 1 Sep 2026 | 2 hours**

| Field                   | Detail                                                                   |
|-------------------------|--------------------------------------------------------------------------|
| Study Date              | Tue, 1 Sep 2026                                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                |
| Calendar                | Day 16 in <a href="#">calendar/mern-mastery-study.ics</a>                |
| Calendar Week           | Week 3 of 26                                                             |
| Month Phase             | Month 1 – JS/TS/Git                                                      |
| Exact Topic             | Promises – then, catch, finally, chaining                                |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a>                |
| Theory Section          | Section 15: Promises                                                     |
| Practice File           | <a href="#">practice/week-03-exercises.md</a>                            |
| Specific Exercises      | Write 3 promise chains: fetch mock, sequential delays, error propagation |
| LeetCode                | None                                                                     |
| Weekend / Project Tasks | None                                                                     |
| Time Split Today        | 0:00-0:20 review event loop                                              |

**What Success Looks Like Today**

Three promise chains work; you explain Promise states (pending/fulfilled/rejected).

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 17 – Wed | 17 2 Sep 2026 | 2 hours**

| Field                   | Detail                                                        |
|-------------------------|---------------------------------------------------------------|
| Study Date              | Wed, 2 Sep 2026                                               |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                     |
| Calendar                | Day 17 in <a href="#">calendar/mern-mastery-study.ics</a>     |
| Calendar Week           | Week 3 of 26                                                  |
| Month Phase             | Month 1 – JS/TS/Git                                           |
| Exact Topic             | async/await, try/catch, Fetch API                             |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a>     |
| Theory Section          | Sections 16-17: async/await, Fetch API & JSON                 |
| Practice File           | <a href="#">practice/week-03-exercises.md</a>                 |
| Specific Exercises      | Fetch API exercise – get JSONPlaceholder posts, handle errors |
| LeetCode                | None                                                          |
| Weekend / Project Tasks | None                                                          |
| Time Split Today        | 0:00-0:25 review promises                                     |

**What Success Looks Like Today**

Async function fetches and displays data with proper try/catch error handling.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 18 – Thu | 3 Sep 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Thu, 3 Sep 2026                                           |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 18 in <a href="#">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 3 of 26                                              |
| Month Phase             | Month 1 – JS/TS/Git                                       |
| Exact Topic             | Promise.all, Promise.race, Promise.allSettled             |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a> |
| Theory Section          | Section 15: Promise combinators subsection                |
| Practice File           | <a href="#">practice/week-03-exercises.md</a>             |
| Specific Exercises      | Exercise 4 (Promise.all polyfill)                         |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:20 review async                                    |

**What Success Looks Like Today**

promiseAll polyfill handles resolve/reject correctly with empty array edge case.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 19 – Fri | 4 Sep 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Fri, 4 Sep 2026                                             |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 19 in <a href="#">calendar/mern-mastery-study.ics</a>   |
| Calendar Week           | Week 3 of 26                                                |
| Month Phase             | Month 1 – JS/TS/Git                                         |
| Exact Topic             | Stack Pattern – Valid Parentheses                           |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a> |
| Theory Section          | Pattern 4: Stack – Valid Parentheses pattern                |
| Practice File           | <a href="#">practice/week-03-exercises.md</a>               |
| Specific Exercises      | Exercise 5 preview (retry with backoff)                     |
| LeetCode                | <a href="#">LC 20: Valid Parentheses</a>                    |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:30 stack theory                                      |

**What Success Looks Like Today**

LC 20 solved with stack in  $O(n)$ ; you explain matching bracket logic aloud.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

## Day 20 – Sat | 5 Sep 2026 | 8 hours

| Field                   | Detail                                                                               |
|-------------------------|--------------------------------------------------------------------------------------|
| Study Date              | Sat, 5 Sep 2026                                                                      |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                              |
| Calendar                | Day 20 in <a href="#">calendar/mern-mastery-study.ics</a>                            |
| Calendar Week           | Week 3 of 26                                                                         |
| Month Phase             | Month 1 – JS/TS/Git                                                                  |
| Exact Topic             | Debounce, Throttle, bind/call/apply                                                  |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a>                            |
| Theory Section          | Section 19: Debounce & Throttle; Section 8: this binding                             |
| Practice File           | <a href="#">practice/week-03-exercises.md</a>                                        |
| Specific Exercises      | Exercise 2 (debounce), Exercise 3 (throttle), Exercise 6 (myBind)                    |
| LeetCode                | <a href="#">LC 206: Reverse Linked List</a> , <a href="#">LC 155: Min Stack</a>      |
| Weekend / Project Tasks | Block 3 (3h): Start CLI Todo App – project structure, todos.json schema, add command |
| Time Split Today        | 0:00-3:00 Ex 2,3,6                                                                   |

### What Success Looks Like Today

Debounce/throttle tested with simulated input; CLI Todo add works.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 21 – Sun | 6 Sep 2026 | 8 hours**

| Field                   | Detail                                                                                       |
|-------------------------|----------------------------------------------------------------------------------------------|
| Study Date              | Sun, 6 Sep 2026                                                                              |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                      |
| Calendar                | Day 21 in <a href="#">calendar/mern-mastery-study.ics</a>                                    |
| Calendar Week           | Week 3 of 26                                                                                 |
| Month Phase             | Month 1 – JS/TS/Git                                                                          |
| Exact Topic             | Finish CLI Todo App + Event Loop Mastery                                                     |
| Theory File             | <a href="#">detailed/theory/01-javascript-complete.md</a>                                    |
| Theory Section          | Section 14 review + Section 18: ES Modules vs CommonJS                                       |
| Practice File           | <a href="#">practice/week-03-exercises.md</a>                                                |
| Specific Exercises      | CLI Todo: list, complete, delete commands; async file I/O                                    |
| LeetCode                | <a href="#">LC 14: Longest Common Prefix</a> , <a href="#">LC 21: Merge Two Sorted Lists</a> |
| Weekend / Project Tasks | Complete CLI Todo App – full CRUD + file persistence in todos.json                           |
| Time Split Today        | 0:00-4:00 CLI Todo                                                                           |

**What Success Looks Like Today**

CLI Todo fully functional; 10 event loop prediction problems scored  $\geq 8/10$ .

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 22 – Mon | 7 Sep 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Mon, 7 Sep 2026                                           |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 22 in <a href="#">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 4 of 26                                              |
| Month Phase             | Month 1 – JS/TS/Git                                       |
| Exact Topic             | TypeScript Setup, Basic Types, Type Inference             |
| Theory File             | <a href="#">detailed/theory/02-typescript-complete.md</a> |
| Theory Section          | Sections 1-2: Setup, Basic Types                          |
| Practice File           | <a href="#">practice/week-04-exercises.md</a>             |
| Specific Exercises      | Exercise 1 (Convert processUsers to typed TS)             |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:25 review async JS                                 |

**What Success Looks Like Today**

TS project compiles with strict:false; processUsers fully typed with interfaces.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 23 – Tue | 8 Sep 2026 | 2 hours**

| Field                   | Detail                                                                  |
|-------------------------|-------------------------------------------------------------------------|
| Study Date              | Tue, 8 Sep 2026                                                         |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                               |
| Calendar                | Day 23 in <a href="#">calendar/mern-mastery-study.ics</a>               |
| Calendar Week           | Week 4 of 26                                                            |
| Month Phase             | Month 1 – JS/TS/Git                                                     |
| Exact Topic             | Interfaces vs Type Aliases, Union & Intersection Types                  |
| Theory File             | <a href="#">detailed/theory/02-typescript-complete.md</a>               |
| Theory Section          | Section 3: Interfaces vs Types                                          |
| Practice File           | <a href="#">practice/week-04-exercises.md</a>                           |
| Specific Exercises      | Define 5 interfaces: User, Product, Order, ApiResponse, PaginatedResult |
| LeetCode                | None                                                                    |
| Weekend / Project Tasks | None                                                                    |
| Time Split Today        | 0:00-0:20 review                                                        |

**What Success Looks Like Today**

You explain when to use interface vs type; 5 interfaces compile with no any.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 24 – Wed | 9 Sep 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Wed, 9 Sep 2026                                           |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 24 in <a href="#">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 4 of 26                                              |
| Month Phase             | Month 1 – JS/TS/Git                                       |
| Exact Topic             | Generics – functions, interfaces, constraints             |
| Theory File             | <a href="#">detailed/theory/02-typescript-complete.md</a> |
| Theory Section          | Section 5: Generics                                       |
| Practice File           | <a href="#">practice/week-04-exercises.md</a>             |
| Specific Exercises      | Exercise 2 (Generic apiCall wrapper)                      |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:25 review                                          |

**What Success Looks Like Today**

apiCall works with multiple endpoints; you explain generic constraint syntax.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 25 – Thu | 10 Sep 2026 | 2 hours**

| Field                   | Detail                                                                              |
|-------------------------|-------------------------------------------------------------------------------------|
| Study Date              | Thu, 10 Sep 2026                                                                    |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                           |
| Calendar                | Day 25 in <a href="#">calendar/mern-mastery-study.ics</a>                           |
| Calendar Week           | Week 4 of 26                                                                        |
| Month Phase             | Month 1 – JS/TS/Git                                                                 |
| Exact Topic             | Utility Types, Type Guards, Discriminated Unions                                    |
| Theory File             | <a href="#">detailed/theory/02-typescript-complete.md</a>                           |
| Theory Section          | Sections 6-7: Utility Types, Type Guards                                            |
| Practice File           | <a href="#">practice/week-04-exercises.md</a>                                       |
| Specific Exercises      | Exercise 3 (ProductCreate/Update/Public), Ex 4 (isAdmin guard), Ex 5 (Result union) |
| LeetCode                | None                                                                                |
| Weekend / Project Tasks | None                                                                                |
| Time Split Today        | 0:00-0:20 review generics                                                           |

**What Success Looks Like Today**

All 3 utility type exercises compile; type guard narrows correctly in if block.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 26 – Fri | 11 Sep 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Fri, 11 Sep 2026                                            |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 26 in <a href="#">calendar/mern-mastery-study.ics</a>   |
| Calendar Week           | Week 4 of 26                                                |
| Month Phase             | Month 1 – JS/TS/Git                                         |
| Exact Topic             | Binary Search Pattern                                       |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a> |
| Theory Section          | Pattern 2: Binary Search                                    |
| Practice File           | <a href="#">practice/week-04-exercises.md</a>               |
| Specific Exercises      | Review TS exercises                                         |
| LeetCode                | <a href="#">LC 704: Binary Search</a>                       |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:30 binary search theory                              |

**What Success Looks Like Today**

LC 704 solved iteratively and recursively; you state  $O(\log n)$  invariant.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 27 – Sat | 12 Sep 2026 | 8 hours**

| Field                   | Detail                                                                                    |
|-------------------------|-------------------------------------------------------------------------------------------|
| Study Date              | Sat, 12 Sep 2026                                                                          |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                   |
| Calendar                | Day 27 in <a href="#">calendar/mern-mastery-study.ics</a>                                 |
| Calendar Week           | Week 4 of 26                                                                              |
| Month Phase             | Month 1 – JS/TS/Git                                                                       |
| Exact Topic             | Git Branching, Merge, Conflicts + TS Catalog API Start                                    |
| Theory File             | <a href="#">detailed/theory/00-git-complete.md</a>                                        |
| Theory Section          | Sections 3-5: Branching, Merging, Conflict Resolution                                     |
| Practice File           | <a href="#">practice/week-04-exercises.md</a>                                             |
| Specific Exercises      | Git Exercises 1-5: init, branches, 5 commits, intentional conflict, .gitignore            |
| LeetCode                | <a href="#">LC 35: Search Insert Position</a> , <a href="#">LC 278: First Bad Version</a> |
| Weekend / Project Tasks | Block 2 (3h): Start TS Catalog API – Express + TS setup, GET /products stub               |
| Time Split Today        | 0:00-3:00 Git drills                                                                      |

**What Success Looks Like Today**

Merge conflict resolved manually; Catalog API server runs; 2 LC done.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 28 – Sun | 13 Sep 2026 | 8 hours**

| Field                   | Detail                                                                             |
|-------------------------|------------------------------------------------------------------------------------|
| Study Date              | Sun, 13 Sep 2026                                                                   |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                            |
| Calendar                | Day 28 in <a href="#">calendar/mern-mastery-study.ics</a>                          |
| Calendar Week           | Week 4 of 26                                                                       |
| Month Phase             | Month 1 – JS/TS/Git                                                                |
| Exact Topic             | Month 1 Capstone – TS Catalog API + Review                                         |
| Theory File             | <a href="#">detailed/theory/02-typescript-complete.md</a>                          |
| Theory Section          | Section 9: tsconfig strict mode                                                    |
| Practice File           | <a href="#">practice/week-04-exercises.md</a>                                      |
| Specific Exercises      | Finish Catalog API: CRUD + Zod validation; 50 JS flashcards self-test              |
| LeetCode                | Review LC 1-20 – list any unsolved in REVIEW.md                                    |
| Weekend / Project Tasks | Finish TS Catalog API – CRUD /products with Zod, error middleware, strict tsconfig |
| Time Split Today        | 0:00-4:00 Catalog API                                                              |

**What Success Looks Like Today**

Catalog API complete with validation; Month 1 flashcards  $\geq 40/50$ ; GitHub month-1 tag created.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

# MONTH 2 – React + Next.js (Days 29–56)

MERN MASTERY · 1 CR PRODUCT PREP · HARSH VARDHAN GUPTA

Day 29 – Mon |  14 Sep 2026 | 2 hours

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Mon, 14 Sep 2026                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 29 in <a href="#">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 5 of 26                                              |
| Month Phase             | Month 2 – React/Next.js                                   |
| Exact Topic             | What is React, JSX, Virtual DOM, Component Model          |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>      |
| Theory Section          | Sections 1–3: Intro, JSX, Components                      |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>          |
| Specific Exercises      | Week 5 Exercise 1: Create Vite + React + TS app           |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:25 Month 1 review                                  |

## What Success Looks Like Today

Vite app runs; you explain JSX vs HTML and why React uses virtual DOM.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 30 – Tue | 15 Sep 2026 | 2 hours**

| Field                   | Detail                                                  |
|-------------------------|---------------------------------------------------------|
| Study Date              | Tue, 15 Sep 2026                                        |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                               |
| Calendar                | Day 30 in <code>calendar/mern-mastery-study.ics</code>  |
| Calendar Week           | Week 5 of 26                                            |
| Month Phase             | Month 2 – React/Next.js                                 |
| Exact Topic             | Props, State, useState Hook                             |
| Theory File             | <code>detailed/theory/03-react-complete.md</code>       |
| Theory Section          | Section 4: Props and State                              |
| Practice File           | <code>practice/week-05-08-exercises.md</code>           |
| Specific Exercises      | Week 5 Exercise 2: Counter, Toggle, Greeting components |
| LeetCode                | None                                                    |
| Weekend / Project Tasks | None                                                    |
| Time Split Today        | 0:00-0:20 review                                        |

**What Success Looks Like Today**

Three components work; you explain unidirectional data flow.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 31 – Wed | 17 16 Sep 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Wed, 16 Sep 2026                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 31 in <a href="#">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 5 of 26                                              |
| Month Phase             | Month 2 – React/Next.js                                   |
| Exact Topic             | Lists, Keys, Conditional Rendering                        |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>      |
| Theory Section          | Section 7: Lists, Keys, Conditional Rendering             |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>          |
| Specific Exercises      | Week 5 Exercise 3: Todo list (add, delete, toggle)        |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:25 review                                          |

**What Success Looks Like Today**

Todo list renders with stable keys; no index-as-key for dynamic lists.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 32 – Thu | 17 Sep 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Thu, 17 Sep 2026                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 32 in <a href="#">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 5 of 26                                              |
| Month Phase             | Month 2 – React/Next.js                                   |
| Exact Topic             | useEffect – mount, update, cleanup                        |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>      |
| Theory Section          | Section 5: useEffect                                      |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>          |
| Specific Exercises      | Week 5 Exercise 4: Document title sync + fetch on mount   |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:20 review                                          |

**What Success Looks Like Today**

useEffect updates title; fetch runs once on mount with cleanup explained.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 33 – Fri | 17 18 Sep 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Fri, 18 Sep 2026                                            |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 33 in <a href="#">calendar/mern-mastery-study.ics</a>   |
| Calendar Week           | Week 5 of 26                                                |
| Month Phase             | Month 2 – React/Next.js                                     |
| Exact Topic             | Linked List – Merge Two Sorted Lists                        |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a> |
| Theory Section          | Pattern 5: Linked List – merge pattern                      |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>            |
| Specific Exercises      | Week 5 Exercise 5: Conditional rendering patterns           |
| LeetCode                | <a href="#">LC 21: Merge Two Sorted Lists</a>               |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:30 LL theory                                         |

**What Success Looks Like Today**

LC 21 solved with dummy head technique; conditional rendering demo complete.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 34 – Sat |  19 Sep 2026 | 8 hours**

| Field                   | Detail                                                                            |
|-------------------------|-----------------------------------------------------------------------------------|
| Study Date              | Sat, 19 Sep 2026                                                                  |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                           |
| Calendar                | Day 34 in <a href="#">calendar/mern-mastery-study.ics</a>                         |
| Calendar Week           | Week 5 of 26                                                                      |
| Month Phase             | Month 2 – React/Next.js                                                           |
| Exact Topic             | Controlled Forms + Vite Project Structure                                         |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>                              |
| Theory Section          | Section 8: Forms – Controlled Components                                          |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>                                  |
| Specific Exercises      | Build registration form with validation messages                                  |
| LeetCode                | <a href="#">LC 26: Remove Duplicates</a> , <a href="#">LC 27: Remove Element</a>  |
| Weekend / Project Tasks | Block 3 (2h): Finalize Vite + React + TS project structure, ESLint, folder layout |
| Time Split Today        | 0:00-4:00 forms                                                                   |

**What Success Looks Like Today**

Registration form validates on submit; project structure documented in README.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 35 – Sun | 20 Sep 2026 | 8 hours**

| Field                   | Detail                                                                                                           |
|-------------------------|------------------------------------------------------------------------------------------------------------------|
| Study Date              | Sun, 20 Sep 2026                                                                                                 |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                                          |
| Calendar                | Day 35 in <a href="#">calendar/mern-mastery-study.ics</a>                                                        |
| Calendar Week           | Week 5 of 26                                                                                                     |
| Month Phase             | Month 2 – React/Next.js                                                                                          |
| Exact Topic             | Multi-Step Form Wizard (Weekend Project)                                                                         |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>                                                             |
| Theory Section          | Section 8: Multi-step form state patterns                                                                        |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>                                                                 |
| Specific Exercises      | Weekend project: 3-step wizard with persisted state between steps                                                |
| LeetCode                | <a href="#">LC 58: Length of Last Word</a> , <a href="#">LC 66: Plus One</a> , <a href="#">LC 67: Add Binary</a> |
| Weekend / Project Tasks | Multi-step form wizard – 3 steps (Personal, Account, Review), state persisted, back/next navigation              |
| Time Split Today        | 0:00-5:00 wizard project                                                                                         |

**What Success Looks Like Today**

Wizard complete with validation per step; 3 LC solved; deployed to GitHub.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 36 – Mon | 21 Sep 2026 | 2 hours**

| Field                   | Detail                                                       |
|-------------------------|--------------------------------------------------------------|
| Study Date              | Mon, 21 Sep 2026                                             |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                    |
| Calendar                | Day 36 in <a href="#">calendar/mern-mastery-study.ics</a>    |
| Calendar Week           | Week 6 of 26                                                 |
| Month Phase             | Month 2 – React/Next.js                                      |
| Exact Topic             | useRef, useMemo, useCallback                                 |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>         |
| Theory Section          | Section 6: useRef, useMemo, useCallback                      |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>             |
| Specific Exercises      | Week 6: Optimize a list component – memoize expensive filter |
| LeetCode                | None                                                         |
| Weekend / Project Tasks | None                                                         |
| Time Split Today        | 0:00-0:25 review                                             |

**What Success Looks Like Today**

You explain when to use each hook; list re-render count reduced with useMemo.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 37 – Tue | 22 Sep 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Tue, 22 Sep 2026                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 37 in <a href="#">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 6 of 26                                              |
| Month Phase             | Month 2 – React/Next.js                                   |
| Exact Topic             | Custom Hooks – useFetch, useDebounce, useLocalStorage     |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>      |
| Theory Section          | Section 6: Custom Hooks                                   |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>          |
| Specific Exercises      | Week 6 Ex 1-3: useFetch, useLocalStorage, useDebounce     |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:20 review                                          |

**What Success Looks Like Today**

Three custom hooks extracted and reused in 2+ components each.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 38 – Wed | 23 Sep 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Wed, 23 Sep 2026                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 38 in <a href="#">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 6 of 26                                              |
| Month Phase             | Month 2 – React/Next.js                                   |
| Exact Topic             | useContext and Context API                                |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>      |
| Theory Section          | Section 6: Context API                                    |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>          |
| Specific Exercises      | Week 6 Ex 4: ThemeProvider (dark/light toggle)            |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:25 review                                          |

**What Success Looks Like Today**

Theme toggles app-wide without prop drilling; you explain Provider pattern.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 39 – Thu | 24 Sep 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Thu, 24 Sep 2026                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 39 in <a href="#">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 6 of 26                                              |
| Month Phase             | Month 2 – React/Next.js                                   |
| Exact Topic             | React Hook Form + Zod Validation                          |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>      |
| Theory Section          | Section 8: React Hook Form + Zod                          |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>          |
| Specific Exercises      | Week 6 Ex 5: Login form with email/password Zod schema    |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:20 review                                          |

**What Success Looks Like Today**

Login form shows field-level errors; Zod schema shared with backend types.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 40 – Fri | 25 Sep 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Fri, 25 Sep 2026                                            |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 40 in <a href="#">calendar/mern-mastery-study.ics</a>   |
| Calendar Week           | Week 6 of 26                                                |
| Month Phase             | Month 2 – React/Next.js                                     |
| Exact Topic             | Dynamic Programming Intro – Climbing Stairs                 |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a> |
| Theory Section          | Pattern 8: Dynamic Programming – 1D intro                   |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>            |
| Specific Exercises      | Review Week 6 exercises                                     |
| LeetCode                | <a href="#">LC 70: Climbing Stairs</a>                      |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:30 DP intro                                          |

**What Success Looks Like Today**

LC 70 solved with bottom-up DP; you explain overlapping subproblems.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 41 – Sat |  26 Sep 2026 | 8 hours**

| Field                   | Detail                                                                                                                                     |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Study Date              | Sat, 26 Sep 2026                                                                                                                           |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                                                                    |
| Calendar                | Day 41 in <a href="#">calendar/mern-mastery-study.ics</a>                                                                                  |
| Calendar Week           | Week 6 of 26                                                                                                                               |
| Month Phase             | Month 2 – React/Next.js                                                                                                                    |
| Exact Topic             | User Directory Project (Weekend)                                                                                                           |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>                                                                                       |
| Theory Section          | Section 6-7: Data fetching patterns                                                                                                        |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>                                                                                           |
| Specific Exercises      | User directory: fetch JSONPlaceholder, card grid, search filter                                                                            |
| LeetCode                | <a href="#">LC 69: Sqrt(x)</a> , <a href="#">LC 83: Remove Duplicates from List</a> , <a href="#">LC 94: Binary Tree Inorder Traversal</a> |
| Weekend / Project Tasks | User directory – JSONPlaceholder API, search, loading/error states                                                                         |
| Time Split Today        | 0:00-5:00 user directory                                                                                                                   |

**What Success Looks Like Today**

User cards render with search; loading skeleton shown during fetch.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 42 – Sun | 27 Sep 2026 | 8 hours**

| Field                   | Detail                                                                                                                            |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Study Date              | Sun, 27 Sep 2026                                                                                                                  |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                                                           |
| Calendar                | Day 42 in <a href="#">calendar/mern-mastery-study.ics</a>                                                                         |
| Calendar Week           | Week 6 of 26                                                                                                                      |
| Month Phase             | Month 2 – React/Next.js                                                                                                           |
| Exact Topic             | Polish User Directory + Tree Problems                                                                                             |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>                                                                              |
| Theory Section          | Section 9: Loading states, error boundaries preview                                                                               |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>                                                                                  |
| Specific Exercises      | Add modal detail view, empty state, error retry to user directory                                                                 |
| LeetCode                | <a href="#">LC 100: Same Tree</a> , <a href="#">LC 101: Symmetric Tree</a> , <a href="#">LC 104: Maximum Depth of Binary Tree</a> |
| Weekend / Project Tasks | Polish user directory – modal, responsive layout, commit to week-06 folder                                                        |
| Time Split Today        | 0:00-4:00 polish UI                                                                                                               |

**What Success Looks Like Today**

Modal opens on click; 3 tree LC solved; project README updated.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 43 – Mon | 28 Sep 2026 | 2 hours**

| Field                   | Detail                                                     |
|-------------------------|------------------------------------------------------------|
| Study Date              | Mon, 28 Sep 2026                                           |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                  |
| Calendar                | Day 43 in <a href="#">calendar/mern-mastery-study.ics</a>  |
| Calendar Week           | Week 7 of 26                                               |
| Month Phase             | Month 2 – React/Next.js                                    |
| Exact Topic             | Performance – React.memo, lazy, Suspense                   |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>       |
| Theory Section          | Section 9: Performance Optimization                        |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>           |
| Specific Exercises      | Week 7 Ex 1: React.memo + useMemo demo with React DevTools |
| LeetCode                | None                                                       |
| Weekend / Project Tasks | None                                                       |
| Time Split Today        | 0:00-0:25 review                                           |

**What Success Looks Like Today**

You measure re-renders before/after memo; lazy route loads on demand.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 44 – Tue | 29 Sep 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Tue, 29 Sep 2026                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 44 in <a href="#">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 7 of 26                                              |
| Month Phase             | Month 2 – React/Next.js                                   |
| Exact Topic             | React Router v6 – Routes, Params, Navigation              |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>      |
| Theory Section          | Section 11: React Router v6                               |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>          |
| Specific Exercises      | Week 7 Ex 3: Home, About, User/:id, 404 routes            |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:20 review                                          |

**What Success Looks Like Today**

Four routes work; URL params extracted; 404 catch-all renders.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 45 – Wed | 30 Sep 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Wed, 30 Sep 2026                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 45 in <a href="#">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 7 of 26                                              |
| Month Phase             | Month 2 – React/Next.js                                   |
| Exact Topic             | Protected Routes and Nested Routes                        |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>      |
| Theory Section          | Section 11: Protected & Nested Routes                     |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>          |
| Specific Exercises      | Week 7 Ex 4: Protected route with auth redirect           |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:25 review                                          |

**What Success Looks Like Today**

Unauthenticated users redirect to /login; nested dashboard layout works.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 46 – Thu | 17 1 Oct 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Thu, 1 Oct 2026                                             |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 46 in <a href="#">calendar/mern-mastery-study.ics</a>   |
| Calendar Week           | Week 7 of 26                                                |
| Month Phase             | Month 2 – React/Next.js                                     |
| Exact Topic             | Error Boundaries + Zustand State Management                 |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>        |
| Theory Section          | Sections 9-10: Error Boundaries, Zustand                    |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>            |
| Specific Exercises      | Week 7 Ex 5: Error boundary; global cart state with Zustand |
| LeetCode                | None                                                        |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:20 review                                            |

**What Success Looks Like Today**

Error boundary catches render error; cart persists across route changes.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 47 – Fri | 17 2 Oct 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Fri, 2 Oct 2026                                             |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 47 in <a href="#">calendar/mern-mastery-study.ics</a>   |
| Calendar Week           | Week 7 of 26                                                |
| Month Phase             | Month 2 – React/Next.js                                     |
| Exact Topic             | Tree DFS – Maximum Depth                                    |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a> |
| Theory Section          | Pattern 6: Trees – DFS                                      |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>            |
| Specific Exercises      | Week 7 Ex 2: Lazy load routes with Suspense                 |
| LeetCode                | <a href="#">LC 104: Maximum Depth of Binary Tree</a>        |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:30 tree DFS                                          |

**What Success Looks Like Today**

LC 104 solved recursively and iteratively; lazy routes working.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 48 – Sat | 3 Oct 2026 | 8 hours**

| Field                   | Detail                                                                                             |
|-------------------------|----------------------------------------------------------------------------------------------------|
| Study Date              | Sat, 3 Oct 2026                                                                                    |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                            |
| Calendar                | Day 48 in <a href="#">calendar/mern-mastery-study.ics</a>                                          |
| Calendar Week           | Week 7 of 26                                                                                       |
| Month Phase             | Month 2 – React/Next.js                                                                            |
| Exact Topic             | E-Commerce Listing Project Start                                                                   |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>                                               |
| Theory Section          | Section 10: Zustand + Context for cart                                                             |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>                                                   |
| Specific Exercises      | E-commerce: product grid, add to cart, cart drawer                                                 |
| LeetCode                | <a href="#">LC 108: Convert Sorted Array to BST</a> , <a href="#">LC 110: Balanced Binary Tree</a> |
| Weekend / Project Tasks | E-commerce listing – mock product API, filter by category, cart in Zustand                         |
| Time Split Today        | 0:00-6:00 e-commerce start                                                                         |

**What Success Looks Like Today**

Products display with filters; cart adds/removes items with count badge.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 49 – Sun | 4 Oct 2026 | 8 hours**

| Field                   | Detail                                                                                  |
|-------------------------|-----------------------------------------------------------------------------------------|
| Study Date              | Sun, 4 Oct 2026                                                                         |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                 |
| Calendar                | Day 49 in <a href="#">calendar/mern-mastery-study.ics</a>                               |
| Calendar Week           | Week 7 of 26                                                                            |
| Month Phase             | Month 2 – React/Next.js                                                                 |
| Exact Topic             | Finish E-Commerce UI                                                                    |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>                                    |
| Theory Section          | Section 9: Performance review for product lists                                         |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>                                        |
| Specific Exercises      | Add sort, cart persist (localStorage), checkout summary page                            |
| LeetCode                | <a href="#">LC 111: Minimum Depth of Binary Tree</a> , <a href="#">LC 112: Path Sum</a> |
| Weekend / Project Tasks | Finish e-commerce – filter, sort, cart persist, responsive mobile layout                |
| Time Split Today        | 0:00-6:00 e-commerce finish                                                             |

**What Success Looks Like Today**

Cart survives refresh; sort/filter work together; 60+ LC total milestone approaching.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 50 – Mon | 5 Oct 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Mon, 5 Oct 2026                                           |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 50 in <a href="#">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 8 of 26                                              |
| Month Phase             | Month 2 – React/Next.js                                   |
| Exact Topic             | Next.js App Router and File Structure                     |
| Theory File             | <a href="#">detailed/theory/04-nextjs-complete.md</a>     |
| Theory Section          | Sections 1-2: App Router, File Conventions                |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>          |
| Specific Exercises      | Week 8 Ex 1: Create Next.js 14 app with App Router        |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:25 review React                                    |

**What Success Looks Like Today**

Next.js app runs; you explain app/ vs pages/ and file-based routing.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 51 – Tue | 6 Oct 2026 | 2 hours**

| Field                   | Detail                                                              |
|-------------------------|---------------------------------------------------------------------|
| Study Date              | Tue, 6 Oct 2026                                                     |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                           |
| Calendar                | Day 51 in <a href="#">calendar/mern-mastery-study.ics</a>           |
| Calendar Week           | Week 8 of 26                                                        |
| Month Phase             | Month 2 – React/Next.js                                             |
| Exact Topic             | Server vs Client Components                                         |
| Theory File             | <a href="#">detailed/theory/04-nextjs-complete.md</a>               |
| Theory Section          | Section 3: Server and Client Components                             |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>                    |
| Specific Exercises      | Week 8 Ex 2-3: Server Component fetch + Client counter on same page |
| LeetCode                | None                                                                |
| Weekend / Project Tasks | None                                                                |
| Time Split Today        | 0:00-0:20 review                                                    |

**What Success Looks Like Today**

Mixed page renders; you explain "use client" boundary and when to use each.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 52 – Wed | 7 Oct 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Wed, 7 Oct 2026                                           |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 52 in <a href="#">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 8 of 26                                              |
| Month Phase             | Month 2 – React/Next.js                                   |
| Exact Topic             | Data Fetching – SSR, SSG, ISR                             |
| Theory File             | <a href="#">detailed/theory/04-nextjs-complete.md</a>     |
| Theory Section          | Section 4: Data Fetching Patterns                         |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>          |
| Specific Exercises      | Week 8: Blog data fetch with fetch cache options          |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:25 review                                          |

**What Success Looks Like Today**

You explain SSR vs SSG vs ISR with revalidate example in code.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 53 – Thu | 8 Oct 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Thu, 8 Oct 2026                                           |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 53 in <a href="#">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 8 of 26                                              |
| Month Phase             | Month 2 – React/Next.js                                   |
| Exact Topic             | API Routes and Middleware                                 |
| Theory File             | <a href="#">detailed/theory/04-nextjs-complete.md</a>     |
| Theory Section          | Sections 5-6: API Routes, Middleware                      |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>          |
| Specific Exercises      | Week 8 Ex 4-5: Dynamic /blog/[slug], POST /api/contact    |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:20 review                                          |

**What Success Looks Like Today**

Dynamic route renders slug; contact API validates body and returns 201.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 54 – Fri | 9 Oct 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Fri, 9 Oct 2026                                             |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 54 in <a href="#">calendar/mern-mastery-study.ics</a>   |
| Calendar Week           | Week 8 of 26                                                |
| Month Phase             | Month 2 – React/Next.js                                     |
| Exact Topic             | Binary Tree – Invert Binary Tree                            |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a> |
| Theory Section          | Pattern 6: Trees – inversion                                |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>            |
| Specific Exercises      | Review Next.js exercises                                    |
| LeetCode                | <a href="#">LC 226: Invert Binary Tree</a>                  |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:30 tree review                                       |

**What Success Looks Like Today**

LC 226 solved; you can invert tree recursively in under 10 minutes.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 55 – Sat | 10 Oct 2026 | 8 hours**

| Field                   | Detail                                                                            |
|-------------------------|-----------------------------------------------------------------------------------|
| Study Date              | Sat, 10 Oct 2026                                                                  |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                           |
| Calendar                | Day 55 in <a href="#">calendar/mern-mastery-study.ics</a>                         |
| Calendar Week           | Week 8 of 26                                                                      |
| Month Phase             | Month 2 – React/Next.js                                                           |
| Exact Topic             | Blog Project with MDX and SEO                                                     |
| Theory File             | <a href="#">detailed/theory/04-nextjs-complete.md</a>                             |
| Theory Section          | Sections 7-9: MDX, Metadata, SEO                                                  |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>                                  |
| Specific Exercises      | Week 8 weekend project: 5 MDX posts, metadata, OG tags                            |
| LeetCode                | <a href="#">LC 118: Pascal's Triangle</a> , <a href="#">LC 136: Single Number</a> |
| Weekend / Project Tasks | Blog with Next.js – 5 MDX posts, SEO metadata, Tailwind styling                   |
| Time Split Today        | 0:00-6:00 blog build                                                              |

**What Success Looks Like Today**

5 posts render; each page has unique title/description; Tailwind styled.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 56 – Sun | 11 Oct 2026 | 8 hours**

| Field                   | Detail                                                                          |
|-------------------------|---------------------------------------------------------------------------------|
| Study Date              | Sun, 11 Oct 2026                                                                |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                         |
| Calendar                | Day 56 in <a href="#">calendar/mern-mastery-study.ics</a>                       |
| Calendar Week           | Week 8 of 26                                                                    |
| Month Phase             | Month 2 – React/Next.js                                                         |
| Exact Topic             | Deploy Blog + Month 2 Review                                                    |
| Theory File             | <a href="#">detailed/theory/04-nextjs-complete.md</a>                           |
| Theory Section          | Section 10: Deployment to Vercel                                                |
| Practice File           | <a href="#">practice/week-05-08-exercises.md</a>                                |
| Specific Exercises      | Month 2 review – list weak React/Next topics                                    |
| LeetCode                | Review all Month 2 LC – target 60+ total solved                                 |
| Weekend / Project Tasks | Deploy blog to Vercel – live URL, custom domain optional, README with demo link |
| Time Split Today        | 0:00-4:00 deploy                                                                |

**What Success Looks Like Today**

Live blog URL on resume draft; Month 2 topics reviewed; 60+ LC milestone hit.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

# MONTH 3 – Backend + Databases (Days 57-84)

MERN MASTERY · 1 CR PRODUCT PREP · HARSH VARDHAN GUPTA

Day 57 – Mon |  17 12 Oct 2026 | 2 hours

| Field                   | Detail                                                        |
|-------------------------|---------------------------------------------------------------|
| Study Date              | Mon, 12 Oct 2026                                              |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                     |
| Calendar                | Day 57 in <a href="#">calendar/mern-mastery-study.ics</a>     |
| Calendar Week           | Week 9 of 26                                                  |
| Month Phase             | Month 3 – Backend/DB                                          |
| Exact Topic             | Node.js Intro, Event Loop, Module System                      |
| Theory File             | <a href="#">detailed/theory/05-nodejs-express-complete.md</a> |
| Theory Section          | Sections 1-2: Node.js Architecture, Event Loop                |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>              |
| Specific Exercises      | Week 9 Ex 1: Hello HTTP server with Node http module          |
| LeetCode                | None                                                          |
| Weekend / Project Tasks | None                                                          |
| Time Split Today        | 0:00-0:25 Month 2 review                                      |

## What Success Looks Like Today

Server responds on port 3000; you explain Node event loop vs browser.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message

- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

## Day 58 – Tue | 13 Oct 2026 | 2 hours

| Field                   | Detail                                                         |
|-------------------------|----------------------------------------------------------------|
| Study Date              | Tue, 13 Oct 2026                                               |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                      |
| Calendar                | Day 58 in <a href="#">calendar/mern-mastery-study.ics</a>      |
| Calendar Week           | Week 9 of 26                                                   |
| Month Phase             | Month 3 – Backend/DB                                           |
| Exact Topic             | Express Basics, Middleware Chain                               |
| Theory File             | <a href="#">detailed/theory/05-nodejs-express-complete.md</a>  |
| Theory Section          | Sections 4-5: Express Setup, Middleware                        |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>               |
| Specific Exercises      | Week 9 Ex 1: Express + JSON parser + request logger middleware |
| LeetCode                | None                                                           |
| Weekend / Project Tasks | None                                                           |
| Time Split Today        | 0:00-0:20 review                                               |

### What Success Looks Like Today

Middleware chain logs requests; JSON body parsed on POST routes.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 59 – Wed | 14 Oct 2026 | 2 hours**

| Field                   | Detail                                                        |
|-------------------------|---------------------------------------------------------------|
| Study Date              | Wed, 14 Oct 2026                                              |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                     |
| Calendar                | Day 59 in <a href="#">calendar/mern-mastery-study.ics</a>     |
| Calendar Week           | Week 9 of 26                                                  |
| Month Phase             | Month 3 – Backend/DB                                          |
| Exact Topic             | REST API Design Principles                                    |
| Theory File             | <a href="#">detailed/theory/05-nodejs-express-complete.md</a> |
| Theory Section          | Section 9: REST API Design                                    |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>              |
| Specific Exercises      | Week 9 Ex 2: CRUD /api/users in-memory array                  |
| LeetCode                | None                                                          |
| Weekend / Project Tasks | None                                                          |
| Time Split Today        | 0:00-0:25 review                                              |

**What Success Looks Like Today**

Full CRUD with proper HTTP verbs and status codes (201, 404, 204).

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 60 – Thu | 15 Oct 2026 | 2 hours**

| Field                   | Detail                                                          |
|-------------------------|-----------------------------------------------------------------|
| Study Date              | Thu, 15 Oct 2026                                                |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                       |
| Calendar                | Day 60 in <a href="#">calendar/mern-mastery-study.ics</a>       |
| Calendar Week           | Week 9 of 26                                                    |
| Month Phase             | Month 3 – Backend/DB                                            |
| Exact Topic             | Zod Validation and Error Handling                               |
| Theory File             | <a href="#">detailed/theory/05-nodejs-express-complete.md</a>   |
| Theory Section          | Sections 7-8: Validation, Error Handling                        |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>                |
| Specific Exercises      | Week 9 Ex 3-4: Zod validation middleware + global error handler |
| LeetCode                | None                                                            |
| Weekend / Project Tasks | None                                                            |
| Time Split Today        | 0:00-0:20 review                                                |

**What Success Looks Like Today**

Invalid payloads return 400 with Zod error details; unhandled errors return 500 JSON.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 61 – Fri | 16 Oct 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Fri, 16 Oct 2026                                            |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 61 in <a href="#">calendar/mern-mastery-study.ics</a>   |
| Calendar Week           | Week 9 of 26                                                |
| Month Phase             | Month 3 – Backend/DB                                        |
| Exact Topic             | Linked List Cycle Detection                                 |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a> |
| Theory Section          | Pattern 5: Linked List – Floyd cycle                        |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>            |
| Specific Exercises      | Review Week 9 API code                                      |
| LeetCode                | <a href="#">LC 141: Linked List Cycle</a>                   |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:30 LL cycle theory                                   |

**What Success Looks Like Today**

LC 141 solved with fast/slow pointers in  $O(1)$  space.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 62 – Sat | 17 Oct 2026 | 8 hours**

| Field                   | Detail                                                                                                                            |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Study Date              | Sat, 17 Oct 2026                                                                                                                  |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                                                           |
| Calendar                | Day 62 in <a href="#">calendar/mern-mastery-study.ics</a>                                                                         |
| Calendar Week           | Week 9 of 26                                                                                                                      |
| Month Phase             | Month 3 – Backend/DB                                                                                                              |
| Exact Topic             | Layered Architecture Refactor                                                                                                     |
| Theory File             | <a href="#">detailed/theory/05-nodejs-express-complete.md</a>                                                                     |
| Theory Section          | Section 6: Layered Architecture (routes/controllers/services)                                                                     |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>                                                                                  |
| Specific Exercises      | Week 9 Ex 5: Refactor users API into routes/controllers/services                                                                  |
| LeetCode                | <a href="#">LC 160: Intersection of Two Linked Lists</a> , <a href="#">LC 167: Two Sum II</a> , <a href="#">LC 155: Min Stack</a> |
| Weekend / Project Tasks | Block 3 (3h): Start Auth API skeleton – register/login route stubs                                                                |
| Time Split Today        | 0:00-5:00 refactor                                                                                                                |

**What Success Looks Like Today**

Clean separation of concerns; 3 LC done; auth routes stubbed.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 63 – Sun | 18 Oct 2026 | 8 hours**

| Field                   | Detail                                                                                  |
|-------------------------|-----------------------------------------------------------------------------------------|
| Study Date              | Sun, 18 Oct 2026                                                                        |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                 |
| Calendar                | Day 63 in <a href="#">calendar/mern-mastery-study.ics</a>                               |
| Calendar Week           | Week 9 of 26                                                                            |
| Month Phase             | Month 3 – Backend/DB                                                                    |
| Exact Topic             | Auth API Skeleton + Node Event Loop Notes                                               |
| Theory File             | <a href="#">detailed/theory/05-nodejs-express-complete.md</a>                           |
| Theory Section          | Section 2: Node Event Loop deep dive                                                    |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>                                        |
| Specific Exercises      | Auth skeleton: POST /register, POST /login route structure                              |
| LeetCode                | Review Week 9 LC list                                                                   |
| Weekend / Project Tasks | Auth API skeleton – route structure, placeholder handlers, write event loop summary doc |
| Time Split Today        | 0:00-5:00 auth skeleton                                                                 |

**What Success Looks Like Today**

Auth routes respond with placeholder; event loop summary written in own words.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 64 – Mon | 19 Oct 2026 | 2 hours**

| Field                   | Detail                                                     |
|-------------------------|------------------------------------------------------------|
| Study Date              | Mon, 19 Oct 2026                                           |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                  |
| Calendar                | Day 64 in <a href="#">calendar/mern-mastery-study.ics</a>  |
| Calendar Week           | Week 10 of 26                                              |
| Month Phase             | Month 3 – Backend/DB                                       |
| Exact Topic             | AuthN vs AuthZ, JWT Structure                              |
| Theory File             | <a href="#">detailed/theory/07-auth-devops-complete.md</a> |
| Theory Section          | Sections 1-2: Authentication vs Authorization, JWT         |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>           |
| Specific Exercises      | Week 10: Draw JWT flow diagram (login → access → refresh)  |
| LeetCode                | None                                                       |
| Weekend / Project Tasks | None                                                       |
| Time Split Today        | 0:00-0:25 review                                           |

**What Success Looks Like Today**

JWT diagram shows header.payload.signature; you explain access vs refresh tokens.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 65 – Tue | 20 Oct 2026 | 2 hours**

| Field                   | Detail                                                     |
|-------------------------|------------------------------------------------------------|
| Study Date              | Tue, 20 Oct 2026                                           |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                  |
| Calendar                | Day 65 in <a href="#">calendar/mern-mastery-study.ics</a>  |
| Calendar Week           | Week 10 of 26                                              |
| Month Phase             | Month 3 – Backend/DB                                       |
| Exact Topic             | bcrypt Password Hashing                                    |
| Theory File             | <a href="#">detailed/theory/07-auth-devops-complete.md</a> |
| Theory Section          | Section 4: Password Hashing with bcrypt                    |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>           |
| Specific Exercises      | Week 10 Ex 1: Hash password, verify, compare timing-safe   |
| LeetCode                | None                                                       |
| Weekend / Project Tasks | None                                                       |
| Time Split Today        | 0:00-0:20 review                                           |

**What Success Looks Like Today**

bcrypt hash/verify works; you explain salt rounds and why not MD5.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 66 – Wed | 21 Oct 2026 | 2 hours**

| Field                   | Detail                                                           |
|-------------------------|------------------------------------------------------------------|
| Study Date              | Wed, 21 Oct 2026                                                 |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                        |
| Calendar                | Day 66 in <a href="#">calendar/mern-mastery-study.ics</a>        |
| Calendar Week           | Week 10 of 26                                                    |
| Month Phase             | Month 3 – Backend/DB                                             |
| Exact Topic             | Access + Refresh Token Flow                                      |
| Theory File             | <a href="#">detailed/theory/07-auth-devops-complete.md</a>       |
| Theory Section          | Section 2: Token Flow – access + refresh rotation                |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>                 |
| Specific Exercises      | Week 10 Ex 2-3: Generate access JWT + refresh in httpOnly cookie |
| LeetCode                | None                                                             |
| Weekend / Project Tasks | None                                                             |
| Time Split Today        | 0:00-0:25 review                                                 |

**What Success Looks Like Today**

Login returns access token; refresh stored in httpOnly cookie; rotation logic stubbed.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 67 – Thu | 22 Oct 2026 | 2 hours**

| Field                   | Detail                                                     |
|-------------------------|------------------------------------------------------------|
| Study Date              | Thu, 22 Oct 2026                                           |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                  |
| Calendar                | Day 67 in <a href="#">calendar/mern-mastery-study.ics</a>  |
| Calendar Week           | Week 10 of 26                                              |
| Month Phase             | Month 3 – Backend/DB                                       |
| Exact Topic             | Auth Middleware and RBAC                                   |
| Theory File             | <a href="#">detailed/theory/07-auth-devops-complete.md</a> |
| Theory Section          | Section 1: Middleware, Role-Based Access Control           |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>           |
| Specific Exercises      | Week 10 Ex 4-5: Auth middleware + admin vs user routes     |
| LeetCode                | None                                                       |
| Weekend / Project Tasks | None                                                       |
| Time Split Today        | 0:00-0:20 review                                           |

**What Success Looks Like Today**

Protected routes return 401 without token; admin route returns 403 for user role.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 68 – Fri | 23 Oct 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Fri, 23 Oct 2026                                            |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 68 in <a href="#">calendar/mern-mastery-study.ics</a>   |
| Calendar Week           | Week 10 of 26                                               |
| Month Phase             | Month 3 – Backend/DB                                        |
| Exact Topic             | Two Pointers – Two Sum II                                   |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a> |
| Theory Section          | Pattern 1: Two Pointers on sorted array                     |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>            |
| Specific Exercises      | Review auth code                                            |
| LeetCode                | <a href="#">LC 167: Two Sum II</a>                          |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:30 two pointers                                      |

**What Success Looks Like Today**

LC 167 solved with left/right pointers in  $O(n)$ .

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 69 – Sat | 24 Oct 2026 | 8 hours**

| Field                   | Detail                                                                                       |
|-------------------------|----------------------------------------------------------------------------------------------|
| Study Date              | Sat, 24 Oct 2026                                                                             |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                      |
| Calendar                | Day 69 in <a href="#">calendar/mern-mastery-study.ics</a>                                    |
| Calendar Week           | Week 10 of 26                                                                                |
| Month Phase             | Month 3 – Backend/DB                                                                         |
| Exact Topic             | Complete Auth Microservice                                                                   |
| Theory File             | <a href="#">detailed/theory/07-auth-devops-complete.md</a>                                   |
| Theory Section          | Sections 1-4: Full auth flow integration                                                     |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>                                             |
| Specific Exercises      | Week 10 weekend: Full refresh token rotation, logout, token blacklist stub                   |
| LeetCode                | <a href="#">LC 169: Majority Element</a> , <a href="#">LC 171: Excel Sheet Column Number</a> |
| Weekend / Project Tasks | Complete auth microservice – register, login, refresh, logout, protected routes              |
| Time Split Today        | 0:00-6:00 auth microservice                                                                  |

**What Success Looks Like Today**

End-to-end auth works with Postman collection; refresh rotation implemented.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 70 – Sun | 25 Oct 2026 | 8 hours**

| Field                   | Detail                                                                                                                                                                       |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Study Date              | Sun, 25 Oct 2026                                                                                                                                                             |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                                                                                                      |
| Calendar                | Day 70 in <a href="#">calendar/mern-mastery-study.ics</a>                                                                                                                    |
| Calendar Week           | Week 10 of 26                                                                                                                                                                |
| Month Phase             | Month 3 – Backend/DB                                                                                                                                                         |
| Exact Topic             | OAuth Concepts + Bit Manipulation LC                                                                                                                                         |
| Theory File             | <a href="#">detailed/theory/07-auth-devops-complete.md</a>                                                                                                                   |
| Theory Section          | Section 3: OAuth 2.0 Overview                                                                                                                                                |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>                                                                                                                             |
| Specific Exercises      | Read NextAuth.js / Passport.js docs – write OAuth flow summary                                                                                                               |
| LeetCode                | <a href="#">LC 190: Reverse Bits</a> , <a href="#">LC 191: Number of 1 Bits</a> , <a href="#">LC 202: Happy Number</a> , <a href="#">LC 203: Remove Linked List Elements</a> |
| Weekend / Project Tasks | OAuth concept study – document Authorization Code flow with diagram                                                                                                          |
| Time Split Today        | 0:00-4:00 OAuth study                                                                                                                                                        |

**What Success Looks Like Today**

OAuth summary written; 4 LC solved; auth microservice README complete.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 71 – Mon | 26 Oct 2026 | 2 hours**

| Field                   | Detail                                                       |
|-------------------------|--------------------------------------------------------------|
| Study Date              | Mon, 26 Oct 2026                                             |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                    |
| Calendar                | Day 71 in <a href="#">calendar/mern-mastery-study.ics</a>    |
| Calendar Week           | Week 11 of 26                                                |
| Month Phase             | Month 3 – Backend/DB                                         |
| Exact Topic             | MongoDB Basics and Atlas Setup                               |
| Theory File             | <a href="#">detailed/theory/06-databases-complete.md</a>     |
| Theory Section          | Section 2: MongoDB – Document Model, Atlas Setup             |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>             |
| Specific Exercises      | Week 11: Create MongoDB Atlas cluster, connect from Node app |
| LeetCode                | None                                                         |
| Weekend / Project Tasks | None                                                         |
| Time Split Today        | 0:00-0:25 review auth                                        |

**What Success Looks Like Today**

Atlas cluster live; Node app connects with connection string in .env.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 72 – Tue | 27 Oct 2026 | 2 hours**

| Field                   | Detail                                                                     |
|-------------------------|----------------------------------------------------------------------------|
| Study Date              | Tue, 27 Oct 2026                                                           |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                  |
| Calendar                | Day 72 in <a href="#">calendar/mern-mastery-study.ics</a>                  |
| Calendar Week           | Week 11 of 26                                                              |
| Month Phase             | Month 3 – Backend/DB                                                       |
| Exact Topic             | Mongoose Schemas and Validation                                            |
| Theory File             | <a href="#">detailed/theory/06-databases-complete.md</a>                   |
| Theory Section          | Section 2: Mongoose – Schemas, Validation, Hooks                           |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>                           |
| Specific Exercises      | Week 11 Ex 1: User schema with email unique, password required, timestamps |
| LeetCode                | None                                                                       |
| Weekend / Project Tasks | None                                                                       |
| Time Split Today        | 0:00-0:20 review                                                           |

**What Success Looks Like Today**

User schema validates on save; duplicate email returns friendly error.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 73 – Wed | 28 Oct 2026 | 2 hours**

| Field                   | Detail                                                               |
|-------------------------|----------------------------------------------------------------------|
| Study Date              | Wed, 28 Oct 2026                                                     |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                            |
| Calendar                | Day 73 in <a href="#">calendar/mern-mastery-study.ics</a>            |
| Calendar Week           | Week 11 of 26                                                        |
| Month Phase             | Month 3 – Backend/DB                                                 |
| Exact Topic             | MongoDB CRUD, Queries, Pagination                                    |
| Theory File             | <a href="#">detailed/theory/06-databases-complete.md</a>             |
| Theory Section          | Section 2: CRUD Operations, Query Operators                          |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>                     |
| Specific Exercises      | Week 11 Ex 2-3: CRUD with Atlas + paginated GET ?<br>page=1&limit=10 |
| LeetCode                | None                                                                 |
| Weekend / Project Tasks | None                                                                 |
| Time Split Today        | 0:00-0:25 review                                                     |

**What Success Looks Like Today**

Pagination returns { data, total, page, totalPages }; queries use projection.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 74 – Thu | 29 Oct 2026 | 2 hours**

| Field                   | Detail                                                          |
|-------------------------|-----------------------------------------------------------------|
| Study Date              | Thu, 29 Oct 2026                                                |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                       |
| Calendar                | Day 74 in <a href="#">calendar/mern-mastery-study.ics</a>       |
| Calendar Week           | Week 11 of 26                                                   |
| Month Phase             | Month 3 – Backend/DB                                            |
| Exact Topic             | Aggregation Pipeline                                            |
| Theory File             | <a href="#">detailed/theory/06-databases-complete.md</a>        |
| Theory Section          | Section 2: Aggregation Framework                                |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>                |
| Specific Exercises      | Week 11 Ex 4: Revenue by month aggregation on orders collection |
| LeetCode                | None                                                            |
| Weekend / Project Tasks | None                                                            |
| Time Split Today        | 0:00-0:20 review                                                |

**What Success Looks Like Today**

Aggregation pipeline uses \$match, \$group, \$sort; results match expected output.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 75 – Fri | 30 Oct 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Fri, 30 Oct 2026                                            |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 75 in <a href="#">calendar/mern-mastery-study.ics</a>   |
| Calendar Week           | Week 11 of 26                                               |
| Month Phase             | Month 3 – Backend/DB                                        |
| Exact Topic             | Reverse Linked List                                         |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a> |
| Theory Section          | Pattern 5: Linked List – reversal                           |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>            |
| Specific Exercises      | Week 11 Ex 5 preview: Index on email – explain query plan   |
| LeetCode                | <a href="#">LC 206: Reverse Linked List</a>                 |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:30 LL reversal                                       |

**What Success Looks Like Today**

LC 206 solved iteratively and recursively; index explain() documented.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 76 – Sat | 31 Oct 2026 | 8 hours**

| Field                   | Detail                                                                                     |
|-------------------------|--------------------------------------------------------------------------------------------|
| Study Date              | Sat, 31 Oct 2026                                                                           |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                    |
| Calendar                | Day 76 in <a href="#">calendar/mern-mastery-study.ics</a>                                  |
| Calendar Week           | Week 11 of 26                                                                              |
| Month Phase             | Month 3 – Backend/DB                                                                       |
| Exact Topic             | Blog API with MongoDB                                                                      |
| Theory File             | <a href="#">detailed/theory/06-databases-complete.md</a>                                   |
| Theory Section          | Section 2: Embedding vs Referencing                                                        |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>                                           |
| Specific Exercises      | Blog API: Post, Comment, Tag schemas – CRUD endpoints                                      |
| LeetCode                | <a href="#">LC 217: Contains Duplicate</a> , <a href="#">LC 219: Contains Duplicate II</a> |
| Weekend / Project Tasks | Blog API with MongoDB – posts, comments, tags, author reference                            |
| Time Split Today        | 0:00-6:00 blog API                                                                         |

**What Success Looks Like Today**

Posts CRUD works; comments embedded or referenced; 2 LC done.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 77 – Sun | 17 1 Nov 2026 | 8 hours**

| Field                   | Detail                                                                                             |
|-------------------------|----------------------------------------------------------------------------------------------------|
| Study Date              | Sun, 1 Nov 2026                                                                                    |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                            |
| Calendar                | Day 77 in <a href="#">calendar/mern-mastery-study.ics</a>                                          |
| Calendar Week           | Week 11 of 26                                                                                      |
| Month Phase             | Month 3 – Backend/DB                                                                               |
| Exact Topic             | Finish Blog API + Tree LC                                                                          |
| Theory File             | <a href="#">detailed/theory/06-databases-complete.md</a>                                           |
| Theory Section          | Section 2: Indexing best practices                                                                 |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>                                                   |
| Specific Exercises      | Finish blog: tag filtering, comment threads, seed script                                           |
| LeetCode                | <a href="#">LC 226: Invert Binary Tree</a> , <a href="#">LC 230: Kth Smallest Element in a BST</a> |
| Weekend / Project Tasks | Finish blog API – tag filter, pagination, Postman collection exported                              |
| Time Split Today        | 0:00-5:00 blog finish                                                                              |

**What Success Looks Like Today**

Blog API feature-complete; Postman collection shared; tree LC done.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 78 – Mon | 17 2 Nov 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Mon, 2 Nov 2026                                           |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 78 in <a href="#">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 12 of 26                                             |
| Month Phase             | Month 3 – Backend/DB                                      |
| Exact Topic             | SQL vs NoSQL, PostgreSQL Setup                            |
| Theory File             | <a href="#">detailed/theory/06-databases-complete.md</a>  |
| Theory Section          | Sections 1, 3: SQL vs NoSQL, PostgreSQL Setup             |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>          |
| Specific Exercises      | Week 12 Ex 1: Create Neon/Supabase PostgreSQL database    |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:25 review Mongo                                    |

**What Success Looks Like Today**

PostgreSQL instance live; you articulate 3 cases for SQL vs MongoDB.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 79 – Tue | 3 Nov 2026 | 2 hours**

| Field                   | Detail                                                    |
|-------------------------|-----------------------------------------------------------|
| Study Date              | Tue, 3 Nov 2026                                           |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                 |
| Calendar                | Day 79 in <a href="#">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 12 of 26                                             |
| Month Phase             | Month 3 – Backend/DB                                      |
| Exact Topic             | Schema Design, JOINS, Normalization                       |
| Theory File             | <a href="#">detailed/theory/06-databases-complete.md</a>  |
| Theory Section          | Section 3: Schema Design, JOINS                           |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>          |
| Specific Exercises      | Week 12 Ex 1: users + orders schema with FK, indexes      |
| LeetCode                | None                                                      |
| Weekend / Project Tasks | None                                                      |
| Time Split Today        | 0:00-0:20 review                                          |

**What Success Looks Like Today**

Schema diagram drawn; FK constraints defined; 3NF explained for your schema.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 80 – Wed | 4 Nov 2026 | 2 hours**

| Field                   | Detail                                                     |
|-------------------------|------------------------------------------------------------|
| Study Date              | Wed, 4 Nov 2026                                            |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                  |
| Calendar                | Day 80 in <a href="#">calendar/mern-mastery-study.ics</a>  |
| Calendar Week           | Week 12 of 26                                              |
| Month Phase             | Month 3 – Backend/DB                                       |
| Exact Topic             | Prisma ORM and Migrations                                  |
| Theory File             | <a href="#">detailed/theory/06-databases-complete.md</a>   |
| Theory Section          | Section 3: Prisma – Schema, Migrations, Client             |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>           |
| Specific Exercises      | Week 12 Ex 2: Prisma setup, initial migration, seed script |
| LeetCode                | None                                                       |
| Weekend / Project Tasks | None                                                       |
| Time Split Today        | 0:00-0:25 review                                           |

**What Success Looks Like Today**

prisma migrate dev runs; seed populates test users and orders.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 81 – Thu | 5 Nov 2026 | 2 hours**

| Field                   | Detail                                                       |
|-------------------------|--------------------------------------------------------------|
| Study Date              | Thu, 5 Nov 2026                                              |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                    |
| Calendar                | Day 81 in <a href="#">calendar/mern-mastery-study.ics</a>    |
| Calendar Week           | Week 12 of 26                                                |
| Month Phase             | Month 3 – Backend/DB                                         |
| Exact Topic             | Redis Caching and Rate Limiting                              |
| Theory File             | <a href="#">detailed/theory/06-databases-complete.md</a>     |
| Theory Section          | Section 4: Redis – Caching, Sessions, Rate Limiting          |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>             |
| Specific Exercises      | Week 12 Ex 4-5: Cache GET /users/:id, rate limit 100 req/min |
| LeetCode                | None                                                         |
| Weekend / Project Tasks | None                                                         |
| Time Split Today        | 0:00-0:20 review                                             |

**What Success Looks Like Today**

Cache hit/miss logged; rate limiter returns 429 after threshold.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 82 – Fri | 6 Nov 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Fri, 6 Nov 2026                                             |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 82 in <a href="#">calendar/mern-mastery-study.ics</a>   |
| Calendar Week           | Week 12 of 26                                               |
| Month Phase             | Month 3 – Backend/DB                                        |
| Exact Topic             | BST – Kth Smallest Element                                  |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a> |
| Theory Section          | Pattern 6: Trees – BST inorder                              |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>            |
| Specific Exercises      | Week 12 Ex 3: JOIN query users with order totals            |
| LeetCode                | <a href="#">LC 230: Kth Smallest Element in a BST</a>       |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:30 BST theory                                        |

**What Success Looks Like Today**

LC 230 solved with inorder traversal; JOIN query returns correct aggregates.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 83 – Sat | 7 Nov 2026 | 8 hours**

| Field                   | Detail                                                                                       |
|-------------------------|----------------------------------------------------------------------------------------------|
| Study Date              | Sat, 7 Nov 2026                                                                              |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                      |
| Calendar                | Day 83 in <a href="#">calendar/mern-mastery-study.ics</a>                                    |
| Calendar Week           | Week 12 of 26                                                                                |
| Month Phase             | Month 3 – Backend/DB                                                                         |
| Exact Topic             | Rebuild Blog with PostgreSQL                                                                 |
| Theory File             | <a href="#">detailed/theory/06-databases-complete.md</a>                                     |
| Theory Section          | Section 3: Prisma relations – one-to-many, many-to-many                                      |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>                                             |
| Specific Exercises      | Week 12 weekend: Same blog API with Prisma + PostgreSQL                                      |
| LeetCode                | <a href="#">LC 238: Product of Array Except Self</a> , <a href="#">LC 242: Valid Anagram</a> |
| Weekend / Project Tasks | Rebuild blog API with PostgreSQL – compare response times and schema with Mongo version      |
| Time Split Today        | 0:00-6:00 PG blog rebuild                                                                    |

**What Success Looks Like Today**

PG blog API feature parity with Mongo version; comparison doc started.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 84 – Sun | 8 Nov 2026 | 8 hours**

| Field                   | Detail                                                                         |
|-------------------------|--------------------------------------------------------------------------------|
| Study Date              | Sun, 8 Nov 2026                                                                |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                        |
| Calendar                | Day 84 in <a href="#">calendar/mern-mastery-study.ics</a>                      |
| Calendar Week           | Week 12 of 26                                                                  |
| Month Phase             | Month 3 – Backend/DB                                                           |
| Exact Topic             | Rate Limiter + Redis Sessions + 100 LC Milestone                               |
| Theory File             | <a href="#">detailed/theory/06-databases-complete.md</a>                       |
| Theory Section          | Section 4: Redis session store                                                 |
| Practice File           | <a href="#">practice/week-09-12-exercises.md</a>                               |
| Specific Exercises      | Write Mongo vs SQL tradeoffs README; Redis session for auth                    |
| LeetCode                | <a href="#">LC 252: Meeting Rooms</a> , <a href="#">LC 268: Missing Number</a> |
| Weekend / Project Tasks | Rate limiter + Redis session integration; Mongo vs SQL comparison README       |
| Time Split Today        | 0:00-4:00 Redis session                                                        |

**What Success Looks Like Today**

100+ LC milestone reached; README compares Mongo vs SQL with real examples from your APIs.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

# MONTH 4 – Projects + DevOps (Days 85-112)

MERN MASTERY · 1 CR PRODUCT PREP · HARSH VARDHAN GUPTA

Day 85 – Mon |  17 9 Nov 2026 | 2 hours

| Field                   | Detail                                                                        |
|-------------------------|-------------------------------------------------------------------------------|
| Study Date              | Mon, 9 Nov 2026                                                               |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                     |
| Calendar                | Day 85 in <a href="#">calendar/mern-mastery-study.ics</a>                     |
| Calendar Week           | Week 13 of 26                                                                 |
| Month Phase             | Month 4 – Projects/DevOps                                                     |
| Exact Topic             | TaskFlow Project Setup – Monorepo, TS, ESLint                                 |
| Theory File             | <a href="#">detailed/theory/05-nodejs-express-complete.md</a>                 |
| Theory Section          | Section 6: Project Structure for production apps                              |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                              |
| Specific Exercises      | TaskFlow: Initialize monorepo (Turborepo or npm workspaces), shared TS config |
| LeetCode                | None                                                                          |
| Weekend / Project Tasks | TaskFlow setup – monorepo structure, ESLint, Prettier, Husky pre-commit       |
| Time Split Today        | 0:00-0:25 review Month 3                                                      |

## What Success Looks Like Today

Monorepo builds; packages/api and packages/web scaffolded; CI-ready structure.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message

- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

## Day 86 – Tue | 10 Nov 2026 | 2 hours

| Field                   | Detail                                                                     |
|-------------------------|----------------------------------------------------------------------------|
| Study Date              | Tue, 10 Nov 2026                                                           |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                  |
| Calendar                | Day 86 in <a href="#">calendar/mern-mastery-study.ics</a>                  |
| Calendar Week           | Week 13 of 26                                                              |
| Month Phase             | Month 4 – Projects/DevOps                                                  |
| Exact Topic             | TaskFlow Database Models – User, Team, Project, Task                       |
| Theory File             | <a href="#">detailed/theory/06-databases-complete.md</a>                   |
| Theory Section          | Section 2: Mongoose schema design for nested entities                      |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                           |
| Specific Exercises      | Define Mongoose schemas with refs: User, Team, Project, Task               |
| LeetCode                | None                                                                       |
| Weekend / Project Tasks | TaskFlow models – all schemas with validation, indexes on teamId/projectId |
| Time Split Today        | 0:00-0:20 review                                                           |

### What Success Looks Like Today

Four models defined with proper refs and validation; ERD diagram in README.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 87 – Wed | 11 Nov 2026 | 2 hours**

| Field                   | Detail                                                            |
|-------------------------|-------------------------------------------------------------------|
| Study Date              | Wed, 11 Nov 2026                                                  |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                         |
| Calendar                | Day 87 in <a href="#">calendar/mern-mastery-study.ics</a>         |
| Calendar Week           | Week 13 of 26                                                     |
| Month Phase             | Month 4 – Projects/DevOps                                         |
| Exact Topic             | TaskFlow Auth Integration – JWT + Refresh                         |
| Theory File             | <a href="#">detailed/theory/07-auth-devops-complete.md</a>        |
| Theory Section          | Sections 1-2: Integrate auth into existing project                |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                  |
| Specific Exercises      | Port auth microservice patterns into TaskFlow API                 |
| LeetCode                | None                                                              |
| Weekend / Project Tasks | TaskFlow auth – register, login, refresh, protect all task routes |
| Time Split Today        | 0:00-0:25 review                                                  |

**What Success Looks Like Today**

Auth flow works end-to-end in TaskFlow; tokens stored securely.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 88 – Thu | 12 Nov 2026 | 2 hours**

| Field                   | Detail                                                        |
|-------------------------|---------------------------------------------------------------|
| Study Date              | Thu, 12 Nov 2026                                              |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                     |
| Calendar                | Day 88 in <a href="#">calendar/mern-mastery-study.ics</a>     |
| Calendar Week           | Week 13 of 26                                                 |
| Month Phase             | Month 4 – Projects/DevOps                                     |
| Exact Topic             | TaskFlow Teams + Invite API                                   |
| Theory File             | <a href="#">detailed/theory/05-nodejs-express-complete.md</a> |
| Theory Section          | Section 9: REST design for nested resources                   |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>              |
| Specific Exercises      | POST /teams, POST /teams/:id/invite, GET /teams/:id/members   |
| LeetCode                | None                                                          |
| Weekend / Project Tasks | Teams API – create team, invite by email, accept invite flow  |
| Time Split Today        | 0:00-0:20 review                                              |

**What Success Looks Like Today**

Team CRUD works; invite generates token; member list returns roles.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 89 – Fri | 13 Nov 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Fri, 13 Nov 2026                                            |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 89 in <a href="#">calendar/mern-mastery-study.ics</a>   |
| Calendar Week           | Week 13 of 26                                               |
| Month Phase             | Month 4 – Projects/DevOps                                   |
| Exact Topic             | Meeting Rooms II – Interval Scheduling                      |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a> |
| Theory Section          | Pattern: Intervals – meeting rooms                          |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>            |
| Specific Exercises      | Review TaskFlow API progress                                |
| LeetCode                | <a href="#">LC 253: Meeting Rooms II</a>                    |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:30 interval theory                                   |

**What Success Looks Like Today**

LC 253 solved with min-heap or sweep line; you explain interval overlap.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 90 – Sat | 14 Nov 2026 | 8 hours**

| Field                   | Detail                                                                                  |
|-------------------------|-----------------------------------------------------------------------------------------|
| Study Date              | Sat, 14 Nov 2026                                                                        |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                 |
| Calendar                | Day 90 in <a href="#">calendar/mern-mastery-study.ics</a>                               |
| Calendar Week           | Week 13 of 26                                                                           |
| Month Phase             | Month 4 – Projects/DevOps                                                               |
| Exact Topic             | TaskFlow Tasks CRUD + Comments + Permissions                                            |
| Theory File             | <a href="#">detailed/theory/05-nodejs-express-complete.md</a>                           |
| Theory Section          | Section 6: Service layer for complex business logic                                     |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                                        |
| Specific Exercises      | Tasks: create, assign, status, priority, due date, comments, RBAC                       |
| LeetCode                | <a href="#">LC 283: Move Zeroes</a> , <a href="#">LC 287: Find the Duplicate Number</a> |
| Weekend / Project Tasks | TaskFlow tasks module – full CRUD, comments, owner/member/viewer permissions            |
| Time Split Today        | 0:00-6:00 tasks module                                                                  |

**What Success Looks Like Today**

Tasks CRUD with role checks; comments thread on tasks; 2 LC done.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 91 – Sun | 15 Nov 2026 | 8 hours**

| Field                   | Detail                                                                                       |
|-------------------------|----------------------------------------------------------------------------------------------|
| Study Date              | Sun, 15 Nov 2026                                                                             |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                      |
| Calendar                | Day 91 in <a href="#">calendar/mern-mastery-study.ics</a>                                    |
| Calendar Week           | Week 13 of 26                                                                                |
| Month Phase             | Month 4 – Projects/DevOps                                                                    |
| Exact Topic             | TaskFlow Real-Time Updates – Socket.io                                                       |
| Theory File             | <a href="#">detailed/theory/05-nodejs-express-complete.md</a>                                |
| Theory Section          | Section 10: WebSockets with Socket.io                                                        |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                                             |
| Specific Exercises      | Socket.io: task update broadcasts to team room                                               |
| LeetCode                | <a href="#">LC 300: Longest Increasing Subsequence</a> , <a href="#">LC 322: Coin Change</a> |
| Weekend / Project Tasks | Socket.io integration – join team room, broadcast task create/update/delete events           |
| Time Split Today        | 0:00-6:00 Socket.io                                                                          |

**What Success Looks Like Today**

Two browser tabs see real-time task updates; socket auth verified.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 92 – Mon | 17 16 Nov 2026 | 2 hours**

| Field                   | Detail                                                                 |
|-------------------------|------------------------------------------------------------------------|
| Study Date              | Mon, 16 Nov 2026                                                       |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                              |
| Calendar                | Day 92 in <code>calendar/mern-mastery-study.ics</code>                 |
| Calendar Week           | Week 14 of 26                                                          |
| Month Phase             | Month 4 – Projects/DevOps                                              |
| Exact Topic             | TaskFlow File Upload – Multer / S3                                     |
| Theory File             | <u><code>detailed/theory/07-auth-devops-complete.md</code></u>         |
| Theory Section          | Section 5: File Storage – local vs S3                                  |
| Practice File           | <u><code>practice/week-13-16-exercises.md</code></u>                   |
| Specific Exercises      | POST /tasks/:id/attachments with Multer or S3 presigned URL            |
| LeetCode                | None                                                                   |
| Weekend / Project Tasks | File upload – attach files to tasks, validate mime type and size limit |
| Time Split Today        | 0:00-0:25 review                                                       |

**What Success Looks Like Today**

Files upload and link to tasks; size/type validation returns 400 for invalid.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 93 – Tue | 17 Nov 2026 | 2 hours**

| Field                   | Detail                                                                           |
|-------------------------|----------------------------------------------------------------------------------|
| Study Date              | Tue, 17 Nov 2026                                                                 |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                        |
| Calendar                | Day 93 in <a href="#">calendar/mern-mastery-study.ics</a>                        |
| Calendar Week           | Week 14 of 26                                                                    |
| Month Phase             | Month 4 – Projects/DevOps                                                        |
| Exact Topic             | TaskFlow Email Notifications                                                     |
| Theory File             | <a href="#">detailed/theory/07-auth-devops-complete.md</a>                       |
| Theory Section          | Section 6: Email – Nodemailer, SendGrid, queues                                  |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                                 |
| Specific Exercises      | Send email on task assignment and invite acceptance                              |
| LeetCode                | None                                                                             |
| Weekend / Project Tasks | Email notifications – task assigned, invite sent (use Ethereum or SendGrid test) |
| Time Split Today        | 0:00-0:20 review                                                                 |

**What Success Looks Like Today**

Emails trigger on events; templates for assignment and invite.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 94 – Wed | 17 18 Nov 2026 | 2 hours**

| Field                   | Detail                                                                       |
|-------------------------|------------------------------------------------------------------------------|
| Study Date              | Wed, 18 Nov 2026                                                             |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                    |
| Calendar                | Day 94 in <a href="#">calendar/mern-mastery-study.ics</a>                    |
| Calendar Week           | Week 14 of 26                                                                |
| Month Phase             | Month 4 – Projects/DevOps                                                    |
| Exact Topic             | TaskFlow Frontend – Dashboard Layout (Next.js)                               |
| Theory File             | <a href="#">detailed/theory/04-nextjs-complete.md</a>                        |
| Theory Section          | Section 3: Layouts, nested layouts in App Router                             |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                             |
| Specific Exercises      | Next.js dashboard: sidebar, header, team switcher, responsive                |
| LeetCode                | None                                                                         |
| Weekend / Project Tasks | TaskFlow frontend – dashboard shell with sidebar navigation and team context |
| Time Split Today        | 0:00-0:25 review                                                             |

**What Success Looks Like Today**

Dashboard layout renders; sidebar collapses on mobile; team switcher works.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 95 – Thu | 17 19 Nov 2026 | 2 hours**

| Field                   | Detail                                                                 |
|-------------------------|------------------------------------------------------------------------|
| Study Date              | Thu, 19 Nov 2026                                                       |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                              |
| Calendar                | Day 95 in <a href="#">calendar/mern-mastery-study.ics</a>              |
| Calendar Week           | Week 14 of 26                                                          |
| Month Phase             | Month 4 – Projects/DevOps                                              |
| Exact Topic             | TaskFlow Frontend – Kanban Task Board                                  |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>                   |
| Theory Section          | Section 9: Drag-and-drop patterns (dnd-kit or native)                  |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                       |
| Specific Exercises      | Kanban columns: Todo, In Progress, Done – drag or click to move status |
| LeetCode                | None                                                                   |
| Weekend / Project Tasks | Kanban board UI – columns by status, task cards, quick edit modal      |
| Time Split Today        | 0:00-0:20 review                                                       |

**What Success Looks Like Today**

Tasks move between columns; status syncs with API; optimistic updates.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 96 – Fri | 20 Nov 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Fri, 20 Nov 2026                                            |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 96 in <a href="#">calendar/mern-mastery-study.ics</a>   |
| Calendar Week           | Week 14 of 26                                               |
| Month Phase             | Month 4 – Projects/DevOps                                   |
| Exact Topic             | Coin Change – Dynamic Programming                           |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a> |
| Theory Section          | Pattern 8: DP – unbounded knapsack / coin change            |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>            |
| Specific Exercises      | Review TaskFlow frontend                                    |
| LeetCode                | <a href="#">LC 322: Coin Change</a>                         |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:30 DP theory                                         |

**What Success Looks Like Today**

LC 322 solved bottom-up; you explain  $dp[i] = \min$  coins for amount  $i$ .

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

## Day 97 – Sat | 21 Nov 2026 | 8 hours

| Field                   | Detail                                                                                               |
|-------------------------|------------------------------------------------------------------------------------------------------|
| Study Date              | Sat, 21 Nov 2026                                                                                     |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                              |
| Calendar                | Day 97 in <a href="#">calendar/mern-mastery-study.ics</a>                                            |
| Calendar Week           | Week 14 of 26                                                                                        |
| Month Phase             | Month 4 – Projects/DevOps                                                                            |
| Exact Topic             | TaskFlow Docker Compose – app + mongo + redis                                                        |
| Theory File             | <a href="#">detailed/theory/07-auth-devops-complete.md</a>                                           |
| Theory Section          | Section 5: Docker & Docker Compose                                                                   |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                                                     |
| Specific Exercises      | Dockerfile for API and web; docker-compose.yml with mongo, redis                                     |
| LeetCode                | <a href="#">LC 347: Top K Frequent Elements</a> , <a href="#">LC 349: Intersection of Two Arrays</a> |
| Weekend / Project Tasks | Docker Compose – full stack runs with docker compose up; .env.example documented                     |
| Time Split Today        | 0:00-6:00 Docker                                                                                     |

### What Success Looks Like Today

docker compose up starts all services; README has Docker instructions.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 98 – Sun | 22 Nov 2026 | 8 hours**

| Field                   | Detail                                                               |
|-------------------------|----------------------------------------------------------------------|
| Study Date              | Sun, 22 Nov 2026                                                     |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                              |
| Calendar                | Day 98 in <a href="#">calendar/mern-mastery-study.ics</a>            |
| Calendar Week           | Week 14 of 26                                                        |
| Month Phase             | Month 4 – Projects/DevOps                                            |
| Exact Topic             | TaskFlow Polish, Bugs, README                                        |
| Theory File             | <a href="#">detailed/theory/07-auth-devops-complete.md</a>           |
| Theory Section          | Section 5: Production checklist                                      |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                     |
| Specific Exercises      | Bug bash: test all flows, fix edge cases, write architecture diagram |
| LeetCode                | Review TaskFlow-week LC list                                         |
| Weekend / Project Tasks | Polish TaskFlow – fix bugs, architecture diagram in README, demo GIF |
| Time Split Today        | 0:00-8:00 polish, bugs, README, architecture diagram                 |

**What Success Looks Like Today**

TaskFlow deployed milestone; README has diagram, features list, setup steps.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 99 – Mon | 23 Nov 2026 | 2 hours**

| Field                   | Detail                                                            |
|-------------------------|-------------------------------------------------------------------|
| Study Date              | Mon, 23 Nov 2026                                                  |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                         |
| Calendar                | Day 99 in <a href="#">calendar/mern-mastery-study.ics</a>         |
| Calendar Week           | Week 15 of 26                                                     |
| Month Phase             | Month 4 – Projects/DevOps                                         |
| Exact Topic             | ShopLite Setup – Product Schema (PostgreSQL)                      |
| Theory File             | <a href="#">detailed/theory/06-databases-complete.md</a>          |
| Theory Section          | Section 3: E-commerce schema – products, categories, inventory    |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                  |
| Specific Exercises      | ShopLite: Prisma schema for Product, Category, Cart, Order        |
| LeetCode                | None                                                              |
| Weekend / Project Tasks | ShopLite setup – PostgreSQL schema, Prisma migrate, seed products |
| Time Split Today        | 0:00-0:25 TaskFlow review                                         |

**What Success Looks Like Today**

ShopLite repo scaffolded; product schema with categories and stock count.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 100 – Tue | 24 Nov 2026 | 2 hours**

| Field                   | Detail                                                                             |
|-------------------------|------------------------------------------------------------------------------------|
| Study Date              | Tue, 24 Nov 2026                                                                   |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                          |
| Calendar                | Day 100 in <a href="#">calendar/mern-mastery-study.ics</a>                         |
| Calendar Week           | Week 15 of 26                                                                      |
| Month Phase             | Month 4 – Projects/DevOps                                                          |
| Exact Topic             | ShopLite Product Catalog API + Admin Routes                                        |
| Theory File             | <a href="#">detailed/theory/05-nodejs-express-complete.md</a>                      |
| Theory Section          | Section 9: REST for catalog with filtering                                         |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                                   |
| Specific Exercises      | GET /products?category=&search=, admin<br>POST/PUT/DELETE /admin/products          |
| LeetCode                | None                                                                               |
| Weekend / Project Tasks | Product catalog API – public listing with filters, admin<br>CRUD protected by RBAC |
| Time Split Today        | 0:00-0:20 review                                                                   |

**What Success Looks Like Today**

Catalog filters by category and search; admin routes require admin role.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 101 – Wed | 25 Nov 2026 | 2 hours**

| Field                   | Detail                                                            |
|-------------------------|-------------------------------------------------------------------|
| Study Date              | Wed, 25 Nov 2026                                                  |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                         |
| Calendar                | Day 101 in <a href="#">calendar/mern-mastery-study.ics</a>        |
| Calendar Week           | Week 15 of 26                                                     |
| Month Phase             | Month 4 – Projects/DevOps                                         |
| Exact Topic             | ShopLite Cart + Redis Session                                     |
| Theory File             | <a href="#">detailed/theory/06-databases-complete.md</a>          |
| Theory Section          | Section 4: Redis for cart sessions                                |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                  |
| Specific Exercises      | Cart: add, update qty, remove, get cart – stored in Redis session |
| LeetCode                | None                                                              |
| Weekend / Project Tasks | Shopping cart – Redis-backed session cart with expiry             |
| Time Split Today        | 0:00-0:25 review                                                  |

**What Success Looks Like Today**

Cart persists across requests; guest cart works without login.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 102 – Thu | 26 Nov 2026 | 2 hours**

| Field                   | Detail                                                                            |
|-------------------------|-----------------------------------------------------------------------------------|
| Study Date              | Thu, 26 Nov 2026                                                                  |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                         |
| Calendar                | Day 102 in <a href="#">calendar/mern-mastery-study.ics</a>                        |
| Calendar Week           | Week 15 of 26                                                                     |
| Month Phase             | Month 4 – Projects/DevOps                                                         |
| Exact Topic             | ShopLite Checkout + Order Creation                                                |
| Theory File             | <a href="#">detailed/theory/06-databases-complete.md</a>                          |
| Theory Section          | Section 3: Transactions – ACID order creation                                     |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                                  |
| Specific Exercises      | POST /checkout – validate stock, create order, decrement inventory in transaction |
| LeetCode                | None                                                                              |
| Weekend / Project Tasks | Checkout flow – order creation with Prisma \$transaction, stock validation        |
| Time Split Today        | 0:00-0:20 review                                                                  |

**What Success Looks Like Today**

Order created atomically; insufficient stock returns 409; order history query works.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 103 – Fri | 27 Nov 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Fri, 27 Nov 2026                                            |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 103 in <a href="#">calendar/mern-mastery-study.ics</a>  |
| Calendar Week           | Week 15 of 26                                               |
| Month Phase             | Month 4 – Projects/DevOps                                   |
| Exact Topic             | Longest Increasing Subsequence                              |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a> |
| Theory Section          | Pattern 8: DP – LIS with binary search optimization         |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>            |
| Specific Exercises      | Review ShopLite checkout code                               |
| LeetCode                | <a href="#">LC 300: Longest Increasing Subsequence</a>      |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:30 LIS theory                                        |

**What Success Looks Like Today**

LC 300 solved  $O(n \log n)$  with patience sorting or DP approach.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 104 – Sat | 28 Nov 2026 | 8 hours**

| Field                   | Detail                                                                                                |
|-------------------------|-------------------------------------------------------------------------------------------------------|
| Study Date              | Sat, 28 Nov 2026                                                                                      |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                               |
| Calendar                | Day 104 in <a href="#">calendar/mern-mastery-study.ics</a>                                            |
| Calendar Week           | Week 15 of 26                                                                                         |
| Month Phase             | Month 4 – Projects/DevOps                                                                             |
| Exact Topic             | ShopLite Payment Integration – Razorpay/Stripe Test                                                   |
| Theory File             | <a href="#">detailed/theory/07-auth-devops-complete.md</a>                                            |
| Theory Section          | Section 6: Payment integration patterns                                                               |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                                                      |
| Specific Exercises      | Stripe/Razorpay test mode: create payment intent, webhook handler stub                                |
| LeetCode                | <a href="#">LC 371: Sum of Two Integers</a> , <a href="#">LC 373: Find K Pairs with Smallest Sums</a> |
| Weekend / Project Tasks | Payment integration – test checkout with Stripe/Razorpay sandbox                                      |
| Time Split Today        | 0:00-6:00 payment integration                                                                         |

**What Success Looks Like Today**

Test payment completes in sandbox; webhook logs payment.succeeded event.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 105 – Sun | 29 Nov 2026 | 8 hours**

| Field                   | Detail                                                                                                   |
|-------------------------|----------------------------------------------------------------------------------------------------------|
| Study Date              | Sun, 29 Nov 2026                                                                                         |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                                  |
| Calendar                | Day 105 in <a href="#">calendar/mern-mastery-study.ics</a>                                               |
| Calendar Week           | Week 15 of 26                                                                                            |
| Month Phase             | Month 4 – Projects/DevOps                                                                                |
| Exact Topic             | ShopLite Order Management + Admin Dashboard                                                              |
| Theory File             | <a href="#">detailed/theory/04-nextjs-complete.md</a>                                                    |
| Theory Section          | Section 4: Server Components for admin data tables                                                       |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                                                         |
| Specific Exercises      | Admin: order list, status update, product management UI                                                  |
| LeetCode                | <a href="#">LC 378: Kth Smallest Element in a Sorted Matrix</a> , <a href="#">LC 392: Is Subsequence</a> |
| Weekend / Project Tasks | Admin dashboard – orders table, status filters, product edit forms                                       |
| Time Split Today        | 0:00-6:00 admin dashboard                                                                                |

**What Success Looks Like Today**

Admin can view/update orders; dashboard shows revenue summary.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 106 – Mon | 30 Nov 2026 | 2 hours**

| Field                   | Detail                                                                    |
|-------------------------|---------------------------------------------------------------------------|
| Study Date              | Mon, 30 Nov 2026                                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                 |
| Calendar                | Day 106 in <a href="#">calendar/mern-mastery-study.ics</a>                |
| Calendar Week           | Week 16 of 26                                                             |
| Month Phase             | Month 4 – Projects/DevOps                                                 |
| Exact Topic             | Jest Unit Tests                                                           |
| Theory File             | <a href="#">detailed/theory/07-auth-devops-complete.md</a>                |
| Theory Section          | Section 7: Testing – Jest unit tests                                      |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                          |
| Specific Exercises      | Unit tests for utility functions: validation, price calc, cart helpers    |
| LeetCode                | None                                                                      |
| Weekend / Project Tasks | Jest unit tests – $\geq 10$ tests for utils and services, coverage report |
| Time Split Today        | 0:00-0:25 review                                                          |

**What Success Looks Like Today**

npm test passes; coverage  $\geq 70\%$  on utils/services.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 107 – Tue | 17 1 Dec 2026 | 2 hours**

| Field                   | Detail                                                            |
|-------------------------|-------------------------------------------------------------------|
| Study Date              | Tue, 1 Dec 2026                                                   |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                         |
| Calendar                | Day 107 in <a href="#">calendar/mern-mastery-study.ics</a>        |
| Calendar Week           | Week 16 of 26                                                     |
| Month Phase             | Month 4 – Projects/DevOps                                         |
| Exact Topic             | Supertest API Integration Tests                                   |
| Theory File             | <a href="#">detailed/theory/07-auth-devops-complete.md</a>        |
| Theory Section          | Section 7: Supertest for API endpoints                            |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                  |
| Specific Exercises      | Integration tests: auth flow, product CRUD, checkout with test DB |
| LeetCode                | None                                                              |
| Weekend / Project Tasks | API integration tests – auth, products, cart, checkout endpoints  |
| Time Split Today        | 0:00-0:20 review                                                  |

**What Success Looks Like Today**

Supertest suite runs against test database; CI-ready test script.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 108 – Wed | 17 2 Dec 2026 | 2 hours**

| Field                   | Detail                                                           |
|-------------------------|------------------------------------------------------------------|
| Study Date              | Wed, 2 Dec 2026                                                  |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                        |
| Calendar                | Day 108 in <a href="#">calendar/mern-mastery-study.ics</a>       |
| Calendar Week           | Week 16 of 26                                                    |
| Month Phase             | Month 4 – Projects/DevOps                                        |
| Exact Topic             | React Testing Library                                            |
| Theory File             | <a href="#">detailed/theory/03-react-complete.md</a>             |
| Theory Section          | Section 12: Testing React Components                             |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                 |
| Specific Exercises      | RTL tests: product card, cart button, login form submission      |
| LeetCode                | None                                                             |
| Weekend / Project Tasks | RTL component tests – 5+ components with user-event interactions |
| Time Split Today        | 0:00-0:25 review                                                 |

**What Success Looks Like Today**

RTL tests simulate user clicks and form submit; tests pass in CI.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 109 – Thu | 17 3 Dec 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Thu, 3 Dec 2026                                             |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 109 in <a href="#">calendar/mern-mastery-study.ics</a>  |
| Calendar Week           | Week 16 of 26                                               |
| Month Phase             | Month 4 – Projects/DevOps                                   |
| Exact Topic             | GitHub Actions CI Pipeline                                  |
| Theory File             | <a href="#">detailed/theory/07-auth-devops-complete.md</a>  |
| Theory Section          | Section 8: CI/CD with GitHub Actions                        |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>            |
| Specific Exercises      | CI workflow: lint, test, build on pull request              |
| LeetCode                | None                                                        |
| Weekend / Project Tasks | GitHub Actions – ci.yml runs lint + test + build on push/PR |
| Time Split Today        | 0:00-0:20 review                                            |

**What Success Looks Like Today**

Green CI badge on README; failed tests block merge.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 110 – Fri | 4 Dec 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Fri, 4 Dec 2026                                             |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 110 in <a href="#">calendar/mern-mastery-study.ics</a>  |
| Calendar Week           | Week 16 of 26                                               |
| Month Phase             | Month 4 – Projects/DevOps                                   |
| Exact Topic             | Top K Frequent Elements – Heap                              |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a> |
| Theory Section          | Pattern 3: Hash Map + Heap                                  |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>            |
| Specific Exercises      | Review CI pipeline                                          |
| LeetCode                | <a href="#">LC 347: Top K Frequent Elements</a>             |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:30 heap theory                                       |

**What Success Looks Like Today**

LC 347 solved with heap or bucket sort; 150 LC milestone approaching.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 111 – Sat | 17 5 Dec 2026 | 8 hours**

| Field                   | Detail                                                                                         |
|-------------------------|------------------------------------------------------------------------------------------------|
| Study Date              | Sat, 5 Dec 2026                                                                                |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                        |
| Calendar                | Day 111 in <a href="#">calendar/mern-mastery-study.ics</a>                                     |
| Calendar Week           | Week 16 of 26                                                                                  |
| Month Phase             | Month 4 – Projects/DevOps                                                                      |
| Exact Topic             | Deploy Both Projects – Vercel + Render                                                         |
| Theory File             | <a href="#">detailed/theory/07-auth-devops-complete.md</a>                                     |
| Theory Section          | Section 8: Deployment – Vercel, Render, env vars                                               |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                                               |
| Specific Exercises      | Deploy TaskFlow frontend (Vercel) + API (Render), ShopLite same pattern                        |
| LeetCode                | <a href="#">LC 399: Evaluate Division</a> , <a href="#">LC 416: Partition Equal Subset Sum</a> |
| Weekend / Project Tasks | Deploy TaskFlow + ShopLite – live URLs, env vars, health check endpoints                       |
| Time Split Today        | 0:00-8:00 deployment marathon                                                                  |

**What Success Looks Like Today**

Both projects have live demo URLs; health checks return 200.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 112 – Sun | 17 6 Dec 2026 | 8 hours**

| Field                   | Detail                                                                                                                 |
|-------------------------|------------------------------------------------------------------------------------------------------------------------|
| Study Date              | Sun, 6 Dec 2026                                                                                                        |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                                                |
| Calendar                | Day 112 in <a href="#">calendar/mern-mastery-study.ics</a>                                                             |
| Calendar Week           | Week 16 of 26                                                                                                          |
| Month Phase             | Month 4 – Projects/DevOps                                                                                              |
| Exact Topic             | Architecture Diagrams + Resume Bullets + 150 LC                                                                        |
| Theory File             | <a href="#">detailed/theory/07-auth-devops-complete.md</a>                                                             |
| Theory Section          | Section 8: Documentation for portfolio                                                                                 |
| Practice File           | <a href="#">practice/week-13-16-exercises.md</a>                                                                       |
| Specific Exercises      | Write resume bullets with metrics for both projects                                                                    |
| LeetCode                | <a href="#">LC 424: Longest Repeating Character Replacement</a> ,<br><a href="#">LC 435: Non-overlapping Intervals</a> |
| Weekend / Project Tasks | Architecture diagrams (draw.io/mermaid), resume bullets, GitHub pinned repos                                           |
| Time Split Today        | 0:00-4:00 diagrams + resume                                                                                            |

**What Success Looks Like Today**

150+ LC milestone; resume has 3 quantified project bullets; diagrams in both READMEs.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

# MONTH 5 – DSA + LLD (Days 113-140)

MERN MASTERY · 1 CR PRODUCT PREP · HARSH VARDHAN GUPTA

Day 113 – Mon |  17 7 Dec 2026 | 2 hours

| Field                   | Detail                                                                         |
|-------------------------|--------------------------------------------------------------------------------|
| Study Date              | Mon, 7 Dec 2026                                                                |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                      |
| Calendar                | Day 113 in <a href="#">calendar/mern-mastery-study.ics</a>                     |
| Calendar Week           | Week 17 of 26                                                                  |
| Month Phase             | Month 5 – DSA/LLD                                                              |
| Exact Topic             | Two Pointers – 3Sum, Container With Most Water                                 |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                    |
| Theory Section          | Pattern 1: Two Pointers – opposite ends, fast/slow                             |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                               |
| Specific Exercises      | Week 17 Mon: Timed 2-problem session – explain approach before coding          |
| LeetCode                | <a href="#">LC 15: 3Sum</a> , <a href="#">LC 11: Container With Most Water</a> |
| Weekend / Project Tasks | None                                                                           |
| Time Split Today        | 0:00-0:30 pattern review                                                       |

## What Success Looks Like Today

Both mediums solved in  $\leq 35$  min each with clean two-pointer approach.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 114 – Tue | 8 Dec 2026 | 2 hours**

| Field                   | Detail                                                                                                                                 |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Study Date              | Tue, 8 Dec 2026                                                                                                                        |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                                                                              |
| Calendar                | Day 114 in <a href="#">calendar/mern-mastery-study.ics</a>                                                                             |
| Calendar Week           | Week 17 of 26                                                                                                                          |
| Month Phase             | Month 5 – DSA/LLD                                                                                                                      |
| Exact Topic             | Sliding Window – Longest Substring, Character Replacement                                                                              |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                                                                            |
| Theory Section          | Pattern 2: Sliding Window – fixed and variable                                                                                         |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                                                                       |
| Specific Exercises      | Week 17 Tue: Draw window state on paper before coding                                                                                  |
| LeetCode                | <a href="#">LC 3: Longest Substring Without Repeating Characters</a> , <a href="#">LC 424: Longest Repeating Character Replacement</a> |
| Weekend / Project Tasks | None                                                                                                                                   |
| Time Split Today        | 0:00-0:30 sliding window theory                                                                                                        |

**What Success Looks Like Today**

Variable window template internalized; both problems solved without hints.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 115 – Wed | 9 Dec 2026 | 2 hours**

| Field                   | Detail                                                                                       |
|-------------------------|----------------------------------------------------------------------------------------------|
| Study Date              | Wed, 9 Dec 2026                                                                              |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                                    |
| Calendar                | Day 115 in <a href="#">calendar/mern-mastery-study.ics</a>                                   |
| Calendar Week           | Week 17 of 26                                                                                |
| Month Phase             | Month 5 – DSA/LLD                                                                            |
| Exact Topic             | Hash Map – Group Anagrams, Longest Consecutive Sequence                                      |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                                  |
| Theory Section          | Pattern 3: Hash Map & Set                                                                    |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                             |
| Specific Exercises      | Week 17 Wed: Write brute force first, then optimize                                          |
| LeetCode                | <a href="#">LC 49: Group Anagrams</a> , <a href="#">LC 128: Longest Consecutive Sequence</a> |
| Weekend / Project Tasks | None                                                                                         |
| Time Split Today        | 0:00-0:30 hash map theory                                                                    |

**What Success Looks Like Today**

LC 128 solved  $O(n)$  with hash set; anagram grouping uses sorted key or count array.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 116 – Thu | 10 Dec 2026 | 2 hours**

| Field                   | Detail                                                                                          |
|-------------------------|-------------------------------------------------------------------------------------------------|
| Study Date              | Thu, 10 Dec 2026                                                                                |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                                       |
| Calendar                | Day 116 in <a href="#">calendar/mern-mastery-study.ics</a>                                      |
| Calendar Week           | Week 17 of 26                                                                                   |
| Month Phase             | Month 5 – DSA/LLD                                                                               |
| Exact Topic             | Subarray Problems – Prefix Sum + Sliding Window                                                 |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                                     |
| Theory Section          | Pattern 2-3: Subarray sum patterns                                                              |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                                |
| Specific Exercises      | Week 17 Thu: Mixed subarray problems                                                            |
| LeetCode                | <a href="#">LC 560: Subarray Sum Equals K</a> , <a href="#">LC 76: Minimum Window Substring</a> |
| Weekend / Project Tasks | None                                                                                            |
| Time Split Today        | 0:00-0:30 subarray theory                                                                       |

**What Success Looks Like Today**

LC 560 with prefix sum map; LC 76 with variable window — both explained aloud.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 117 – Fri | 11 Dec 2026 | 2 hours**

| Field                   | Detail                                                                        |
|-------------------------|-------------------------------------------------------------------------------|
| Study Date              | Fri, 11 Dec 2026                                                              |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                     |
| Calendar                | Day 117 in <a href="#">calendar/mern-mastery-study.ics</a>                    |
| Calendar Week           | Week 17 of 26                                                                 |
| Month Phase             | Month 5 – DSA/LLD                                                             |
| Exact Topic             | Revision Day – 2 Weakest Patterns                                             |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                   |
| Theory Section          | Review Patterns 1-3 – skim failed problems                                    |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                              |
| Specific Exercises      | Week 17 Fri: Redo 2 hardest problems from Mon-Thu without looking at old code |
| LeetCode                | Self-selected – pick 2 problems you failed or took >45 min                    |
| Weekend / Project Tasks | None                                                                          |
| Time Split Today        | 0:00-0:30 identify weak patterns                                              |

**What Success Looks Like Today**

Both redo problems solved faster than first attempt; gaps logged in REVIEW.md.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 118 – Sat | 12 Dec 2026 | 8 hours**

| Field                   | Detail                                                                                                                                                 |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Study Date              | Sat, 12 Dec 2026                                                                                                                                       |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                                                                                |
| Calendar                | Day 118 in <a href="#">calendar/mern-mastery-study.ics</a>                                                                                             |
| Calendar Week           | Week 17 of 26                                                                                                                                          |
| Month Phase             | Month 5 – DSA/LLD                                                                                                                                      |
| Exact Topic             | Sliding Window Hard Batch                                                                                                                              |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                                                                                            |
| Theory Section          | Pattern 2: Advanced sliding window                                                                                                                     |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                                                                                       |
| Specific Exercises      | Week 17 Sat: 3 hard sliding window problems timed                                                                                                      |
| LeetCode                | <a href="#">LC 239: Sliding Window Maximum</a> , <a href="#">LC 438: Find All Anagrams in a String</a> , <a href="#">LC 567: Permutation in String</a> |
| Weekend / Project Tasks | None                                                                                                                                                   |
| Time Split Today        | 0:00-2:00 LC 239                                                                                                                                       |

**What Success Looks Like Today**

3/3 sliding window hards solved; deque technique for LC 239 understood.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 119 – Sun | 17 13 Dec 2026 | 8 hours**

| Field                   | Detail                                                                             |
|-------------------------|------------------------------------------------------------------------------------|
| Study Date              | Sun, 13 Dec 2026                                                                   |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                            |
| Calendar                | Day 119 in <a href="#">calendar/mern-mastery-study.ics</a>                         |
| Calendar Week           | Week 17 of 26                                                                      |
| Month Phase             | Month 5 – DSA/LLD                                                                  |
| Exact Topic             | LLD: LRU Cache – Full Implementation                                               |
| Theory File             | <a href="#">detailed/theory/09-lld-complete.md</a>                                 |
| Theory Section          | LLD Problem 1: LRU Cache – requirements, class diagram                             |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                   |
| Specific Exercises      | Week 17 Sun: LRU Cache in TypeScript – HashMap + Doubly Linked List                |
| LeetCode                | Optional warm-up: LC 146 LRU Cache                                                 |
| Weekend / Project Tasks | LLD: LRU Cache – get/put O(1), capacity eviction, unit tests, explain aloud 10 min |
| Time Split Today        | 0:00-1:00 requirements + diagram                                                   |

**What Success Looks Like Today**

LRU Cache passes LC 146 equivalent tests; class diagram and extensibility explained.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 120 – Mon | 14 Dec 2026 | 2 hours**

| Field                   | Detail                                                                                                               |
|-------------------------|----------------------------------------------------------------------------------------------------------------------|
| Study Date              | Mon, 14 Dec 2026                                                                                                     |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                                                            |
| Calendar                | Day 120 in <a href="#">calendar/mern-mastery-study.ics</a>                                                           |
| Calendar Week           | Week 18 of 26                                                                                                        |
| Month Phase             | Month 5 – DSA/LLD                                                                                                    |
| Exact Topic             | Binary Search – Rotated Array, Find Minimum                                                                          |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                                                          |
| Theory Section          | Pattern 2: Binary Search – rotated, boundary                                                                         |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                                                     |
| Specific Exercises      | Week 18 Mon: Binary search template practice                                                                         |
| LeetCode                | <a href="#">LC 33: Search in Rotated Sorted Array</a> , <a href="#">LC 153: Find Minimum in Rotated Sorted Array</a> |
| Weekend / Project Tasks | None                                                                                                                 |
| Time Split Today        | 0:00-0:30 BS theory                                                                                                  |

**What Success Looks Like Today**

Both rotated array problems solved with unified binary search template.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 121 – Tue | 15 Dec 2026 | 2 hours**

| Field                   | Detail                                                                                  |
|-------------------------|-----------------------------------------------------------------------------------------|
| Study Date              | Tue, 15 Dec 2026                                                                        |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                               |
| Calendar                | Day 121 in <a href="#">calendar/mern-mastery-study.ics</a>                              |
| Calendar Week           | Week 18 of 26                                                                           |
| Month Phase             | Month 5 – DSA/LLD                                                                       |
| Exact Topic             | Binary Search on Answer – 2D Matrix, Koko Eating Bananas                                |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                             |
| Theory Section          | Pattern 2: Binary Search on Answer Space                                                |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                        |
| Specific Exercises      | Week 18 Tue: Identify monotonic predicate before coding                                 |
| LeetCode                | <a href="#">LC 74: Search a 2D Matrix</a> , <a href="#">LC 875: Koko Eating Bananas</a> |
| Weekend / Project Tasks | None                                                                                    |
| Time Split Today        | 0:00-0:30 BS-on-answer theory                                                           |

**What Success Looks Like Today**

LC 875 solved with BS on answer; you explain predicate function clearly.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 122 – Wed | 17 16 Dec 2026 | 2 hours**

| Field                   | Detail                                                                                             |
|-------------------------|----------------------------------------------------------------------------------------------------|
| Study Date              | Wed, 16 Dec 2026                                                                                   |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                                          |
| Calendar                | Day 122 in <a href="#">calendar/mern-mastery-study.ics</a>                                         |
| Calendar Week           | Week 18 of 26                                                                                      |
| Month Phase             | Month 5 – DSA/LLD                                                                                  |
| Exact Topic             | Stack – Daily Temperatures, Largest Rectangle                                                      |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                                        |
| Theory Section          | Pattern 4: Monotonic Stack                                                                         |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                                   |
| Specific Exercises      | Week 18 Wed: Monotonic stack template                                                              |
| LeetCode                | <a href="#">LC 739: Daily Temperatures</a> , <a href="#">LC 84: Largest Rectangle in Histogram</a> |
| Weekend / Project Tasks | None                                                                                               |
| Time Split Today        | 0:00-0:30 stack theory                                                                             |

**What Success Looks Like Today**

Monotonic stack template applied to both; LC 84 attempted (study solution if stuck 40+ min).

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 123 – Thu | 17 Dec 2026 | 2 hours**

| Field                   | Detail                                                                                                  |
|-------------------------|---------------------------------------------------------------------------------------------------------|
| Study Date              | Thu, 17 Dec 2026                                                                                        |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                                               |
| Calendar                | Day 123 in <a href="#">calendar/mern-mastery-study.ics</a>                                              |
| Calendar Week           | Week 18 of 26                                                                                           |
| Month Phase             | Month 5 – DSA/LLD                                                                                       |
| Exact Topic             | Linked List Hard – Reverse k-Group, Copy with Random Pointer                                            |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                                             |
| Theory Section          | Pattern 5: Linked List – advanced                                                                       |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                                        |
| Specific Exercises      | Week 18 Thu: Draw pointer mutations on paper first                                                      |
| LeetCode                | <a href="#">LC 25: Reverse Nodes in k-Group</a> , <a href="#">LC 138: Copy List with Random Pointer</a> |
| Weekend / Project Tasks | None                                                                                                    |
| Time Split Today        | 0:00-0:30 LL theory                                                                                     |

**What Success Looks Like Today**

At least 1/2 hard LL solved; pointer diagram drawn before coding.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 124 – Fri | 18 Dec 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Fri, 18 Dec 2026                                            |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 124 in <a href="#">calendar/mern-mastery-study.ics</a>  |
| Calendar Week           | Week 18 of 26                                               |
| Month Phase             | Month 5 – DSA/LLD                                           |
| Exact Topic             | Revision Day – Week 18 Weak Spots                           |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a> |
| Theory Section          | Patterns 2, 4, 5 review                                     |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>            |
| Specific Exercises      | Week 18 Fri: Redo 2 weakest from Mon-Thu                    |
| LeetCode                | Self-selected revision problems                             |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:30 review notes                                      |

**What Success Looks Like Today**

Weak patterns identified; redo time improved  $\geq 25\%$  vs first attempt.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 125 – Sat | 17 19 Dec 2026 | 8 hours**

| Field                   | Detail                                                                                                                                               |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Study Date              | Sat, 19 Dec 2026                                                                                                                                     |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                                                                              |
| Calendar                | Day 125 in <a href="#">calendar/mern-mastery-study.ics</a>                                                                                           |
| Calendar Week           | Week 18 of 26                                                                                                                                        |
| Month Phase             | Month 5 – DSA/LLD                                                                                                                                    |
| Exact Topic             | Intervals – Merge, Insert, Min Arrows                                                                                                                |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                                                                                          |
| Theory Section          | Pattern: Intervals – merge and sweep                                                                                                                 |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                                                                                     |
| Specific Exercises      | Week 18 Sat: 3 interval problems timed                                                                                                               |
| LeetCode                | <a href="#">LC 56: Merge Intervals</a> , <a href="#">LC 57: Insert Interval</a> , <a href="#">LC 452: Minimum Number of Arrows to Burst Balloons</a> |
| Weekend / Project Tasks | None                                                                                                                                                 |
| Time Split Today        | 0:00-2:00 LC 56                                                                                                                                      |

**What Success Looks Like Today**

3 interval problems solved; merge template reusable for scheduling problems.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 126 – Sun | 20 Dec 2026 | 8 hours**

| Field                   | Detail                                                                         |
|-------------------------|--------------------------------------------------------------------------------|
| Study Date              | Sun, 20 Dec 2026                                                               |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                        |
| Calendar                | Day 126 in <a href="#">calendar/mern-mastery-study.ics</a>                     |
| Calendar Week           | Week 18 of 26                                                                  |
| Month Phase             | Month 5 – DSA/LLD                                                              |
| Exact Topic             | LLD: Rate Limiter + Parking Lot                                                |
| Theory File             | <a href="#">detailed/theory/09-lld-complete.md</a>                             |
| Theory Section          | LLD Problem 2: Rate Limiter; LLD Problem 3: Parking Lot                        |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                               |
| Specific Exercises      | Week 18 Sun: Token bucket rate limiter + multi-floor parking lot               |
| LeetCode                | None                                                                           |
| Weekend / Project Tasks | LLD: Rate Limiter (token bucket) + Parking Lot (OOP design with Vehicle types) |
| Time Split Today        | 0:00-4:00 Rate Limiter                                                         |

**What Success Looks Like Today**

Both LLD problems coded in TypeScript; 10-min requirements + diagram each; extensibility discussed.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

## Day 127 – Mon | 21 Dec 2026 | 2 hours

| Field                   | Detail                                                                                                                                        |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Study Date              | Mon, 21 Dec 2026                                                                                                                              |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                                                                                     |
| Calendar                | Day 127 in <a href="#">calendar/mern-mastery-study.ics</a>                                                                                    |
| Calendar Week           | Week 19 of 26                                                                                                                                 |
| Month Phase             | Month 5 – DSA/LLD                                                                                                                             |
| Exact Topic             | Tree BFS – Level Order, Construct from Preorder/Inorder                                                                                       |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                                                                                   |
| Theory Section          | Pattern 6: Trees – BFS level order                                                                                                            |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                                                                              |
| Specific Exercises      | Week 19 Mon: BFS with queue template                                                                                                          |
| LeetCode                | <a href="#">LC 102: Binary Tree Level Order Traversal</a> , <a href="#">LC 105: Construct Binary Tree from Preorder and Inorder Traversal</a> |
| Weekend / Project Tasks | None                                                                                                                                          |
| Time Split Today        | 0:00-0:30 tree BFS theory                                                                                                                     |

### What Success Looks Like Today

BFS template applied; LC 105 solved with recursive divide on inorder index.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 128 – Tue | 22 Dec 2026 | 2 hours**

| Field                   | Detail                                                                                                                |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Study Date              | Tue, 22 Dec 2026                                                                                                      |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                                                             |
| Calendar                | Day 128 in <a href="#">calendar/mern-mastery-study.ics</a>                                                            |
| Calendar Week           | Week 19 of 26                                                                                                         |
| Month Phase             | Month 5 – DSA/LLD                                                                                                     |
| Exact Topic             | Tree Properties – Right Side View, LCA                                                                                |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                                                           |
| Theory Section          | Pattern 6: Trees – DFS variants                                                                                       |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                                                      |
| Specific Exercises      | Week 19 Tue: DFS with state passing                                                                                   |
| LeetCode                | <a href="#">LC 199: Binary Tree Right Side View</a> , <a href="#">LC 236: Lowest Common Ancestor of a Binary Tree</a> |
| Weekend / Project Tasks | None                                                                                                                  |
| Time Split Today        | 0:00-0:30 tree DFS theory                                                                                             |

**What Success Looks Like Today**

LC 236 solved with recursive LCA; right side view uses BFS or DFS with depth tracking.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 129 – Wed | 23 Dec 2026 | 2 hours**

| Field                   | Detail                                                                          |
|-------------------------|---------------------------------------------------------------------------------|
| Study Date              | Wed, 23 Dec 2026                                                                |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                       |
| Calendar                | Day 129 in <a href="#">calendar/mern-mastery-study.ics</a>                      |
| Calendar Week           | Week 19 of 26                                                                   |
| Month Phase             | Month 5 – DSA/LLD                                                               |
| Exact Topic             | Graph BFS/DFS – Number of Islands, Clone Graph                                  |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                     |
| Theory Section          | Pattern 7: Graphs – BFS/DFS on grid and adjacency list                          |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                |
| Specific Exercises      | Week 19 Wed: Graph traversal templates                                          |
| LeetCode                | <a href="#">LC 200: Number of Islands</a> , <a href="#">LC 133: Clone Graph</a> |
| Weekend / Project Tasks | None                                                                            |
| Time Split Today        | 0:00-0:30 graph theory                                                          |

**What Success Looks Like Today**

Both graph problems solved; you explain visited set vs in-place marking tradeoff.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 130 – Thu | 24 Dec 2026 | 2 hours**

| Field                   | Detail                                                                               |
|-------------------------|--------------------------------------------------------------------------------------|
| Study Date              | Thu, 24 Dec 2026                                                                     |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                            |
| Calendar                | Day 130 in <a href="#">calendar/mern-mastery-study.ics</a>                           |
| Calendar Week           | Week 19 of 26                                                                        |
| Month Phase             | Month 5 – DSA/LLD                                                                    |
| Exact Topic             | Topological Sort – Course Schedule I & II                                            |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                          |
| Theory Section          | Pattern 7: Graphs – Topological Sort (Kahn's / DFS)                                  |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                     |
| Specific Exercises      | Week 19 Thu: Draw DAG before coding topo sort                                        |
| LeetCode                | <a href="#">LC 207: Course Schedule</a> , <a href="#">LC 210: Course Schedule II</a> |
| Weekend / Project Tasks | None                                                                                 |
| Time Split Today        | 0:00-0:30 topo sort theory                                                           |

**What Success Looks Like Today**

Cycle detection works for LC 207; LC 210 returns valid ordering.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 131 – Fri | 25 Dec 2026 | 2 hours**

| Field                   | Detail                                                      |
|-------------------------|-------------------------------------------------------------|
| Study Date              | Fri, 25 Dec 2026                                            |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                   |
| Calendar                | Day 131 in <a href="#">calendar/mern-mastery-study.ics</a>  |
| Calendar Week           | Week 19 of 26                                               |
| Month Phase             | Month 5 – DSA/LLD                                           |
| Exact Topic             | Revision Day – Trees + Graphs                               |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a> |
| Theory Section          | Patterns 6–7 review                                         |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>            |
| Specific Exercises      | Week 19 Fri: Redo 2 weakest tree/graph problems             |
| LeetCode                | Self-selected revision                                      |
| Weekend / Project Tasks | None                                                        |
| Time Split Today        | 0:00-0:30 flashcards                                        |

**What Success Looks Like Today**

Tree and graph templates written on flashcard; 2 redo problems solved.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

## Day 132 – Sat | 26 Dec 2026 | 8 hours

| Field                   | Detail                                                                                                                                 |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Study Date              | Sat, 26 Dec 2026                                                                                                                       |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                                                                |
| Calendar                | Day 132 in <a href="#">calendar/mern-mastery-study.ics</a>                                                                             |
| Calendar Week           | Week 19 of 26                                                                                                                          |
| Month Phase             | Month 5 – DSA/LLD                                                                                                                      |
| Exact Topic             | Graph Hard Batch – Word Ladder, Surrounded Regions, Pacific Atlantic                                                                   |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                                                                            |
| Theory Section          | Pattern 7: Multi-source BFS, boundary DFS                                                                                              |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                                                                       |
| Specific Exercises      | Week 19 Sat: 3 graph mediums timed                                                                                                     |
| LeetCode                | <a href="#">LC 127: Word Ladder</a> , <a href="#">LC 130: Surrounded Regions</a> , <a href="#">LC 417: Pacific Atlantic Water Flow</a> |
| Weekend / Project Tasks | None                                                                                                                                   |
| Time Split Today        | 0:00-2:00 LC 127                                                                                                                       |

### What Success Looks Like Today

2/3 graph mediums solved; multi-source BFS understood for LC 417.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 133 – Sun | 27 Dec 2026 | 8 hours**

| Field                   | Detail                                                                                                         |
|-------------------------|----------------------------------------------------------------------------------------------------------------|
| Study Date              | Sun, 27 Dec 2026                                                                                               |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                                        |
| Calendar                | Day 133 in <a href="#">calendar/mern-mastery-study.ics</a>                                                     |
| Calendar Week           | Week 19 of 26                                                                                                  |
| Month Phase             | Month 5 – DSA/LLD                                                                                              |
| Exact Topic             | LLD: Splitwise + Vending Machine                                                                               |
| Theory File             | <a href="#">detailed/theory/09-lld-complete.md</a>                                                             |
| Theory Section          | LLD Problem 4: Splitwise; LLD Problem 5: Vending Machine                                                       |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                                               |
| Specific Exercises      | Week 19 Sun: Expense splitting + vending machine state machine                                                 |
| LeetCode                | None                                                                                                           |
| Weekend / Project Tasks | LLD: Splitwise (add expense, split equally/exact, balance sheet) + Vending Machine (states, inventory, change) |
| Time Split Today        | 0:00-4:00 Splitwise                                                                                            |

**What Success Looks Like Today**

Both systems coded with clear class responsibilities; state pattern used in vending machine.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 134 – Mon | 28 Dec 2026 | 2 hours**

| Field                   | Detail                                                                         |
|-------------------------|--------------------------------------------------------------------------------|
| Study Date              | Mon, 28 Dec 2026                                                               |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                      |
| Calendar                | Day 134 in <a href="#">calendar/mern-mastery-study.ics</a>                     |
| Calendar Week           | Week 20 of 26                                                                  |
| Month Phase             | Month 5 – DSA/LLD                                                              |
| Exact Topic             | 1D Dynamic Programming – House Robber I & II                                   |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                    |
| Theory Section          | Pattern 8: DP – 1D linear                                                      |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                               |
| Specific Exercises      | Week 20 Mon: DP recurrence before code                                         |
| LeetCode                | <a href="#">LC 198: House Robber</a> , <a href="#">LC 213: House Robber II</a> |
| Weekend / Project Tasks | None                                                                           |
| Time Split Today        | 0:00-0:30 1D DP theory                                                         |

**What Success Looks Like Today**

Both house robber variants solved; recurrence relation written on paper first.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 135 – Tue | 29 Dec 2026 | 2 hours**

| Field                   | Detail                                                                       |
|-------------------------|------------------------------------------------------------------------------|
| Study Date              | Tue, 29 Dec 2026                                                             |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                    |
| Calendar                | Day 135 in <a href="#">calendar/mern-mastery-study.ics</a>                   |
| Calendar Week           | Week 20 of 26                                                                |
| Month Phase             | Month 5 – DSA/LLD                                                            |
| Exact Topic             | Coin Change Pattern – Unbounded Knapsack                                     |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                  |
| Theory Section          | Pattern 8: DP – coin change variants                                         |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                             |
| Specific Exercises      | Week 20 Tue: Compare top-down vs bottom-up                                   |
| LeetCode                | <a href="#">LC 322: Coin Change</a> , <a href="#">LC 518: Coin Change II</a> |
| Weekend / Project Tasks | None                                                                         |
| Time Split Today        | 0:00-0:30 coin change theory                                                 |

**What Success Looks Like Today**

LC 322 and LC 518 solved bottom-up; you explain difference (min vs count ways).

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 136 – Wed | 30 Dec 2026 | 2 hours**

| Field                   | Detail                                                                                     |
|-------------------------|--------------------------------------------------------------------------------------------|
| Study Date              | Wed, 30 Dec 2026                                                                           |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                                  |
| Calendar                | Day 136 in <a href="#">calendar/mern-mastery-study.ics</a>                                 |
| Calendar Week           | Week 20 of 26                                                                              |
| Month Phase             | Month 5 – DSA/LLD                                                                          |
| Exact Topic             | String DP – LCS and Edit Distance                                                          |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                                |
| Theory Section          | Pattern 8: DP – 2D string DP                                                               |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                           |
| Specific Exercises      | Week 20 Wed: Fill DP table on paper for small input                                        |
| LeetCode                | <a href="#">LC 1143: Longest Common Subsequence</a> , <a href="#">LC 72: Edit Distance</a> |
| Weekend / Project Tasks | None                                                                                       |
| Time Split Today        | 0:00-0:30 string DP theory                                                                 |

**What Success Looks Like Today**

2D DP table drawn manually for "abc"/"adc"; both problems solved.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 137 – Thu | 31 Dec 2026 | 2 hours**

| Field                   | Detail                                                                       |
|-------------------------|------------------------------------------------------------------------------|
| Study Date              | Thu, 31 Dec 2026                                                             |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                    |
| Calendar                | Day 137 in <a href="#">calendar/mern-mastery-study.ics</a>                   |
| Calendar Week           | Week 20 of 26                                                                |
| Month Phase             | Month 5 – DSA/LLD                                                            |
| Exact Topic             | DP Hard – Burst Balloons or Decode Ways                                      |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                  |
| Theory Section          | Pattern 8: DP – interval and counting                                        |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                             |
| Specific Exercises      | Week 20 Thu: Pick harder DP – spend full 2h on one problem                   |
| LeetCode                | <a href="#">LC 312: Burst Balloons</a> OR <a href="#">LC 91: Decode Ways</a> |
| Weekend / Project Tasks | None                                                                         |
| Time Split Today        | 0:00-0:30 hard DP theory                                                     |

**What Success Looks Like Today**

One hard DP solved or thoroughly studied with solution after 45 min attempt.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 138 – Fri | 17 1 Jan 2027 | 2 hours**

| Field                   | Detail                                                          |
|-------------------------|-----------------------------------------------------------------|
| Study Date              | Fri, 1 Jan 2027                                                 |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                       |
| Calendar                | Day 138 in <a href="#">calendar/mern-mastery-study.ics</a>      |
| Calendar Week           | Week 20 of 26                                                   |
| Month Phase             | Month 5 – DSA/LLD                                               |
| Exact Topic             | Mixed DP Revision – 5 Problems                                  |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>     |
| Theory Section          | Pattern 8: Full DP review                                       |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                |
| Specific Exercises      | Week 20 Fri: 5 mixed DP problems – 20 min each                  |
| LeetCode                | Pick 5 from: LC 70, 139, 198, 322, 518, 1143 (unsolved or weak) |
| Weekend / Project Tasks | None                                                            |
| Time Split Today        | 0:00-0:30 DP cheat sheet                                        |

**What Success Looks Like Today**

≥3/5 mixed DP solved in 20 min each; DP pattern cheat sheet updated.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 139 – Sat | 17 2 Jan 2027 | 8 hours**

| Field                   | Detail                                                                                                                                     |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Study Date              | Sat, 2 Jan 2027                                                                                                                            |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                                                                    |
| Calendar                | Day 139 in <a href="#">calendar/mern-mastery-study.ics</a>                                                                                 |
| Calendar Week           | Week 20 of 26                                                                                                                              |
| Month Phase             | Month 5 – DSA/LLD                                                                                                                          |
| Exact Topic             | DP Hard Batch – Word Break, Max Product, Partition                                                                                         |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                                                                                |
| Theory Section          | Pattern 8: Advanced DP                                                                                                                     |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                                                                           |
| Specific Exercises      | Week 20 Sat: 3 DP mediums/hards                                                                                                            |
| LeetCode                | <a href="#">LC 139: Word Break</a> , <a href="#">LC 152: Maximum Product Subarray</a> , <a href="#">LC 416: Partition Equal Subset Sum</a> |
| Weekend / Project Tasks | None                                                                                                                                       |
| Time Split Today        | 0:00-2:00 LC 139                                                                                                                           |

**What Success Looks Like Today**

2/3 DP batch solved; word break uses DP + set lookup.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 140 – Sun | 3 Jan 2027 | 8 hours**

| Field                   | Detail                                                                                       |
|-------------------------|----------------------------------------------------------------------------------------------|
| Study Date              | Sun, 3 Jan 2027                                                                              |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                      |
| Calendar                | Day 140 in <a href="#">calendar/mern-mastery-study.ics</a>                                   |
| Calendar Week           | Week 20 of 26                                                                                |
| Month Phase             | Month 5 – DSA/LLD                                                                            |
| Exact Topic             | LLD: BookMyShow + 200 LC Milestone                                                           |
| Theory File             | <a href="#">detailed/theory/09-lld-complete.md</a>                                           |
| Theory Section          | LLD Problem 6: BookMyShow Seat Booking                                                       |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                             |
| Specific Exercises      | Week 20 Sun: Seat locking, show, theatre, booking flow                                       |
| LeetCode                | Review LC count – target 200+ total solved                                                   |
| Weekend / Project Tasks | LLD: BookMyShow – Theatre, Show, Seat (locked/booked/available), concurrent booking handling |
| Time Split Today        | 0:00-4:00 BookMyShow LLD                                                                     |

**What Success Looks Like Today**

200+ LC milestone; BookMyShow core booking flow coded; 6 LLD problems complete.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

# MONTH 6 – System Design + Interviews (Days 141–168)

MERN MASTERY · 1 CR PRODUCT PREP · HARSH VARDHAN GUPTA

Day 141 – Mon |  17 4 Jan 2027 | 2 hours

| Field                   | Detail                                                                                                  |
|-------------------------|---------------------------------------------------------------------------------------------------------|
| Study Date              | Mon, 4 Jan 2027                                                                                         |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                                               |
| Calendar                | Day 141 in <a href="calendar/mern-mastery-study.ics">calendar/mern-mastery-study.ics</a>                |
| Calendar Week           | Week 21 of 26                                                                                           |
| Month Phase             | Month 6 – System Design                                                                                 |
| Exact Topic             | System Design Framework + Capacity Estimation                                                           |
| Theory File             | <a href="detailed/theory/10-system-design-complete.md">detailed/theory/10-system-design-complete.md</a> |
| Theory Section          | Interview Framework (RADIO/RESHADED) + Estimation                                                       |
| Practice File           | <a href="practice/week-21-24-exercises.md">practice/week-21-24-exercises.md</a>                         |
| Specific Exercises      | Week 21 Mon: Practice estimation on URL Shortener (QPS, storage)                                        |
| LeetCode                | 1 medium timed (self-selected from weak list)                                                           |
| Weekend / Project Tasks | None                                                                                                    |
| Time Split Today        | 0:00-0:45 SD framework                                                                                  |

## What Success Looks Like Today

You can run RADIO framework in 5 min; back-of-envelope QPS/storage for any system.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message

- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

## Day 142 – Tue | 17 5 Jan 2027 | 2 hours

| Field                   | Detail                                                                  |
|-------------------------|-------------------------------------------------------------------------|
| Study Date              | Tue, 5 Jan 2027                                                         |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                               |
| Calendar                | Day 142 in <a href="#">calendar/mern-mastery-study.ics</a>              |
| Calendar Week           | Week 21 of 26                                                           |
| Month Phase             | Month 6 – System Design                                                 |
| Exact Topic             | HLD: URL Shortener                                                      |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>            |
| Theory Section          | Design 1: URL Shortener – Base62, caching, redirect                     |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                        |
| Specific Exercises      | Week 21 Tue: Draw full diagram – client, API, DB, cache, CDN            |
| LeetCode                | 1 medium timed                                                          |
| Weekend / Project Tasks | SD deliverable: URL Shortener diagram + component list + API design doc |
| Time Split Today        | 0:00-0:45 design draw                                                   |

### What Success Looks Like Today

Complete URL Shortener design in 45 min; Base62 encoding explained; cache strategy defined.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 143 – Wed | 6 Jan 2027 | 2 hours**

| Field                   | Detail                                                                  |
|-------------------------|-------------------------------------------------------------------------|
| Study Date              | Wed, 6 Jan 2027                                                         |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                               |
| Calendar                | Day 143 in <a href="#">calendar/mern-mastery-study.ics</a>              |
| Calendar Week           | Week 21 of 26                                                           |
| Month Phase             | Month 6 – System Design                                                 |
| Exact Topic             | HLD: Paste Bin + Rate Limiter                                           |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>            |
| Theory Section          | Design: Paste Bin (TTL storage) + Distributed Rate Limiter              |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                        |
| Specific Exercises      | Week 21 Wed: Paste Bin TTL strategy + token bucket rate limiter design  |
| LeetCode                | 1 medium timed                                                          |
| Weekend / Project Tasks | SD deliverable: Paste Bin + Rate Limiter combined notes with Redis role |
| Time Split Today        | 0:00-0:45 Paste Bin                                                     |

**What Success Looks Like Today**

Paste Bin TTL eviction explained; rate limiter uses Redis token bucket with sliding window option.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 144 – Thu | 7 Jan 2027 | 2 hours**

| Field                   | Detail                                                                      |
|-------------------------|-----------------------------------------------------------------------------|
| Study Date              | Thu, 7 Jan 2027                                                             |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                   |
| Calendar                | Day 144 in <a href="#">calendar/mern-mastery-study.ics</a>                  |
| Calendar Week           | Week 21 of 26                                                               |
| Month Phase             | Month 6 – System Design                                                     |
| Exact Topic             | Caching, Load Balancing, Key-Value Store Intro                              |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                |
| Theory Section          | Core Concepts: Caching, Load Balancing, Consistent Hashing                  |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                            |
| Specific Exercises      | Week 21 Thu: Key-Value store high-level design sketch                       |
| LeetCode                | 1 medium timed                                                              |
| Weekend / Project Tasks | SD deliverable: Key-Value store sketch with consistent hashing ring diagram |
| Time Split Today        | 0:00-0:45 caching/LB theory                                                 |

**What Success Looks Like Today**

Cache aside vs write-through explained; consistent hashing diagram drawn from memory.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 145 – Fri | 8 Jan 2027 | 2 hours**

| Field                   | Detail                                                       |
|-------------------------|--------------------------------------------------------------|
| Study Date              | Fri, 8 Jan 2027                                              |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                    |
| Calendar                | Day 145 in <a href="#">calendar/mern-mastery-study.ics</a>   |
| Calendar Week           | Week 21 of 26                                                |
| Month Phase             | Month 6 – System Design                                      |
| Exact Topic             | SD Revision + 2 Timed Mediums                                |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a> |
| Theory Section          | Review Designs 1 + core concepts                             |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>             |
| Specific Exercises      | Week 21 Fri: Redraw URL Shortener blind in 30 min            |
| LeetCode                | 2 mediums timed (35 min each)                                |
| Weekend / Project Tasks | None                                                         |
| Time Split Today        | 0:00-0:30 blind URL Shortener                                |

**What Success Looks Like Today**

URL Shortener redrawn without notes in  $\leq 30$  min; 2 LC mediums under time.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 146 – Sat | 17 9 Jan 2027 | 8 hours**

| Field                   | Detail                                                                   |
|-------------------------|--------------------------------------------------------------------------|
| Study Date              | Sat, 9 Jan 2027                                                          |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                  |
| Calendar                | Day 146 in <a href="#">calendar/mern-mastery-study.ics</a>               |
| Calendar Week           | Week 21 of 26                                                            |
| Month Phase             | Month 6 – System Design                                                  |
| Exact Topic             | URL Shortener Blind Redesign + 3 LC                                      |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>             |
| Theory Section          | Design 1 review – deep dive on failure modes                             |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                         |
| Specific Exercises      | Week 21 Sat: Full 45-min blind SD + 3 LC mediums                         |
| LeetCode                | 3 self-selected mediums (timed 35 min each, pick weak patterns)          |
| Weekend / Project Tasks | SD: URL Shortener blind redesign – record yourself explaining for 10 min |
| Time Split Today        | 0:00-0:45 blind SD                                                       |

**What Success Looks Like Today**

Blind SD in 45 min with no major gaps; 3 LC mediums solved.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 147 – Sun | 10 Jan 2027 | 8 hours**

| Field                   | Detail                                                                      |
|-------------------------|-----------------------------------------------------------------------------|
| Study Date              | Sun, 10 Jan 2027                                                            |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                     |
| Calendar                | Day 147 in <a href="#">calendar/mern-mastery-study.ics</a>                  |
| Calendar Week           | Week 21 of 26                                                               |
| Month Phase             | Month 6 – System Design                                                     |
| Exact Topic             | Feed Systems Intro + 3 LC                                                   |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                |
| Theory Section          | Design 2: Instagram/Twitter Feed – fan-out on write vs read                 |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                            |
| Specific Exercises      | Week 21 Sun: Feed system intro diagram + 3 LC mediums                       |
| LeetCode                | 3 mediums timed                                                             |
| Weekend / Project Tasks | SD deliverable: Feed system fan-out vs fan-in comparison doc with tradeoffs |
| Time Split Today        | 0:00-2:00 feed systems study                                                |

**What Success Looks Like Today**

Fan-out on write vs read tradeoffs explained with celebrity user edge case; 3 LC done.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 148 – Mon | 11 Jan 2027 | 2 hours**

| Field                   | Detail                                                                                 |
|-------------------------|----------------------------------------------------------------------------------------|
| Study Date              | Mon, 11 Jan 2027                                                                       |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                              |
| Calendar                | Day 148 in <a href="#">calendar/mern-mastery-study.ics</a>                             |
| Calendar Week           | Week 22 of 26                                                                          |
| Month Phase             | Month 6 – System Design                                                                |
| Exact Topic             | HLD: Twitter / X Feed                                                                  |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                           |
| Theory Section          | Design 2: Instagram/Twitter Feed (timeline generation)                                 |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                       |
| Specific Exercises      | Week 22 Mon: Twitter feed – home timeline, tweet creation, follow graph                |
| LeetCode                | 1 medium timed                                                                         |
| Weekend / Project Tasks | SD deliverable: Twitter feed full diagram with timeline service, fan-out worker, cache |
| Time Split Today        | 0:00-0:45 Twitter feed design                                                          |

**What Success Looks Like Today**

Home timeline generation explained for normal and celebrity users; tweet write path clear.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 149 – Tue | 12 Jan 2027 | 2 hours**

| Field                   | Detail                                                                   |
|-------------------------|--------------------------------------------------------------------------|
| Study Date              | Tue, 12 Jan 2027                                                         |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                |
| Calendar                | Day 149 in <a href="#">calendar/mern-mastery-study.ics</a>               |
| Calendar Week           | Week 22 of 26                                                            |
| Month Phase             | Month 6 – System Design                                                  |
| Exact Topic             | HLD: Instagram                                                           |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>             |
| Theory Section          | Design 2: Instagram – photos, feed, stories overview                     |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                         |
| Specific Exercises      | Week 22 Tue: Instagram – media storage, CDN, feed ranking basics         |
| LeetCode                | 1 medium timed                                                           |
| Weekend / Project Tasks | SD deliverable: Instagram design with media upload flow and CDN strategy |
| Time Split Today        | 0:00-0:45 Instagram design                                               |

**What Success Looks Like Today**

Photo upload → S3 → CDN → feed pipeline drawn; metadata vs blob storage separated.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 150 – Wed | 13 Jan 2027 | 2 hours**

| Field                   | Detail                                                                          |
|-------------------------|---------------------------------------------------------------------------------|
| Study Date              | Wed, 13 Jan 2027                                                                |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                       |
| Calendar                | Day 150 in <a href="#">calendar/mern-mastery-study.ics</a>                      |
| Calendar Week           | Week 22 of 26                                                                   |
| Month Phase             | Month 6 – System Design                                                         |
| Exact Topic             | HLD: Notification System                                                        |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                    |
| Theory Section          | Design 4: Notification System – push, email, SMS, queues                        |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                |
| Specific Exercises      | Week 22 Wed: Notification system – priority queues, delivery guarantees         |
| LeetCode                | 1 medium timed                                                                  |
| Weekend / Project Tasks | SD deliverable: Notification system with Kafka queue, worker pools, retry logic |
| Time Split Today        | 0:00-0:45 notification SD                                                       |

**What Success Looks Like Today**

At-least-once delivery explained; priority queue for urgent vs batch notifications.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 151 – Thu | 14 Jan 2027 | 2 hours**

| Field                   | Detail                                                                     |
|-------------------------|----------------------------------------------------------------------------|
| Study Date              | Thu, 14 Jan 2027                                                           |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                  |
| Calendar                | Day 151 in <a href="#">calendar/mern-mastery-study.ics</a>                 |
| Calendar Week           | Week 22 of 26                                                              |
| Month Phase             | Month 6 – System Design                                                    |
| Exact Topic             | Message Queues, Kafka Basics                                               |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>               |
| Theory Section          | Core Concepts: Message Queues, Kafka, Pub/Sub                              |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                           |
| Specific Exercises      | Week 22 Thu: Read Kafka docs – producers, consumers, partitions, offsets   |
| LeetCode                | 1 medium timed                                                             |
| Weekend / Project Tasks | SD notes: Kafka vs RabbitMQ comparison for notification and feed use cases |
| Time Split Today        | 0:00-0:45 Kafka theory                                                     |

**What Success Looks Like Today**

Kafka partitions and consumer groups explained; comparison doc written.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 152 – Fri | 15 Jan 2027 | 2 hours**

| Field                   | Detail                                                       |
|-------------------------|--------------------------------------------------------------|
| Study Date              | Fri, 15 Jan 2027                                             |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                    |
| Calendar                | Day 152 in <a href="#">calendar/mern-mastery-study.ics</a>   |
| Calendar Week           | Week 22 of 26                                                |
| Month Phase             | Month 6 – System Design                                      |
| Exact Topic             | SD Revision + 2 Timed Mediums                                |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a> |
| Theory Section          | Review Designs 2, 4                                          |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>             |
| Specific Exercises      | Week 22 Fri: Redraw notification system blind                |
| LeetCode                | 2 mediums timed                                              |
| Weekend / Project Tasks | None                                                         |
| Time Split Today        | 0:00-0:30 blind notification SD                              |

**What Success Looks Like Today**

Notification system redrawn in 30 min; 2 LC mediums under time limit.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 153 – Sat | 16 Jan 2027 | 8 hours**

| Field                   | Detail                                                                            |
|-------------------------|-----------------------------------------------------------------------------------|
| Study Date              | Sat, 16 Jan 2027                                                                  |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                           |
| Calendar                | Day 153 in <a href="#">calendar/mern-mastery-study.ics</a>                        |
| Calendar Week           | Week 22 of 26                                                                     |
| Month Phase             | Month 6 – System Design                                                           |
| Exact Topic             | Notification SD Deep Dive + 3 LC                                                  |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                      |
| Theory Section          | Design 4: Notification – scale, dedup, preferences                                |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                  |
| Specific Exercises      | Week 22 Sat: Notification SD 45 min + 3 LC                                        |
| LeetCode                | 3 mediums timed                                                                   |
| Weekend / Project Tasks | SD: Notification system with user preference filtering and deduplication strategy |
| Time Split Today        | 0:00-0:45 notification deep dive                                                  |

**What Success Looks Like Today**

Dedup and user opt-out handled in design; 3 LC mediums solved.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 154 – Sun | 17 Jan 2027 | 8 hours**

| Field                   | Detail                                                             |
|-------------------------|--------------------------------------------------------------------|
| Study Date              | Sun, 17 Jan 2027                                                   |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                            |
| Calendar                | Day 154 in <a href="#">calendar/mern-mastery-study.ics</a>         |
| Calendar Week           | Week 22 of 26                                                      |
| Month Phase             | Month 6 – System Design                                            |
| Exact Topic             | Feed Design Fan-Out + 3 LC                                         |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>       |
| Theory Section          | Design 2: Feed – hybrid fan-out, cache warming                     |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                   |
| Specific Exercises      | Week 22 Sun: Hybrid fan-out design + 3 LC mediums                  |
| LeetCode                | 3 mediums timed                                                    |
| Weekend / Project Tasks | SD: Hybrid fan-out feed design – when to use write vs read fan-out |
| Time Split Today        | 0:00-2:00 feed fan-out design                                      |

**What Success Looks Like Today**

Hybrid approach justified with user follower count thresholds; 3 LC done.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 155 – Mon | 18 Jan 2027 | 2 hours**

| Field                   | Detail                                                                           |
|-------------------------|----------------------------------------------------------------------------------|
| Study Date              | Mon, 18 Jan 2027                                                                 |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                        |
| Calendar                | Day 155 in <a href="#">calendar/mern-mastery-study.ics</a>                       |
| Calendar Week           | Week 23 of 26                                                                    |
| Month Phase             | Month 6 – System Design                                                          |
| Exact Topic             | HLD: WhatsApp / Chat System                                                      |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                     |
| Theory Section          | Design 3: Chat System (WhatsApp-like)                                            |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                 |
| Specific Exercises      | Week 23 Mon: Chat – 1:1 messaging, message sync, delivery receipts               |
| LeetCode                | 1 medium timed                                                                   |
| Weekend / Project Tasks | SD deliverable: Chat system with WebSocket gateway, message store, sync protocol |
| Time Split Today        | 0:00-0:45 chat SD                                                                |

**What Success Looks Like Today**

1:1 chat message flow drawn; online/offline message queue explained.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 156 – Tue | 17 19 Jan 2027 | 2 hours**

| Field                   | Detail                                                                   |
|-------------------------|--------------------------------------------------------------------------|
| Study Date              | Tue, 19 Jan 2027                                                         |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                |
| Calendar                | Day 156 in <a href="#">calendar/mern-mastery-study.ics</a>               |
| Calendar Week           | Week 23 of 26                                                            |
| Month Phase             | Month 6 – System Design                                                  |
| Exact Topic             | WebSockets, Online Presence, Last Seen                                   |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>             |
| Theory Section          | Design 3: Online Presence + Last Seen                                    |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                         |
| Specific Exercises      | Week 23 Tue: Presence service – heartbeat, connection registry           |
| LeetCode                | 1 medium timed                                                           |
| Weekend / Project Tasks | SD deliverable: Presence system with Redis pub/sub and heartbeat timeout |
| Time Split Today        | 0:00-0:45 presence theory                                                |

**What Success Looks Like Today**

Presence heartbeat interval and stale detection explained; last seen storage strategy defined.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 157 – Wed | 20 Jan 2027 | 2 hours**

| Field                   | Detail                                                                     |
|-------------------------|----------------------------------------------------------------------------|
| Study Date              | Wed, 20 Jan 2027                                                           |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                  |
| Calendar                | Day 157 in <a href="#">calendar/mern-mastery-study.ics</a>                 |
| Calendar Week           | Week 23 of 26                                                              |
| Month Phase             | Month 6 – System Design                                                    |
| Exact Topic             | HLD: YouTube (Lite) + Google Docs Intro                                    |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>               |
| Theory Section          | Design: Video Streaming (YouTube lite) + Collaborative editing basics      |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                           |
| Specific Exercises      | Week 23 Wed: Video upload, transcoding pipeline, adaptive bitrate          |
| LeetCode                | 1 medium timed                                                             |
| Weekend / Project Tasks | SD deliverable: YouTube lite – upload, transcode, CDN delivery, view count |
| Time Split Today        | 0:00-0:45 video SD                                                         |

**What Success Looks Like Today**

Video transcoding pipeline drawn; adaptive bitrate streaming explained at high level.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 158 – Thu | 21 Jan 2027 | 2 hours**

| Field                   | Detail                                                               |
|-------------------------|----------------------------------------------------------------------|
| Study Date              | Thu, 21 Jan 2027                                                     |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                            |
| Calendar                | Day 158 in <a href="#">calendar/mern-mastery-study.ics</a>           |
| Calendar Week           | Week 23 of 26                                                        |
| Month Phase             | Month 6 – System Design                                              |
| Exact Topic             | CDN, Video Storage, Blob vs Metadata                                 |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>         |
| Theory Section          | Core Concepts: CDN, Object Storage, Video chunking                   |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                     |
| Specific Exercises      | Week 23 Thu: CDN cache hierarchy, edge locations, cache invalidation |
| LeetCode                | 1 medium timed                                                       |
| Weekend / Project Tasks | SD notes: CDN strategy for static assets vs video segments           |
| Time Split Today        | 0:00-0:45 CDN theory                                                 |

**What Success Looks Like Today**

CDN edge caching vs origin explained; video chunked into HLS/DASH segments.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 159 – Fri | 22 Jan 2027 | 2 hours**

| Field                   | Detail                                                       |
|-------------------------|--------------------------------------------------------------|
| Study Date              | Fri, 22 Jan 2027                                             |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                    |
| Calendar                | Day 159 in <a href="#">calendar/mern-mastery-study.ics</a>   |
| Calendar Week           | Week 23 of 26                                                |
| Month Phase             | Month 6 – System Design                                      |
| Exact Topic             | SD Revision + 2 Timed Mediums                                |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a> |
| Theory Section          | Review Designs 3 + video                                     |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>             |
| Specific Exercises      | Week 23 Fri: Redraw chat system blind                        |
| LeetCode                | 2 mediums timed                                              |
| Weekend / Project Tasks | None                                                         |
| Time Split Today        | 0:00-0:30 blind chat SD                                      |

**What Success Looks Like Today**

Chat system redrawn in 30 min; 2 LC mediums under time.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 160 – Sat | 23 Jan 2027 | 8 hours**

| Field                   | Detail                                                              |
|-------------------------|---------------------------------------------------------------------|
| Study Date              | Sat, 23 Jan 2027                                                    |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                             |
| Calendar                | Day 160 in <a href="#">calendar/mern-mastery-study.ics</a>          |
| Calendar Week           | Week 23 of 26                                                       |
| Month Phase             | Month 6 – System Design                                             |
| Exact Topic             | Chat System SD Deep Dive + 3 LC                                     |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>        |
| Theory Section          | Design 3: Chat – group chat, message ordering, idempotency          |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                    |
| Specific Exercises      | Week 23 Sat: Group chat extension + 3 LC mediums                    |
| LeetCode                | 3 mediums timed                                                     |
| Weekend / Project Tasks | SD: Group chat design with message ordering and idempotent delivery |
| Time Split Today        | 0:00-0:45 group chat extension                                      |

**What Success Looks Like Today**

Group chat fan-out to members designed; message dedup with client-side IDs.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 161 – Sun | 24 Jan 2027 | 8 hours**

| Field                   | Detail                                                                    |
|-------------------------|---------------------------------------------------------------------------|
| Study Date              | Sun, 24 Jan 2027                                                          |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                   |
| Calendar                | Day 161 in <a href="#">calendar/mern-mastery-study.ics</a>                |
| Calendar Week           | Week 23 of 26                                                             |
| Month Phase             | Month 6 – System Design                                                   |
| Exact Topic             | Video Streaming SD + 3 LC                                                 |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>              |
| Theory Section          | Design: Video streaming – view count, recommendations intro               |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                          |
| Specific Exercises      | Week 23 Sun: Full video SD 45 min + 3 LC mediums                          |
| LeetCode                | 3 mediums timed                                                           |
| Weekend / Project Tasks | SD: Complete YouTube lite design with view counter (approximate counting) |
| Time Split Today        | 0:00-0:45 video SD complete                                               |

**What Success Looks Like Today**

Video SD complete with approximate view count (HyperLogLog or batch); 3 LC done.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 162 – Mon | 25 Jan 2027 | 2 hours**

| Field                   | Detail                                                                          |
|-------------------------|---------------------------------------------------------------------------------|
| Study Date              | Mon, 25 Jan 2027                                                                |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                       |
| Calendar                | Day 162 in <a href="#">calendar/mern-mastery-study.ics</a>                      |
| Calendar Week           | Week 24 of 26                                                                   |
| Month Phase             | Month 6 – System Design                                                         |
| Exact Topic             | HLD: Uber / Ride Matching (Basic)                                               |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                    |
| Theory Section          | Design: Uber – geospatial indexing, matching, ETA                               |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                |
| Specific Exercises      | Week 24 Mon: Uber – driver location, ride request, matching algorithm           |
| LeetCode                | 1 medium timed                                                                  |
| Weekend / Project Tasks | SD deliverable: Uber basic – geohash/quadtree for nearby drivers, matching flow |
| Time Split Today        | 0:00-0:45 Uber SD                                                               |

**What Success Looks Like Today**

Geospatial index choice justified; ride state machine (requested → matched → completed) drawn.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 163 – Tue | 26 Jan 2027 | 2 hours**

| Field                   | Detail                                                                              |
|-------------------------|-------------------------------------------------------------------------------------|
| Study Date              | Tue, 26 Jan 2027                                                                    |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                           |
| Calendar                | Day 163 in <a href="#">calendar/mern-mastery-study.ics</a>                          |
| Calendar Week           | Week 24 of 26                                                                       |
| Month Phase             | Month 6 – System Design                                                             |
| Exact Topic             | HLD: E-commerce Order System                                                        |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                        |
| Theory Section          | Design 5: E-commerce Order System                                                   |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                    |
| Specific Exercises      | Week 24 Tue: Order flow – cart, inventory, payment, fulfillment                     |
| LeetCode                | 1 medium timed                                                                      |
| Weekend / Project Tasks | SD deliverable: E-commerce order system with inventory reservation and saga pattern |
| Time Split Today        | 0:00-0:45 e-commerce SD                                                             |

**What Success Looks Like Today**

Order saga with compensating transactions explained; inventory lock during checkout.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 164 – Wed | 27 Jan 2027 | 2 hours**

| Field                   | Detail                                                                          |
|-------------------------|---------------------------------------------------------------------------------|
| Study Date              | Wed, 27 Jan 2027                                                                |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                       |
| Calendar                | Day 164 in <a href="#">calendar/mern-mastery-study.ics</a>                      |
| Calendar Week           | Week 24 of 26                                                                   |
| Month Phase             | Month 6 – System Design                                                         |
| Exact Topic             | HLD: Payment Gateway Flow                                                       |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                    |
| Theory Section          | Design: Payment flow – idempotency, webhooks, reconciliation                    |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                |
| Specific Exercises      | Week 24 Wed: Payment gateway – PCI scope, tokenization, webhook retry           |
| LeetCode                | 1 medium timed                                                                  |
| Weekend / Project Tasks | SD deliverable: Payment flow diagram with idempotency keys and webhook handling |
| Time Split Today        | 0:00-0:45 payment SD                                                            |

**What Success Looks Like Today**

Payment idempotency and webhook at-least-once handling explained; PCI compliance scope noted.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 165 – Thu | 28 Jan 2027 | 2 hours**

| Field                   | Detail                                                                               |
|-------------------------|--------------------------------------------------------------------------------------|
| Study Date              | Thu, 28 Jan 2027                                                                     |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                            |
| Calendar                | Day 165 in <a href="#">calendar/mern-mastery-study.ics</a>                           |
| Calendar Week           | Week 24 of 26                                                                        |
| Month Phase             | Month 6 – System Design                                                              |
| Exact Topic             | STAR Behavioral Stories – Write 8                                                    |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                         |
| Theory Section          | Interview prep: Behavioral (STAR method)                                             |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                     |
| Specific Exercises      | Week 24 Thu: Write 8 STAR stories (see behavioral table in practice file)            |
| LeetCode                | 1 medium timed                                                                       |
| Weekend / Project Tasks | Deliverable: 8 STAR stories document – 1 page each with Situation/Task/Action/Result |
| Time Split Today        | 0:00-1:30 write STAR stories                                                         |

**What Success Looks Like Today**

8 STAR stories written with quantified results; each deliverable in 2 min spoken format.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 166 – Fri | 29 Jan 2027 | 2 hours**

| Field                   | Detail                                                                             |
|-------------------------|------------------------------------------------------------------------------------|
| Study Date              | Fri, 29 Jan 2027                                                                   |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                          |
| Calendar                | Day 166 in <a href="#">calendar/mern-mastery-study.ics</a>                         |
| Calendar Week           | Week 24 of 26                                                                      |
| Month Phase             | Month 6 – System Design                                                            |
| Exact Topic             | Full Mock Interview – DSA + HLD + Behavioral                                       |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                       |
| Theory Section          | Full interview simulation guide                                                    |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                   |
| Specific Exercises      | Week 24 Fri: 45 min DSA + 45 min HLD + 30 min behavioral (use Pramp or self-timer) |
| LeetCode                | 1 unseen medium as mock DSA round                                                  |
| Weekend / Project Tasks | Mock interview log – record gaps, timing, communication feedback                   |
| Time Split Today        | 0:00-0:45 mock DSA                                                                 |

**What Success Looks Like Today**

Full 2-hour mock completed; gap list written with action items for Week 25.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 167 – Sat | 30 Jan 2027 | 8 hours**

| Field                   | Detail                                                                                 |
|-------------------------|----------------------------------------------------------------------------------------|
| Study Date              | Sat, 30 Jan 2027                                                                       |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                |
| Calendar                | Day 167 in <a href="#">calendar/mern-mastery-study.ics</a>                             |
| Calendar Week           | Week 24 of 26                                                                          |
| Month Phase             | Month 6 – System Design                                                                |
| Exact Topic             | Apply to 5 Companies + Resume Polish                                                   |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                           |
| Theory Section          | Job search strategy for foreign product companies                                      |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                       |
| Specific Exercises      | Week 24 Sat: Apply Adobe, Salesforce, Microsoft, Amazon, Uber – tailored cover letters |
| LeetCode                | 2 mediums timed                                                                        |
| Weekend / Project Tasks | Applications: 5 company applications submitted with customized resume bullets          |
| Time Split Today        | 0:00-4:00 applications                                                                 |

**What Success Looks Like Today**

5 applications submitted; resume PDF updated with live project URLs and LC count.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 168 – Sun | 31 Jan 2027 | 8 hours**

| Field                   | Detail                                                                    |
|-------------------------|---------------------------------------------------------------------------|
| Study Date              | Sun, 31 Jan 2027                                                          |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                   |
| Calendar                | Day 168 in <a href="#">calendar/mern-mastery-study.ics</a>                |
| Calendar Week           | Week 24 of 26                                                             |
| Month Phase             | Month 6 – System Design                                                   |
| Exact Topic             | GitHub Audit + LinkedIn + 220 LC Milestone                                |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>              |
| Theory Section          | Portfolio audit checklist                                                 |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                          |
| Specific Exercises      | Week 24 Sun: GitHub pinned repos, README audit, LinkedIn headline + About |
| LeetCode                | Review – target 220+ total LC solved                                      |
| Weekend / Project Tasks | Portfolio: GitHub pinned repos, LinkedIn updated, portfolio site optional |
| Time Split Today        | 0:00-4:00 GitHub + LinkedIn                                               |

**What Success Looks Like Today**

220+ LC milestone; LinkedIn live; GitHub shows 2 deployed projects with diagrams; 10 HLD systems designed.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

# WEEKS 25-26 – Active Interviewing (Days 169-182)

MERN MASTERY · 1 CR PRODUCT PREP · HARSH VARDHAN GUPTA

Day 169 – Mon |  17 1 Feb 2027 | 2 hours

| Field                   | Detail                                                                                   |
|-------------------------|------------------------------------------------------------------------------------------|
| Study Date              | Mon, 1 Feb 2027                                                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                                |
| Calendar                | Day 169 in <a href="calendar/mern-mastery-study.ics">calendar/mern-mastery-study.ics</a> |
| Calendar Week           | Week 25 of 26                                                                            |
| Month Phase             | Weeks 25-26 – Active Interviewing                                                        |
| Exact Topic             | Active Interviewing – DSA + SD + Applications (Day 169)                                  |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                              |
| Theory Section          | Review weakest pattern from REVIEW.md                                                    |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                         |
| Specific Exercises      | Daily interview loop: 2 LC timed + 1 SD redraw                                           |
| LeetCode                | <a href="#">LC 42: Trapping Rain Water</a> , <a href="#">LC 36: Valid Sudoku</a>         |
| Weekend / Project Tasks | Apply to 2 companies: Google, Atlassian – submit applications + track in spreadsheet     |
| Time Split Today        | 0:00-0:30 SD redraw                                                                      |

## What Success Looks Like Today

2 LC solved; 1 SD (URL Shortener) redrawn in 30 min; 2 applications submitted.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message

- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

## Day 170 – Tue | 2 Feb 2027 | 2 hours

| Field                   | Detail                                                                             |
|-------------------------|------------------------------------------------------------------------------------|
| Study Date              | Tue, 2 Feb 2027                                                                    |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                          |
| Calendar                | Day 170 in <a href="#">calendar/mern-mastery-study.ics</a>                         |
| Calendar Week           | Week 25 of 26                                                                      |
| Month Phase             | Weeks 25-26 – Active Interviewing                                                  |
| Exact Topic             | Active Interviewing – DSA + SD + Applications (Day 170)                            |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                       |
| Theory Section          | Design 4: Notification System review                                               |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                   |
| Specific Exercises      | Daily interview loop                                                               |
| LeetCode                | <a href="#">LC 48: Rotate Image</a> , <a href="#">LC 54: Spiral Matrix</a>         |
| Weekend / Project Tasks | Apply to 2 companies: Meta, Apple – customize resume for frontend/backend emphasis |
| Time Split Today        | 0:00-0:35 notification SD                                                          |

### What Success Looks Like Today

2 LC solved; Notification SD in 35 min; 2 applications submitted.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

## Day 171 – Wed | 3 Feb 2027 | 2 hours

| Field                   | Detail                                                                          |
|-------------------------|---------------------------------------------------------------------------------|
| Study Date              | Wed, 3 Feb 2027                                                                 |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                       |
| Calendar                | Day 171 in <a href="#">calendar/mern-mastery-study.ics</a>                      |
| Calendar Week           | Week 25 of 26                                                                   |
| Month Phase             | Weeks 25-26 – Active Interviewing                                               |
| Exact Topic             | Active Interviewing – DSA + SD + Applications (Day 171)                         |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                    |
| Theory Section          | Design 3: Chat System review                                                    |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                |
| Specific Exercises      | Daily interview loop                                                            |
| LeetCode                | <a href="#">LC 73: Set Matrix Zeroes</a> , <a href="#">LC 289: Game of Life</a> |
| Weekend / Project Tasks | Apply to 2 companies: Netflix, Spotify – highlight streaming/project experience |
| Time Split Today        | 0:00-0:35 chat SD                                                               |

### What Success Looks Like Today

2 LC solved; Chat SD in 35 min; 2 applications submitted.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 172 – Thu | 4 Feb 2027 | 2 hours**

| Field                   | Detail                                                                          |
|-------------------------|---------------------------------------------------------------------------------|
| Study Date              | Thu, 4 Feb 2027                                                                 |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                       |
| Calendar                | Day 172 in <a href="#">calendar/mern-mastery-study.ics</a>                      |
| Calendar Week           | Week 25 of 26                                                                   |
| Month Phase             | Weeks 25-26 – Active Interviewing                                               |
| Exact Topic             | Active Interviewing – DSA + SD + Applications (Day 172)                         |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                    |
| Theory Section          | Design 5: E-commerce Order System review                                        |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                |
| Specific Exercises      | Daily interview loop                                                            |
| LeetCode                | <a href="#">LC 79: Word Search</a> , <a href="#">LC 200: Number of Islands</a>  |
| Weekend / Project Tasks | Apply to 2 companies: Stripe, Razorpay – emphasize ShopLite payment integration |
| Time Split Today        | 0:00-0:35 e-commerce SD                                                         |

**What Success Looks Like Today**

2 LC solved; E-commerce SD in 35 min; 2 applications submitted.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 173 – Fri | 5 Feb 2027 | 2 hours**

| Field                   | Detail                                                                                  |
|-------------------------|-----------------------------------------------------------------------------------------|
| Study Date              | Fri, 5 Feb 2027                                                                         |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                               |
| Calendar                | Day 173 in <a href="#">calendar/mern-mastery-study.ics</a>                              |
| Calendar Week           | Week 25 of 26                                                                           |
| Month Phase             | Weeks 25-26 – Active Interviewing                                                       |
| Exact Topic             | Active Interviewing – DSA + SD + Applications (Day 173)                                 |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                            |
| Theory Section          | Design 2: Twitter Feed review                                                           |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                        |
| Specific Exercises      | Daily interview loop – Week 25 checkpoint                                               |
| LeetCode                | <a href="#">LC 721: Accounts Merge</a> , <a href="#">LC 684: Redundant Connection</a>   |
| Weekend / Project Tasks | Apply to 2 companies: Goldman Sachs, DE Shaw – track all 10 applications in spreadsheet |
| Time Split Today        | 0:00-0:35 feed SD                                                                       |

**What Success Looks Like Today**

10 total applications submitted this week; 2 LC; Feed SD in 35 min; application tracker complete.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 174 – Sat | 6 Feb 2027 | 8 hours**

| Field                   | Detail                                                                 |
|-------------------------|------------------------------------------------------------------------|
| Study Date              | Sat, 6 Feb 2027                                                        |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                |
| Calendar                | Day 174 in <a href="#">calendar/mern-mastery-study.ics</a>             |
| Calendar Week           | Week 25 of 26                                                          |
| Month Phase             | Weeks 25-26 – Active Interviewing                                      |
| Exact Topic             | Mock Interviews – Pramp / Peer (Day 174)                               |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>           |
| Theory Section          | Mock interview debrief template                                        |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                       |
| Specific Exercises      | Schedule 2 mock interviews on Pramp or with peer – 1 DSA + 1 SD each   |
| LeetCode                | Use mock platform problems – do not pick familiar LC                   |
| Weekend / Project Tasks | Mock log: record interviewer feedback, timing, communication score 1-5 |
| Time Split Today        | Weekend: 0:00-4:00 mock 1 (DSA+SD)                                     |

**What Success Looks Like Today**

2 mock interviews completed; debrief notes written with 3 improvement actions each.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 175 – Sun | 7 Feb 2027 | 8 hours**

| Field                   | Detail                                                                     |
|-------------------------|----------------------------------------------------------------------------|
| Study Date              | Sun, 7 Feb 2027                                                            |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                    |
| Calendar                | Day 175 in <a href="#">calendar/mern-mastery-study.ics</a>                 |
| Calendar Week           | Week 25 of 26                                                              |
| Month Phase             | Weeks 25-26 – Active Interviewing                                          |
| Exact Topic             | Mock Interviews – Pramp / Peer (Day 175)                                   |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>               |
| Theory Section          | Behavioral mock – STAR story delivery                                      |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                           |
| Specific Exercises      | 2 more mocks: focus on behavioral + system design communication            |
| LeetCode                | Warm-up: 1 easy + 1 medium before mocks                                    |
| Weekend / Project Tasks | Behavioral mock: deliver 4 STAR stories in 30 min session; record yourself |
| Time Split Today        | Weekend: 0:00-1:00 LC warm-up                                              |

**What Success Looks Like Today**

2 mocks done; behavioral stories delivered smoothly under 2 min each; gap list updated.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 176 – Mon | 8 Feb 2027 | 2 hours**

| Field                   | Detail                                                                                                                                                                                   |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Study Date              | Mon, 8 Feb 2027                                                                                                                                                                          |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                                                                                                                                |
| Calendar                | Day 176 in <a href="#">calendar/mern-mastery-study.ics</a>                                                                                                                               |
| Calendar Week           | Week 26 of 26                                                                                                                                                                            |
| Month Phase             | Weeks 25-26 – Active Interviewing                                                                                                                                                        |
| Exact Topic             | Weak Pattern Revision – Two Pointers + Sliding Window                                                                                                                                    |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                                                                                                                              |
| Theory Section          | Patterns 1-2 deep review from REVIEW.md failures                                                                                                                                         |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                                                                                                                         |
| Specific Exercises      | Redo 4 failed problems from Patterns 1-2                                                                                                                                                 |
| LeetCode                | <a href="#">LC 16: 3Sum Closest</a> , <a href="#">LC 209: Minimum Size Subarray Sum</a> , <a href="#">LC 904: Fruit Into Baskets</a> , <a href="#">LC 1004: Max Consecutive Ones III</a> |
| Weekend / Project Tasks | Update REVIEW.md – mark revised patterns as confident or still weak                                                                                                                      |
| Time Split Today        | Weekend: 0:00-4:00 4 revision LC                                                                                                                                                         |

**What Success Looks Like Today**

≥3/4 revision problems solved faster than first attempt; pattern templates rewritten.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 177 – Tue | 9 Feb 2027 | 2 hours**

| Field                   | Detail                                                                                                                                                                            |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Study Date              | Tue, 9 Feb 2027                                                                                                                                                                   |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                                                                                                                         |
| Calendar                | Day 177 in <a href="#">calendar/mern-mastery-study.ics</a>                                                                                                                        |
| Calendar Week           | Week 26 of 26                                                                                                                                                                     |
| Month Phase             | Weeks 25-26 – Active Interviewing                                                                                                                                                 |
| Exact Topic             | Weak Pattern Revision – Trees + Graphs + DP                                                                                                                                       |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                                                                                                                       |
| Theory Section          | Patterns 6-8 deep review from REVIEW.md failures                                                                                                                                  |
| Practice File           | <a href="#">practice/week-17-20-exercises.md</a>                                                                                                                                  |
| Specific Exercises      | Redo 4 failed problems from Patterns 6-8                                                                                                                                          |
| LeetCode                | <a href="#">LC 297: Serialize and Deserialize Binary Tree</a> , <a href="#">LC 126: Word Ladder II</a> , <a href="#">LC 62: Unique Paths</a> , <a href="#">LC 494: Target Sum</a> |
| Weekend / Project Tasks | Update DP cheat sheet with 5 new patterns from revision session                                                                                                                   |
| Time Split Today        | Weekend: 0:00-4:00 4 revision LC                                                                                                                                                  |

**What Success Looks Like Today**

≥2/4 hard revisions solved or thoroughly studied; graph/DP templates updated.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 178 – Wed | 10 Feb 2027 | 2 hours**

| Field                   | Detail                                                                                                  |
|-------------------------|---------------------------------------------------------------------------------------------------------|
| Study Date              | Wed, 10 Feb 2027                                                                                        |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                                               |
| Calendar                | Day 178 in <a href="#">calendar/mern-mastery-study.ics</a>                                              |
| Calendar Week           | Week 26 of 26                                                                                           |
| Month Phase             | Weeks 25-26 – Active Interviewing                                                                       |
| Exact Topic             | Weak Pattern Revision – System Design Gaps                                                              |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                                            |
| Theory Section          | Review all 10 HLD designs – identify 3 weakest                                                          |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                                        |
| Specific Exercises      | Redraw 3 weakest SDs blind – 30 min each                                                                |
| LeetCode                | <a href="#">LC 380: Design Add and Search DS</a> , <a href="#">LC 208: Implement Trie (Prefix Tree)</a> |
| Weekend / Project Tasks | SD gap doc: for each weak design, write 5 likely follow-up questions + answers                          |
| Time Split Today        | Weekend: 0:00-1:30 SD 1                                                                                 |

**What Success Looks Like Today**

3 weakest SDs redrawn without major gaps; follow-up Q&A doc complete.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 179 – Thu | 11 Feb 2027 | 2 hours**

| Field                   | Detail                                                                                      |
|-------------------------|---------------------------------------------------------------------------------------------|
| Study Date              | Thu, 11 Feb 2027                                                                            |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                                   |
| Calendar                | Day 179 in <a href="#">calendar/mern-mastery-study.ics</a>                                  |
| Calendar Week           | Week 26 of 26                                                                               |
| Month Phase             | Weeks 25-26 – Active Interviewing                                                           |
| Exact Topic             | Offer Negotiation Research – Compensation Basics                                            |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                                |
| Theory Section          | Appendix: Salary negotiation for India → foreign product companies                          |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                            |
| Specific Exercises      | Research: levels.fyi, Glassdoor, Blind – compile target company ranges                      |
| LeetCode                | 1 medium timed (maintain sharpness)                                                         |
| Weekend / Project Tasks | Negotiation prep doc: target range (30 LPA+), competing offers strategy, benefits checklist |
| Time Split Today        | 0:00-1:00 research levels.fyi                                                               |

**What Success Looks Like Today**

Salary range document for top 5 target companies; negotiation scripts written for HR call.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 180 – Fri | 12 Feb 2027 | 2 hours**

| Field                   | Detail                                                                                         |
|-------------------------|------------------------------------------------------------------------------------------------|
| Study Date              | Fri, 12 Feb 2027                                                                               |
| Time Slots (IST)        | 19:30-20:30 · 21:00-22:00                                                                      |
| Calendar                | Day 180 in <a href="#">calendar/mern-mastery-study.ics</a>                                     |
| Calendar Week           | Week 26 of 26                                                                                  |
| Month Phase             | Weeks 25-26 – Active Interviewing                                                              |
| Exact Topic             | Offer Negotiation – Mock HR Call + Email Templates                                             |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                                   |
| Theory Section          | Negotiation: counter-offer email templates, equity basics                                      |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                               |
| Specific Exercises      | Write counter-offer email template; practice mock HR negotiation aloud                         |
| LeetCode                | 1 medium timed                                                                                 |
| Weekend / Project Tasks | Deliverables: counter-offer email template, benefits comparison spreadsheet, mock HR recording |
| Time Split Today        | 0:00-0:45 email templates                                                                      |

**What Success Looks Like Today**

Counter-offer email drafted; mock HR negotiation practiced 2x aloud; equity/ESOP basics understood.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 181 – Sat | 13 Feb 2027 | 8 hours**

| Field                   | Detail                                                                                                 |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Study Date              | Sat, 13 Feb 2027                                                                                       |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                                |
| Calendar                | Day 181 in <a href="#">calendar/mern-mastery-study.ics</a>                                             |
| Calendar Week           | Week 26 of 26                                                                                          |
| Month Phase             | Weeks 25-26 – Active Interviewing                                                                      |
| Exact Topic             | Continue Interviews – OA + Phone Screen Prep                                                           |
| Theory File             | <a href="#">detailed/theory/08-dsa-patterns-complete.md</a>                                            |
| Theory Section          | OA patterns: timed tests, machine coding prep                                                          |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                                       |
| Specific Exercises      | Respond to any recruiter emails; complete pending OAs within 48h                                       |
| LeetCode                | <a href="#">LC 528: Random Pick with Weight</a> , <a href="#">LC 295: Find Median from Data Stream</a> |
| Weekend / Project Tasks | Interview tracker: log every OA, phone screen, onsite with dates and outcomes                          |
| Time Split Today        | 0:00-0:30 check applications                                                                           |

**What Success Looks Like Today**

All pending OAs completed; interview tracker up to date; 2 LC maintained.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

**Day 182 – Sun | 14 Feb 2027 | 8 hours**

| Field                   | Detail                                                                                                 |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Study Date              | Sun, 14 Feb 2027                                                                                       |
| Time Slots (IST)        | 09:00-12:00 · 15:00-17:30 · 18:00-20:30                                                                |
| Calendar                | Day 182 in <a href="#">calendar/mern-mastery-study.ics</a>                                             |
| Calendar Week           | Week 26 of 26                                                                                          |
| Month Phase             | Weeks 25-26 – Active Interviewing                                                                      |
| Exact Topic             | Program Graduation – Final Audit + Continue Until Offer                                                |
| Theory File             | <a href="#">detailed/theory/10-system-design-complete.md</a>                                           |
| Theory Section          | Final portfolio audit – all milestones checklist                                                       |
| Practice File           | <a href="#">practice/week-21-24-exercises.md</a>                                                       |
| Specific Exercises      | Complete final audit against all milestones; plan post-182 daily routine                               |
| LeetCode                | 1 medium + 1 hard (celebrate 182 days – push toward 230+ total)                                        |
| Weekend / Project Tasks | Final deliverable: 182-day completion certificate (self-signed), post-program 2hr/day maintenance plan |
| Time Split Today        | Weekend: 0:00-2:00 final LC                                                                            |

**What Success Looks Like Today**

All milestones checked; 230+ LC; 2 deployed projects live; actively interviewing with offer pipeline; post-182 plan written.

- ☐ Theory section read and highlighted
- ☐ Practice exercises complete (or timed LC done)
- ☐ Code committed to GitHub with meaningful message
- ☐ 2-minute aloud explanation of today's main concept recorded or spoken

## Appendix A – LeetCode Master Index by Day

| Day | Problem(s)                                                                                                                                                  |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | <a href="#">LC 1: Two Sum</a>                                                                                                                               |
| 2   | None (save LC for Fri/Sat)                                                                                                                                  |
| 5   | <a href="#">LC 1: Two Sum</a>                                                                                                                               |
| 6   | <a href="#">LC 242: Valid Anagram</a> , <a href="#">LC 121: Best Time to Buy and Sell Stock</a>                                                             |
| 7   | <a href="#">LC 217: Contains Duplicate</a> , <a href="#">LC 53: Maximum Subarray</a>                                                                        |
| 12  | <a href="#">LC 88: Merge Sorted Array</a>                                                                                                                   |
| 13  | <a href="#">LC 27: Move Zeroes</a> , <a href="#">LC 977: Squares of a Sorted Array</a>                                                                      |
| 14  | <a href="#">LC 238: Product of Array Except Self</a> , <a href="#">LC 9: Palindrome Number</a> , <a href="#">LC 26: Remove Duplicates from Sorted Array</a> |
| 19  | <a href="#">LC 20: Valid Parentheses</a>                                                                                                                    |
| 20  | <a href="#">LC 206: Reverse Linked List</a> , <a href="#">LC 155: Min Stack</a>                                                                             |
| 21  | <a href="#">LC 14: Longest Common Prefix</a> , <a href="#">LC 21: Merge Two Sorted Lists</a>                                                                |
| 26  | <a href="#">LC 704: Binary Search</a>                                                                                                                       |
| 27  | <a href="#">LC 35: Search Insert Position</a> , <a href="#">LC 278: First Bad Version</a>                                                                   |
| 28  | Review LC 1-20 – list any unsolved in REVIEW.md                                                                                                             |
| 33  | <a href="#">LC 21: Merge Two Sorted Lists</a>                                                                                                               |
| 34  | <a href="#">LC 26: Remove Duplicates</a> , <a href="#">LC 27: Remove Element</a>                                                                            |
| 35  | <a href="#">LC 58: Length of Last Word</a> , <a href="#">LC 66: Plus One</a> , <a href="#">LC 67: Add Binary</a>                                            |
| 40  | <a href="#">LC 70: Climbing Stairs</a>                                                                                                                      |
| 41  | <a href="#">LC 69: Sqrt(x)</a> , <a href="#">LC 83: Remove Duplicates from List</a> , <a href="#">LC 94: Binary Tree Inorder Traversal</a>                  |

|    |                                                                                                                                                                              |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 42 | <a href="#">LC 100: Same Tree</a> , <a href="#">LC 101: Symmetric Tree</a> , <a href="#">LC 104: Maximum Depth of Binary Tree</a>                                            |
| 47 | <a href="#">LC 104: Maximum Depth of Binary Tree</a>                                                                                                                         |
| 48 | <a href="#">LC 108: Convert Sorted Array to BST</a> , <a href="#">LC 110: Balanced Binary Tree</a>                                                                           |
| 49 | <a href="#">LC 111: Minimum Depth of Binary Tree</a> , <a href="#">LC 112: Path Sum</a>                                                                                      |
| 54 | <a href="#">LC 226: Invert Binary Tree</a>                                                                                                                                   |
| 55 | <a href="#">LC 118: Pascal's Triangle</a> , <a href="#">LC 136: Single Number</a>                                                                                            |
| 56 | Review all Month 2 LC – target 60+ total solved                                                                                                                              |
| 61 | <a href="#">LC 141: Linked List Cycle</a>                                                                                                                                    |
| 62 | <a href="#">LC 160: Intersection of Two Linked Lists</a> , <a href="#">LC 167: Two Sum II</a> , <a href="#">LC 155: Min Stack</a>                                            |
| 63 | Review Week 9 LC list                                                                                                                                                        |
| 68 | <a href="#">LC 167: Two Sum II</a>                                                                                                                                           |
| 69 | <a href="#">LC 169: Majority Element</a> , <a href="#">LC 171: Excel Sheet Column Number</a>                                                                                 |
| 70 | <a href="#">LC 190: Reverse Bits</a> , <a href="#">LC 191: Number of 1 Bits</a> , <a href="#">LC 202: Happy Number</a> , <a href="#">LC 203: Remove Linked List Elements</a> |
| 75 | <a href="#">LC 206: Reverse Linked List</a>                                                                                                                                  |
| 76 | <a href="#">LC 217: Contains Duplicate</a> , <a href="#">LC 219: Contains Duplicate II</a>                                                                                   |
| 77 | <a href="#">LC 226: Invert Binary Tree</a> , <a href="#">LC 230: Kth Smallest Element in a BST</a>                                                                           |
| 82 | <a href="#">LC 230: Kth Smallest Element in a BST</a>                                                                                                                        |
| 83 | <a href="#">LC 238: Product of Array Except Self</a> , <a href="#">LC 242: Valid Anagram</a>                                                                                 |
| 84 | <a href="#">LC 252: Meeting Rooms</a> , <a href="#">LC 268: Missing Number</a>                                                                                               |
| 89 | <a href="#">LC 253: Meeting Rooms II</a>                                                                                                                                     |

|     |                                                                                                                                                        |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 90  | <a href="#">LC 283: Move Zeroes</a> , <a href="#">LC 287: Find the Duplicate Number</a>                                                                |
| 91  | <a href="#">LC 300: Longest Increasing Subsequence</a> , <a href="#">LC 322: Coin Change</a>                                                           |
| 96  | <a href="#">LC 322: Coin Change</a>                                                                                                                    |
| 97  | <a href="#">LC 347: Top K Frequent Elements</a> , <a href="#">LC 349: Intersection of Two Arrays</a>                                                   |
| 98  | Review TaskFlow-week LC list                                                                                                                           |
| 103 | <a href="#">LC 300: Longest Increasing Subsequence</a>                                                                                                 |
| 104 | <a href="#">LC 371: Sum of Two Integers</a> , <a href="#">LC 373: Find K Pairs with Smallest Sums</a>                                                  |
| 105 | <a href="#">LC 378: Kth Smallest Element in a Sorted Matrix</a> , <a href="#">LC 392: Is Subsequence</a>                                               |
| 110 | <a href="#">LC 347: Top K Frequent Elements</a>                                                                                                        |
| 111 | <a href="#">LC 399: Evaluate Division</a> , <a href="#">LC 416: Partition Equal Subset Sum</a>                                                         |
| 112 | <a href="#">LC 424: Longest Repeating Character Replacement</a> , <a href="#">LC 435: Non-overlapping Intervals</a>                                    |
| 113 | <a href="#">LC 15: 3Sum</a> , <a href="#">LC 11: Container With Most Water</a>                                                                         |
| 114 | <a href="#">LC 3: Longest Substring Without Repeating Characters</a> , <a href="#">LC 424: Longest Repeating Character Replacement</a>                 |
| 115 | <a href="#">LC 49: Group Anagrams</a> , <a href="#">LC 128: Longest Consecutive Sequence</a>                                                           |
| 116 | <a href="#">LC 560: Subarray Sum Equals K</a> , <a href="#">LC 76: Minimum Window Substring</a>                                                        |
| 117 | Self-selected – pick 2 problems you failed or took >45 min                                                                                             |
| 118 | <a href="#">LC 239: Sliding Window Maximum</a> , <a href="#">LC 438: Find All Anagrams in a String</a> , <a href="#">LC 567: Permutation in String</a> |
| 119 | Optional warm-up: <a href="#">LC 146 LRU Cache</a>                                                                                                     |
| 120 | <a href="#">LC 33: Search in Rotated Sorted Array</a> , <a href="#">LC 153: Find Minimum in Rotated Sorted Array</a>                                   |

|     |                                                                                                                                                                      |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 121 | <a href="#">LC 74: Search a 2D Matrix</a> , <a href="#">LC 875: Koko Eating Bananas</a>                                                                              |
| 122 | <a href="#">LC 739: Daily Temperatures</a> , <a href="#">LC 84: Largest Rectangle in Histogram</a>                                                                   |
| 123 | <a href="#">LC 25: Reverse Nodes in k-Group</a> , <a href="#">LC 138: Copy List with Random Pointer</a>                                                              |
| 124 | Self-selected revision problems                                                                                                                                      |
| 125 | <a href="#">LC 56: Merge Intervals</a> , <a href="#">LC 57: Insert Interval</a> , <a href="#">LC 452: Minimum Number of Arrows to Burst Balloons</a>                 |
| 127 | <a href="#">LC 102: Binary Tree Level Order Traversal</a> , <a href="#">LC 105: Construct Binary Tree from Preorder and Inorder Traversal</a>                        |
| 128 | <a href="#">LC 199: Binary Tree Right Side View</a> , <a href="#">LC 236: Lowest Common Ancestor of a Binary Tree</a>                                                |
| 129 | <a href="#">LC 200: Number of Islands</a> , <a href="#">LC 133: Clone Graph</a>                                                                                      |
| 130 | <a href="#">LC 207: Course Schedule</a> , <a href="#">LC 210: Course Schedule II</a>                                                                                 |
| 131 | Self-selected revision                                                                                                                                               |
| 132 | <a href="#">LC 127: Word Ladder</a> , <a href="#">LC 130: Surrounded Regions</a> , <a href="#">LC 417: Pacific Atlantic Water Flow</a>                               |
| 134 | <a href="#">LC 198: House Robber</a> , <a href="#">LC 213: House Robber II</a>                                                                                       |
| 135 | <a href="#">LC 322: Coin Change</a> , <a href="#">LC 518: Coin Change II</a>                                                                                         |
| 136 | <a href="#">LC 1143: Longest Common Subsequence</a> , <a href="#">LC 72: Edit Distance</a>                                                                           |
| 137 | <a href="#">LC 312: Burst Balloons</a> OR <a href="#">LC 91: Decode Ways</a>                                                                                         |
| 138 | Pick 5 from: <a href="#">LC 70</a> , <a href="#">139</a> , <a href="#">198</a> , <a href="#">322</a> , <a href="#">518</a> , <a href="#">1143</a> (unsolved or weak) |
| 139 | <a href="#">LC 139: Word Break</a> , <a href="#">LC 152: Maximum Product Subarray</a> , <a href="#">LC 416: Partition Equal Subset Sum</a>                           |
| 140 | Review LC count – target 200+ total solved                                                                                                                           |
| 141 | 1 medium timed (self-selected from weak list)                                                                                                                        |
| 142 | 1 medium timed                                                                                                                                                       |

|     |                                                                 |
|-----|-----------------------------------------------------------------|
| 143 | 1 medium timed                                                  |
| 144 | 1 medium timed                                                  |
| 145 | 2 mediums timed (35 min each)                                   |
| 146 | 3 self-selected mediums (timed 35 min each, pick weak patterns) |
| 147 | 3 mediums timed                                                 |
| 148 | 1 medium timed                                                  |
| 149 | 1 medium timed                                                  |
| 150 | 1 medium timed                                                  |
| 151 | 1 medium timed                                                  |
| 152 | 2 mediums timed                                                 |
| 153 | 3 mediums timed                                                 |
| 154 | 3 mediums timed                                                 |
| 155 | 1 medium timed                                                  |
| 156 | 1 medium timed                                                  |
| 157 | 1 medium timed                                                  |
| 158 | 1 medium timed                                                  |
| 159 | 2 mediums timed                                                 |
| 160 | 3 mediums timed                                                 |
| 161 | 3 mediums timed                                                 |
| 162 | 1 medium timed                                                  |
| 163 | 1 medium timed                                                  |
| 164 | 1 medium timed                                                  |
| 165 | 1 medium timed                                                  |
| 166 | 1 unseen medium as mock DSA round                               |
| 167 | 2 mediums timed                                                 |
| 168 | Review – target 220+ total LC solved                            |

|     |                                                                                                                                                                                          |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 169 | <a href="#">LC 42: Trapping Rain Water</a> , <a href="#">LC 36: Valid Sudoku</a>                                                                                                         |
| 170 | <a href="#">LC 48: Rotate Image</a> , <a href="#">LC 54: Spiral Matrix</a>                                                                                                               |
| 171 | <a href="#">LC 73: Set Matrix Zeroes</a> , <a href="#">LC 289: Game of Life</a>                                                                                                          |
| 172 | <a href="#">LC 79: Word Search</a> , <a href="#">LC 200: Number of Islands</a>                                                                                                           |
| 173 | <a href="#">LC 721: Accounts Merge</a> , <a href="#">LC 684: Redundant Connection</a>                                                                                                    |
| 174 | Use mock platform problems – do not pick familiar LC                                                                                                                                     |
| 175 | Warm-up: 1 easy + 1 medium before mocks                                                                                                                                                  |
| 176 | <a href="#">LC 16: 3Sum Closest</a> , <a href="#">LC 209: Minimum Size Subarray Sum</a> , <a href="#">LC 904: Fruit Into Baskets</a> , <a href="#">LC 1004: Max Consecutive Ones III</a> |
| 177 | <a href="#">LC 297: Serialize and Deserialize Binary Tree</a> , <a href="#">LC 126: Word Ladder II</a> , <a href="#">LC 62: Unique Paths</a> , <a href="#">LC 494: Target Sum</a>        |
| 178 | <a href="#">LC 380: Design Add and Search DS</a> , <a href="#">LC 208: Implement Trie (Prefix Tree)</a>                                                                                  |
| 179 | 1 medium timed (maintain sharpness)                                                                                                                                                      |
| 180 | 1 medium timed                                                                                                                                                                           |
| 181 | <a href="#">LC 528: Random Pick with Weight</a> , <a href="#">LC 295: Find Median from Data Stream</a>                                                                                   |
| 182 | 1 medium + 1 hard (celebrate 182 days – push toward 230+ total)                                                                                                                          |

## Appendix B – Project Deliverables Timeline

| Day     | Project Milestone                             |
|---------|-----------------------------------------------|
| 6       | GitHub repo <code>mern-mastery</code> created |
| 20-21   | CLI Todo App complete                         |
| 27-28   | TS Catalog API complete                       |
| 35      | Multi-step form wizard                        |
| 41-42   | User directory app                            |
| 48-49   | E-commerce listing UI                         |
| 55-56   | Next.js blog deployed to Vercel               |
| 69-70   | Auth microservice complete                    |
| 76-77   | Blog API (MongoDB)                            |
| 83-84   | Blog API (PostgreSQL) + Redis                 |
| 90-98   | TaskFlow complete + Docker                    |
| 99-111  | ShopLite complete + deployed                  |
| 119     | LRU Cache LLD                                 |
| 126     | Rate Limiter + Parking Lot LLD                |
| 133     | Splitwise + Vending Machine LLD               |
| 140     | BookMyShow LLD                                |
| 141-168 | 10 HLD system designs                         |
| 167-173 | 10+ job applications                          |
| 182     | Program graduation audit                      |

## Appendix C – Daily Checklist Template (Printable)

Date: \_\_\_\_\_ Day #: \_\_\_\_\_ Hours planned: \_\_\_\_\_

- ☐ Theory section read (file: \_\_\_\_\_)
- ☐ Exercises complete (file: \_\_\_\_\_)
- ☐ LeetCode solved: #\_\_\_\_\_ Time: \_\_\_\_\_ min
- ☐ Project task advanced: \_\_\_\_\_
- ☐ Git commit hash: \_\_\_\_\_
- ☐ 2-min aloud explanation done
- ☐ Tomorrow preview read (Day \_\_\_\_\_)

Notes / blockers:

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*End of 182-Day Complete Roadmap · MERN Mastery Bootcamp · Target: 7.5 → 30+ LPA*

— LeetCode Master Index by Day

| Day | Problem(s) |
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